Algebraic Theory of Fields

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In the present essay the concept "field" is taken in the same abstract and general manner as in *H. Weber's* investigations *on the general foundations of Galois equation theory*[1], namely as a system of elements with two operations, addition and multiplication, which are subject to the associative and commutative laws, are connected by the distributive law, and admit unrestricted and unique inverse operations[2]. Whereas in *Weber*, however, the aim is a general treatment of *Galois* theory independent of the numerical meaning of the elements, for us the concept of field itself stands at the center of interest. *To obtain a survey of all possible types of fields and to establish, in their basic outlines, their relations to one another* may be taken as the program of this work[3]. Since in this connection the distinctions belonging to arithmetic in the narrower sense between integral and fractional quantities were not to be pursued further, the title *Algebraic Theory of Fields* was chosen.

The tendency just indicated also prescribes the path which we have to take. We shall begin with the formation of the simplest fields and then consider the methods by which one passes from a given field, by extension, to a more comprehensive one. The first task leads to the concept of the *prime field* and to a distinction among fields according to their *characteristic*. In every field there is contained a smallest field, or prime field; it is either of the type of the field of rational numbers (characteristic 0) or of the type of the system of residue classes modulo a prime number p (characteristic p). In many questions the characteristic is without significance; in others, however, a deep difference appears between fields of characteristic 0 and those of prime characteristic. An analogous distinction is prompted by the consideration of the so-called *simple* extensions, which can be obtained by adjoining a single element. According as such an element does or does not satisfy an algebraic equation over the ground field, it is called *algebraic* or *transcendental* (with respect to the ground field). By an *algebraic extension* we mean, quite generally, any extension all of whose elements are algebraic, whereas every other extension is called *transcendental*. Among the algebraic extensions are the *finite* ones, characterized by the fact that all elements of the extension field can be represented linearly and homogeneously, with coefficients from the ground field, by means of a finite number of elements of the extension field. In the literature, usually only these extensions are called algebraic. The investigations belonging here form the content of *Galois* theory.

Following the procedure of *Cauchy*, who showed that the introduction of the complex numbers can be avoided by considerations of congruences modulo $x^2 + 1$[4], *Kronecker* proved that any introduction of irrationalities in the treatment of algebraic problems[5] can likewise be avoided if one replaces equations by congruences modulo systems of moduli. We use the *Kroneckerian* idea first of all to prove that, given a field $\mathfrak{K}$ and a function $\varphi(x)$ irreducible in $\mathfrak{K}$, one can find an extension field which arises from $\mathfrak{K}$ by adjoining a root of $\varphi(x)$. In doing so, however, by introducing a symbol through which we represent all elements of the extension field, and which itself becomes a root of $\varphi(x)$, we pass back from the notation of *congruences* to *equations*. The further investigations are built synthetically upon this first result. The method followed here may be regarded as just as *purely arithmetical* as the method of congruences, to which the transition is of course always possible and, although sometimes inconvenient, by no means to be missed.

[1] Math. Ann. 43, p. 521.

[2] Only division by zero is to be excluded.

[3] I was led to these general investigations especially by *Hensel's Theory of Algebraic Numbers* (Leipzig, 1908), in which the field of p-adic numbers forms the point of departure: a field which is to be counted neither among function fields nor among number fields in the ordinary sense of the word.

[4] Exerc. d'anal. et de phys. math. 4 (1847), p. 87.

[5] For *Kronecker* it is always only a matter of problems requiring *finite* extensions.

From the domain of finite extensions, it is especially to be emphasized that *Galois* theory, as we find it for algebraic number fields, does not exist in full scope for all fields. A large part of that theory rests on the theorem that every equation with vanishing discriminant (multiple roots) must be reducible. But precisely this theorem is not valid in general. This gives us occasion to divide fields into *perfect* and *imperfect* ones. To the former belong all fields of characteristic 0; among the fields of characteristic p, however, only those within which extraction of the p-th root is possible without exception. I have gone somewhat further into the properties of imperfect fields, with their laws partly diverging from ordinary *Galois* theory, than would have been strictly necessary for the continuous investigation. The fields with only finitely many elements still required special treatment. Although in them, as in all other perfect fields, the theorems of *Galois* theory hold without restriction[6], the proof must nevertheless be shaped partly differently.

Under the title "*a fundamental theorem of general arithmetic*" *Kronecker* published in this journal a paper[7] which deals with determining a prime modulus system for which a given polynomial in one variable can be represented as a product of linear factors. That the existence of such a prime modulus system is of fundamental importance for general arithmetic is beyond question; the reference to the fundamental theorem of algebra, [by which *Kronecker* justifies calling his theorem a fundamental theorem, does not appear quite apt. The fundamental theorem of algebra gives more: in the domain of general arithmetic, the corresponding statement would be the proof of a prime modulus system for which *every* function of the rationality domain under consideration decomposes into linear factors. The treatment of this problem lay outside the circle of *Kronecker's* investigations insofar as its solution requires the introduction of *infinitely many* indeterminates. If we translate the problem into our manner of speaking, it is a question of extending a given field to an *algebraically closed* one, that is, to a field in which every polynomial in one indeterminate decomposes into linear factors. This task belongs to the domain of *infinite* algebraic extensions; in § 16 it will first be solved for the case of a prime field of prime characteristic. The method used there is also applicable in other cases, for instance in the natural rationality domains considered by *Kronecker* alone; it requires no appeal to principles which would not be acknowledged by all mathematicians. Matters stand otherwise with the proof that our problem, if one further requires that the extension field contain no superfluous elements, has essentially[8] only *one* solution. This proof cannot be carried out even for the field of rational numbers, for example, without use of the *principle of choice*. This principle also appears unavoidable if one wants to prove the existence of an algebraically closed extension for *every arbitrary* field. The *well-ordering theorem* of *Zermelo*, resting on the principle of choice, forms an essential auxiliary means for this proof[9]. — Many mathematicians still stand opposed to the principle of choice. With the increasing recognition *that there are questions in mathematics which cannot be decided without this principle*, resistance to it should diminish more and more. On the other hand, in the interest of purity of method it appears appropriate to avoid the principle named as far as the nature of the question does not require its use. I have endeavored to make this boundary stand out sharply.

This also applies to the discussions of the last section, which deals with *transcendental* extensions. Among them the simplest appear to be the *purely transcendental* ones. A purely transcendental extension amounts essentially to the adjunction of indeterminates, which may form a finite or also an infinite set of arbitrary cardinality. The well-ordering theorem leads immediately to the recognition that every extension can be decomposed into a purely transcendental and an algebraic one.

[6]This also holds, for example, with respect to the questions settled by *Weber* for fields with infinitely many elements, but left open for finite fields; see the conclusion of the cited work.

[7]Vol. 100, p. 490. = Works III, I, p. 209.

[8]For the precise explanation of this expression, see § 5.

[9]The first application of the well-ordering theorem to an arithmetical problem is found, to my knowledge, in *Hamel*, Mathem. Ann. 60, p. 459.

From here it is easy to characterize the position which the fields considered by *Kronecker* occupy within the totality of fields. In *Kronecker*, the concept of field develops in three stages[10]. The point of departure is the "*absolute rationality domain*," that is, the field of rational numbers, hence a particular prime field. From it one arrives, by adjoining a finite number of indeterminates, hence by a special purely transcendental extension, at the "*natural rationality domain*"; from this, finally, by adjoining an algebraic quantity, that is, by a special algebraic extension, at the "*genus domain*." — If, instead, one starts from an *arbitrary* prime field, undertakes an *arbitrary* purely transcendental extension and then an *arbitrary* algebraic extension, one thereby has a procedure by which one can arrive at *every* field.

The present essay treats only the *basic outlines* of a general theory of fields. I intend to let more extensive investigations, as well as applications to geometry, number theory, and the theory of functions, follow in several further papers.

Overview of contents: I. Foundations (§§ 1–6); II. Algebraic extensions (§§ 7–15); III. Infinite algebraic extensions (§§ 16–21); IV. Transcendental extensions (§§ 22–24). — (Detailed table of contents at the end of the work.)

§ 1.

Systems with double composition. Isomorphism. Fields.

If $\mathfrak{S}$ denotes a system of elements, then by a *law of composition of the system* $\mathfrak{S}$ we mean any specification by which to each pair a, b of elements of this system, which may also be equal, an element c of the same system is assigned. — By a *system with double composition* we mean one for which two laws of composition are given, which we call *addition* and *multiplication.* By addition, to every two elements a, b their *sum* $a + b$ is assigned; by multiplication, their *product* $a \cdot b$ is assigned.

Isomorphism. Two systems $\mathfrak{S}_1, \mathfrak{S}_2$ with double composition are called *isomorphic* or *of the same (composition-) type* if it is possible to put their elements into a one-to-one correspondence in such a way that, to the sum and the product of any two elements of one system, there are always assigned in the other the sum and the product of the corresponding elements; the correspondence itself is called an isomorphic one, or an isomorphism.

Fields. — Up to now we have subjected the two laws of composition to no further condition than that of uniqueness. If we introduce further conditions, we arrive at special kinds of systems with double composition. Among these, the *rationality domains (Kronecker)* or *fields (Dedekind)* are of particular importance: systems satisfying the following 7 conditions:

1. *The associative law of addition:*

$$(a + b) + c = a + (b + c),$$

2. *the commutative law of addition:*

$$a + b = b + a,$$

3. *the associative law of multiplication:*

$$(a \cdot b) \cdot c = a \cdot (b \cdot c),$$

[10] Festschrift for *Kummer's* doctoral jubilee, §§ 1–3. This journal, vol. 92, p. 1. = Works II, p. 237.

4. *the commutative law of multiplication:*

$$a \cdot b = b \cdot a,$$

5. *the distributive law:*

$$a(b + c) = ab + ac,$$

6. *the law of unrestricted and unique subtraction:*

If a, b are elements of the system, then in it there is one and only one element x for which $a + x = b$. — Let this element be denoted by $b - a$ and called the difference of a and b, so that in general

$$a + (b - a) = b$$

holds.

Before setting out the seventh and last condition, we draw from the preceding ones a few simple consequences. Let a, b be any two elements of our system. If we put $a - a = x$, $b - b = y$, then, by 6., the equations $a + x = a$, $b = b + y$ hold; and when we add them, then by applying 1. and 2. we obtain $(a + b) + x = (a + b) + y$, and from this by 6. $x = y$, or $a - a = b - b$. Thus there is a distinguished element — we call it zero (0) — which is represented by every difference of the form $a - a$. — If again a, b denote arbitrary elements of the system, then $a \cdot b = a(b + 0) = a \cdot b + a \cdot 0$, whence $a \cdot 0 = a \cdot b - a \cdot b = 0$, i.e. a product in which one factor is 0 is itself 0. — We now let the last condition follow:

7. *the law of unrestricted and unique division:*

The system contains, besides 0, at least one further element; and if a is an element different from 0 and b is any element of the system, then in it there is one and only one element x for which $a \cdot x = b$. — We call x the quotient of b by a, in symbols $x = \frac{b}{a}$ or $x = b : a$.

If all 7 conditions are satisfied, the system is called a field. Examples of fields are the systems of rational, real, complex numbers, and so forth. The significance of the stated conditions rests on the fact that all laws prevailing in the named fields, insofar as they can be expressed by equations between aggregates built up rationally from their elements, appear as consequences of those conditions and therefore recur in all fields. Without carrying this out in every detail, we will nevertheless emphasize here some of the essential points: if a, b are two non-zero elements of a field $\mathfrak{K}$, then by 7. $a \cdot b$ is different from $a \cdot 0 = 0$; that is, a product is equal to 0 only when at least one fac- tor vanishes. — If a and b are different from 0, and one determines x and y by the conditions $a \cdot x = a$, $b \cdot y = b$, then $(ab)x = b(ax) = ba = (by)a = (ab)y$, $(ab)x = (ab)y$, hence by 7. $x = y$. If we therefore denote the solution of the equation $ax = a$ by ε, then ε, as the unique element of $\mathfrak{K}$, has the property that, multiplied by any element of $\mathfrak{K}$, it leaves that element unchanged, and it is represented by every quotient of the form $\frac{a}{a}$ $(a \neq 0)$. This distinguished element we call the *unit*; we shall always denote it by ε.

Among the theorems whose proof can be conducted solely on the basis of the 7 stated conditions are also the fundamental theorems from determinant theory. In particular, for every field $\mathfrak{K}$ the following holds:

1) n linear equations with n unknowns

$$\sum_{k=1}^{n} a_{ik} x_k = c_i, \qquad (i = 1, \ldots, n)$$

in which the coefficients a_{ik}, c_i are given elements of $\mathfrak{K}$, have, if the determinant $|a_{ik}|$ is not 0, one and only one system of solutions in $\mathfrak{K}$.

2) n homogeneous linear equations with $n+1$ unknowns

$$\sum_{k=0}^{n} a_{ik}x_k = 0 \qquad (i = 1, \dots, n)$$

have at least one system of solutions whose elements are not all 0.

Finally, at this point let it also be noted that there are fields which possess only a *finite* number of elements. Thus the p residue classes modulo a prime number p form a system with double composition which satisfies all 7 conditions, hence is a field in our sense.

§ 2.
Subfields, extension, adjunction.

A subsystem $\mathfrak{K}'$ of a field $\mathfrak{K}$ can, by means of the laws of composition prevailing in $\mathfrak{K}$, itself represent a field. It is plainly necessary and sufficient for this that, whenever α and β are elements of $\mathfrak{K}'$, also $\alpha+\beta$, $\alpha-\beta$, $\alpha\cdot\beta$ and, if $\beta \neq 0$, $\frac{\alpha}{\beta}$ belong to $\mathfrak{K}'$. We then say: the field $\mathfrak{K}$ contains the field $\mathfrak{K}'$, $\mathfrak{K}'$ is contained in $\mathfrak{K}$; and we call $\mathfrak{K}'$ a subfield of $\mathfrak{K}$, and $\mathfrak{K}$ an extension field or extension of $\mathfrak{K}'$. As a rule, among the subfields and extension fields of a field we shall count the field itself; where this is not the case, it will be apparent from the expression used ("proper extension," "proper subfield," "only a subfield") or from the context. — One now immediately verifies the validity of the following theorem:

1. If $\mathfrak{S}$ denotes a system of finitely or infinitely many subfields of a field $\mathfrak{K}$, then in $\mathfrak{K}$ there are elements (for example 0, ε) which occur in all fields of the system $\mathfrak{S}$. The totality of these elements again constitutes a field.

We shall call this field the *intersection* of the field-system $\mathfrak{S}$. — Alongside Theorem 1 there stands a second theorem, no less important:

2. If $\mathfrak{U}$ denotes a system of finitely or infinitely many fields with the property that any two fields $\mathfrak{K}_1$, $\mathfrak{K}_2$ from $\mathfrak{U}$ are always contained in at least one field $\mathfrak{K}_3$ of the system $\mathfrak{U}$, then there is a definite field $\mathfrak{K}$ (not necessarily occurring in $\mathfrak{U}$) which has all fields of the system $\mathfrak{U}$ as subfields and contains no elements other than those which occur in fields of $\mathfrak{U}$.

Proof: Let $\mathfrak{K}$ denote the system of all elements which occur in fields of the system $\mathfrak{U}$. Suppose one has a finite number of elements $a_1, a_2, \dots, a_n$ from $\mathfrak{K}$, and that $\mathfrak{K}_i$ $(i = 1, \dots, n)$ is a field from $\mathfrak{U}$ which contains a_i. Then in $\mathfrak{U}$ there is a field $\mathfrak{K}_2'$ containing $\mathfrak{K}_1$ and $\mathfrak{K}_2$, a field $\mathfrak{K}_3'$ containing $\mathfrak{K}_2'$ and $\mathfrak{K}_3$, and so forth. Thus there are fields in $\mathfrak{U}$ which contain all the fields $\mathfrak{K}_i$ $(i = 1, \dots, n)$ and therefore all the elements a_i. If now a, b are any two elements from $\mathfrak{K}$, and $\mathfrak{K}'$ is any field from $\mathfrak{U}$ which contains a and b, then $a+b$ and $a\cdot b$ have in $\mathfrak{K}'$ a definite meaning. If $\mathfrak{K}''$ is any other field belonging to $\mathfrak{U}$ and containing a and b, then $a+b$ and $a\cdot b$ likewise have in it a definite meaning. But that $a+b$ and $a\cdot b$ can designate in $\mathfrak{K}''$ nothing other than they do in $\mathfrak{K}'$ is a consequence of the assumption that there is a field containing $\mathfrak{K}'$ and $\mathfrak{K}''$. Thus the sum and product of two elements from $\mathfrak{K}$ are unambiguously defined, i.e. $\mathfrak{K}$ is a system with double composition. From the circumstance that every two elements from $\mathfrak{K}$ occur simultaneously in fields from $\mathfrak{U}$, it follows now that composition laws 2. and 4. of § 1 hold; further, that at least one element x_1, and, if $a \neq 0$, at least one element x_2, occurs in $\mathfrak{K}$ for which $a + x_1 = b$, $a \cdot x_2 = b$. Since three elements from $\mathfrak{K}$ also occur in a field from $\mathfrak{U}$, it follows that composition laws 1., 3., and 5. are also satisfied. Finally, since the same also holds for 4 elements a, b, x, y from $\mathfrak{K}$, the equations $a + x = b$, $a + y = b$ and, if $a \neq 0$, also the equations $a \cdot x = b$, $a \cdot y = b$, can hold only when $x = y$. The system $\mathfrak{K}$ is therefore a field, and it is immediately evident that $\mathfrak{K}$ contains all fields from $\mathfrak{U}$ and possesses only such elements as occur in the fields from $\mathfrak{U}$; and likewise that a field possessing these two properties must necessarily be identical with $\mathfrak{K}$.

If the field $\mathfrak{K}$ is contained in the field Ω, and $\mathfrak{S}$ is any system of elements from Ω, then by $\mathfrak{K}(\mathfrak{S})$ we denote the intersection of all those fields between[11] $\mathfrak{K}$ and Ω which contain all elements of the system $\mathfrak{S}$. Of the field $\mathfrak{K}(\mathfrak{S})$ we say that it has arisen from $\mathfrak{K}$ by *adjunction* of the system $\mathfrak{S}$. The field $\mathfrak{K}(\mathfrak{S})$ can be equal to Ω, and is, for example, equal to Ω when $\mathfrak{S} = \Omega$. But $\mathfrak{K}(\mathfrak{S})$ can also be equal to $\mathfrak{K}$; this case occurs when all elements of $\mathfrak{S}$ belong to $\mathfrak{K}$. If from a field $\mathfrak{K}$, by adjunction of a system $\mathfrak{S}_1$, there arises a second field, and from this, by adjunction of a system $\mathfrak{S}_2$, a third, and so forth, then, if $\mathfrak{S}_1, \mathfrak{S}_2, \ldots, \mathfrak{S}_n$ have been adjoined successively, we denote the last field obtained by $\mathfrak{K}(\mathfrak{S}_1, \mathfrak{S}_2, \ldots, \mathfrak{S}_n)$. As is easily seen, the final result does not depend on the order of the systems $\mathfrak{S}$. — Suppose that from a field $\mathfrak{K}$, by adjunction of a system $\mathfrak{S}$ of infinitely many elements, the field $\mathfrak{K}(\mathfrak{S})$ arises. Let $\mathfrak{U}$ denote the system of those fields which arise from $\mathfrak{K}$ by adjunction of one finite subsystem of $\mathfrak{S}$ at a time. Then all fields from $\mathfrak{U}$ are contained in $\mathfrak{K}(\mathfrak{S})$. If $\mathfrak{K}_1, \mathfrak{K}_2$ are any two fields from $\mathfrak{U}$, then $\mathfrak{K}_1 = \mathfrak{K}(\mathfrak{S}_1)$, $\mathfrak{K}_2 = \mathfrak{K}(\mathfrak{S}_2)$, where $\mathfrak{S}_1, \mathfrak{S}_2$ are finite subsystems of $\mathfrak{S}$. If $\mathfrak{S}_3$ denotes that subsystem of $\mathfrak{S}$ which contains all elements from $\mathfrak{S}_1$ and $\mathfrak{S}_2$, but no other element, then $\mathfrak{S}_3$ is also finite; the field $\mathfrak{K}_3 = \mathfrak{K}(\mathfrak{S}_3)$, which contains $\mathfrak{K}_1$ and $\mathfrak{K}_2$, therefore also belongs to $\mathfrak{U}$. Thus the system $\mathfrak{U}$ satisfies the hypothesis of Theorem 2; hence there exists a definite field Ω which contains all fields of the system $\mathfrak{U}$, therefore also $\mathfrak{K}$ and all elements from $\mathfrak{S}$, hence also $\mathfrak{K}(\mathfrak{S})$, but which also contains only such elements as occur in fields of the system $\mathfrak{U}$. Since, however, all these elements already occur in $\mathfrak{K}(\mathfrak{S})$, the field in question is identical with $\mathfrak{K}(\mathfrak{S})$. We therefore obtain the theorem:

3. If $\mathfrak{S}$ is an infinite system, then every element of the field $\mathfrak{K}(\mathfrak{S})$ — and likewise every system of finitely many elements from $\mathfrak{K}(\mathfrak{S})$ — occurs in a field which arises from $\mathfrak{K}$ by adjunction of a finite subsystem of $\mathfrak{S}$.

§ 3.

Integral domains.

By an *integral domain* we shall here mean a system with double composition which satisfies conditions 1. through 6. of § 1, does not consist only of the single element 0, possesses a unit, and in which the product of two elements different from 0 is always again different from 0. By a unit is meant here an element ε for which the equation $x\varepsilon = x$ is satisfied by all elements of the system; it is immediately seen that no more than one unit can exist.

Every field is at the same time an integral domain, whereas the system of integers proves that not every integral domain is a field. — The considerations made in § 2 about fields extend without further ado to integral domains. A subsystem $\mathfrak{J}'$ of an integral domain $\mathfrak{J}$ represents an integral domain if it contains the unit, and if the sum, difference, and product of two elements from $\mathfrak{J}'$ always again belong to $\mathfrak{J}'$. Theorems 1. and 2. of § 2 also hold for integral domains, as one immediately sees; in them one has only to replace the word (sub-)field everywhere by (sub-)integral domain. Adjunction for integral domains we denote by square brackets, so that $\mathfrak{J}[\mathfrak{S}]$ represents an integral domain $\overline{\mathfrak{J}}$ which contains the integral domain $\mathfrak{J}$ and the system of elements $\mathfrak{S}$, and is so constituted that $\mathfrak{S}$ is not already contained in an integral domain different from $\overline{\mathfrak{J}}$ and lying between $\mathfrak{J}$ and $\overline{\mathfrak{J}} = \mathfrak{J}[\mathfrak{S}]$. If $\mathfrak{J}$ is a field, then according to this one must distinguish between $\mathfrak{J}[\mathfrak{S}]$ and $\mathfrak{J}(\mathfrak{S})$. — Now, as is immediately evident, Theorem 3 of § 2 can also be extended to integral domains.

If a and b are elements of an integral domain $\mathfrak{J}$, we say: b is *divisible by* a, a *goes into* b, is a *divisor* of b, and so forth, if in $\mathfrak{J}$ there is an element x for which the equation $ax = b$ is satisfied. If b and c are divisible by a, then the same holds for $b + c$, $b - c$, and for $b \cdot d$, where d is any element from $\mathfrak{J}$. The concept of divisibility is, however, a relative one, depending on the integral domain under consideration: if the element b is not divisible by a in $\mathfrak{J}$, it may nevertheless become divisible by a in an enlarged domain.

[11]A field lies "between" $\mathfrak{K}$ and Ω if it contains $\mathfrak{K}$ and is contained in Ω.

Formation of quotients: — Let an integral domain $\mathfrak{J}$ be contained in a field Ω. If by the individual letters we denote elements from $\mathfrak{J}$, then in Ω, according to the laws holding for every field, we have:

1) if one of the equations $a = \frac{a'}{b'}$, $\frac{a'}{b'} = a$, $ab' = a'$ holds, then the other two also hold,

2) from $\frac{a}{b} = \frac{a'}{b'}$ follows $ab' = a'b$, and conversely,

3) $\dfrac{a}{b} + \dfrac{a'}{b'} = \dfrac{ab' + a'b}{bb'}$, $\dfrac{a}{b} \cdot \dfrac{a'}{b'} = \dfrac{aa'}{bb'}$,

4) $\dfrac{a}{b} - \dfrac{a'}{b'} = \dfrac{ab' - a'b}{bb'}$, $\dfrac{a}{b} : \dfrac{a'}{b'} = \dfrac{ab'}{a'b}$.

Of course it is assumed here that the denominators are different from 0, while otherwise the letters denote arbitrary elements from $\mathfrak{J}$. From 3) and 4) it is seen that those elements of Ω which can be represented as quotients of elements from $\mathfrak{J}$ form a field $\mathfrak{K}$; from 1) — since according to it $\frac{a}{\varepsilon} = a$ — it is seen that $\mathfrak{K}$ contains the integral domain $\mathfrak{J}$. We say: $\mathfrak{K}$ arises from $\mathfrak{J}$ by *formation of quotients*. Such a field must therefore exist as soon as $\mathfrak{J}$ can be contained at all in a field Ω. To test this possibility, one therefore only has to investigate whether it is legitimate to define, for expressions of the form $\frac{a}{b}$, the concepts *equality, sum, product* by 1), 2), and 3), and whether in this way one arrives at a field containing $\mathfrak{J}$. That this is in fact the case for every integral domain $\mathfrak{J}$ is easily seen. If, for example, we examine the definition of equality, then from the assumption $\frac{a}{b} = \frac{a'}{b'}$, $\frac{a'}{b'} = \frac{a''}{b''}$, there result by 2) the equations $ab' = a'b$, $a'b'' = a''b'$; from these, according to the laws valid in $\mathfrak{J}$, by multiplication, $ab'a'b'' = a'ba''b'$, hence, since $b' \neq 0$, $ab''a' = a''ba'$, and, if $a' \neq 0$, $ab'' = a''b$. If, however, $a' = 0$, then from $ab' = a'b$, $a'b'' = a''b'$, $b \neq 0$, $b' \neq 0$, it follows that $a = 0$, $a'' = 0$, and hence again that $ab'' = a''b$. In every case it follows from $\frac{a}{b} = \frac{a'}{b'}$, $\frac{a'}{b'} = \frac{a''}{b''}$ that also $\frac{a}{b} = \frac{a''}{b''}$; and the corresponding assertion holds, on the basis of 1), also when one considers not 3 quotients but perhaps only two and a single element from $\mathfrak{J}$, and so forth. Further, from $\frac{a}{b} = \frac{a'}{b'}$ it follows, on the basis of 2), that conversely $\frac{a'}{b'} = \frac{a}{b}$. Finally, by 2), always $\frac{a}{b} = \frac{a}{b}$. As far as the definitions of sum and product are concerned, one has to prove that they are unambiguous, i.e. that the result is not changed when one summand or factor is replaced by one equal to it. Once this has been done, it is proved that a system with double composition is present, and it remains to show that all 7 conditions stated in § 1 are satisfied. On the basis of 1), the field then contains the integral domain $\mathfrak{J}$. — All these investigations are disposed of in a manner similar to the proof of the legitimacy of the definition of equality given by 1) and 2), and it is therefore unnecessary to enter more closely into the details of the proof.

The field $\mathfrak{K}$ which arises from $\mathfrak{J}$ by formation of quotients is, of course, identical with $\mathfrak{J}$ if $\mathfrak{J}$ is a field. This case always occurs when $\mathfrak{J}$ contains only finitely many, say m, elements. For if then $a\,(\neq 0)$ and b are elements of $\mathfrak{J}$, then as x runs through all m elements of $\mathfrak{J}$, so too does ax; hence it is once equal to b, i.e. condition 7 of § 1 is satisfied, and $\mathfrak{J}$ is a field. — From isomorphic integral domains there arise, by formation of quotients, isomorphic fields, as is immediately evident.

§ 4.

Prime fields. Characteristic.

In order to gain an insight into the various possible types of fields, we must first start from the formation of the simplest fields, and then investigate how, from fields once present, new ones can be derived by extension. The first task leads to the concept of the *prime field*. By this we understand a field which has no subfield apart from itself. The following theorem holds:

1. In every field $\mathfrak{K}$ there is contained one and only one prime field.

Namely, let $\mathfrak{U}$ denote the system of all subfields of $\mathfrak{K}$, and $\mathfrak{P}$ their intersection. $\mathfrak{P}$ is itself a field belonging to $\mathfrak{U}$; and if $\mathfrak{P}'$ is a subfield of $\mathfrak{P}$, then $\mathfrak{P}'$ likewise belongs to $\mathfrak{U}$, hence must contain $\mathfrak{P}$, and therefore is identical with $\mathfrak{P}$. Thus $\mathfrak{P}$ is a prime field. Conversely, if $\mathfrak{T}$ is a subfield of $\mathfrak{K}$, then $\mathfrak{P}$ is a subfield of $\mathfrak{T}$; if therefore $\mathfrak{T}$ is to be a prime field, $\mathfrak{T}$ must be identical with $\mathfrak{P}$.

In the same way we define, as a *prime integral domain*, one which contains no integral domain apart from itself. The intersection of all integral domains contained in an integral domain $\mathfrak{J}$ represents the unique prime integral domain contained in $\mathfrak{J}$.

Let $\mathfrak{K}$ now be a system which for the time being we shall assume only to be an integral domain, and let α be an element of $\mathfrak{K}$. We put

$$\alpha = 1\alpha, \quad \alpha + \alpha = 2\alpha, \quad 2\alpha + \alpha = 3\alpha, \dots,$$

where $1, 2, 3, \dots$ are to be understood as abbreviating symbols, but not as elements of $\mathfrak{K}$. The elements $\alpha, 2\alpha, 3\alpha, \dots$ are called the *natural multiples* of α. If, further, $-n$ denotes a negative integer, we put

$$-(n\alpha) = (-n)\alpha.$$

By this, for every integer m the element $m\alpha$ is defined. The elements $m\alpha$ are called the *integral multiples* of α. For $\alpha = \varepsilon$, if m, m' denote integers, the equations

$$m\varepsilon + m'\varepsilon = (m + m')\varepsilon, \quad m\varepsilon - m'\varepsilon = (m - m')\varepsilon, \quad (m\varepsilon) \cdot (m'\varepsilon) = (mm')\varepsilon \tag{1.}$$

result at once; from them it follows that the integral multiples of ε form an integral domain $\mathfrak{J}$. Since every integral domain contained in $\mathfrak{K}$ contains the unit ε, and hence also the integral multiples of ε, $\mathfrak{J}$ represents the prime integral domain contained in $\mathfrak{K}$. If $\mathfrak{K}$ is a field and $\mathfrak{P}$ is the prime field contained in $\mathfrak{K}$, then $\mathfrak{P}$ contains all elements of $\mathfrak{J}$, and hence also all quotients of elements of $\mathfrak{J}$. But since these already form a field, the prime field $\mathfrak{P}$ is exhausted by them.

Two cases are now conceivable: either the integral multiples of ε are all distinct from one another, i.e. $m\varepsilon = m'\varepsilon$ only when $m = m'$, or that equation can also hold without this being so. In the first case one has a one-to-one correspondence between $\mathfrak{J}$ and the system of integers, which by (1.) proves to be an isomorphism. If $\mathfrak{K}$ is a field, it follows further from this that the prime field $\mathfrak{P}$ contained in $\mathfrak{K}$ is isomorphic to the field of rational numbers. In the second case there are natural multiples of ε which are equal to 0. If p is the smallest natural number for which $p \cdot \varepsilon = 0$, and if m, m' are natural numbers $< p$, then from $m\varepsilon \neq 0$, $m'\varepsilon \neq 0$ it follows that $(mm')\varepsilon = (m\varepsilon) \cdot (m'\varepsilon) \neq 0 = p\varepsilon$, and hence $mm' \neq p$. Thus p is not the product of any two smaller numbers, and is therefore prime. We call p the *characteristic* of $\mathfrak{K}$, while in the case first discussed we assign characteristic 0 to $\mathfrak{K}$. *Accordingly every integral domain has a characteristic, which is either 0 or a prime number.* If the characteristic is a prime number p, it further follows that $m\varepsilon = m'\varepsilon$ holds if and only if $m \equiv m' \pmod{p}$. In connection with equations (1.), this shows that here the prime integral domain $\mathfrak{J}$ is isomorphic to the system of residue classes modulo the prime number p; and since this system is already a field, here $\mathfrak{J} = \mathfrak{P}$. Thus we obtain:

2. The only prime fields which exist are the fields of the type of the field of rational numbers (which from now on we denote by $\mathfrak{P}_0$) and those of the type of the residue-class system modulo a prime number p (from now on denoted by $\mathfrak{P}_p$).

From the preceding it follows further:

3. All subfields of a field $\mathfrak{K}$ have the same characteristic as $\mathfrak{K}$.

If α is any element of a field $\mathfrak{K}$, and m is an integer, then $m\alpha = (m\varepsilon) \cdot \alpha$; thus, if $\alpha \neq 0$, then $m\alpha$ is equal to 0 only when $m\varepsilon = 0$. Hence:

4. An integral multiple $m\alpha$ of a non-zero element α in the field $\mathfrak{K}$ is equal to 0 if and only if either $m = 0$, or the characteristic of $\mathfrak{K}$ is a prime number and m is divisible by this prime number.

Of course these theorems also hold for integral domains.

§ 5.

Equivalent extensions. Simple extensions. Transcendental and algebraic elements. Adjunction of a transcendental element. Integral and rational functions.

If Ω and Ω' are extension fields of a field $\mathfrak{K}$, we say that Ω and Ω' represent *equivalent extensions* of $\mathfrak{K}$ if Ω and Ω' can be mapped isomorphically onto one another in such a way that every element of $\mathfrak{K}$ corresponds to itself. Thus the equivalence of two extensions Ω, Ω' presupposes an isomorphism; but non-equivalent extensions of a field may very well give isomorphic fields. — If the problem is to extend a field so that the newly arisen field satisfies certain conditions, and if it turns out that all extensions satisfying the imposed conditions are equivalent, then we say that *the extension problem has essentially only one solution.* In this sense, for example, the problem of extending the field $\mathfrak{R}$ of real numbers so that every element of the new field satisfies an algebraic equation in $\mathfrak{R}$, but not every element is already contained in $\mathfrak{R}$, is solvable in essentially only one way.

The simplest extensions of a field $\mathfrak{K}$ are those which arise from $\mathfrak{K}$ by adjoining a single element. We call these *simple extensions*, and, if Ω is such an extension, we call every element α of Ω for which $\mathfrak{K}(\alpha) = \Omega$ a *primitive element* of Ω relative to $\mathfrak{K}$. — Let the field $\mathfrak{K}(x)$ arise by adjoining an element x from Ω. Between $\mathfrak{K}$ and $\mathfrak{K}(x)$ lies the integral domain $\mathfrak{K}[x]$ (§ 3). It contains all elements of the form

$$c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n,$$

where the coefficients c are arbitrary elements of $\mathfrak{K}$. For more convenient notation we think of the named elements of $\mathfrak{K}[x]$ as written in the form of infinite series

$$\sum c_k x^k, \tag{1.}$$

where of course it is always assumed that from some index k on all coefficients are equal to 0. Then the equations

$$\begin{cases} \sum a_k x^k + \sum b_k x^k = \sum (a_k + b_k) x^k, \\ \left(\sum a_k x^k\right)\left(\sum b_k x^k\right) = \sum c_k x^k, \quad c_k = a_0 b_k + a_1 b_{k-1} + \cdots + a_k b_0, \end{cases} \tag{2.}$$

and

$$\sum a_k x^k - \sum b_k x^k = \sum (a_k - b_k) x^k \tag{3.}$$

hold. They show that the elements of the form (1.) reproduce themselves under addition, subtraction, and multiplication. Since all elements of $\mathfrak{K}$, in particular also ε, are among the elements of the form (1.), these elements form an integral domain containing $\mathfrak{K}$; and since it also contains x, it must be identical with $\mathfrak{K}[x]$. The field $\mathfrak{K}(x)$ contains the integral domain $\mathfrak{K}[x]$, hence also a definite field $\mathfrak{L}$ arising from $\mathfrak{K}[x]$ by formation of quotients (§ 3). But since $\mathfrak{L}$ contains both $\mathfrak{K}$ and x, $\mathfrak{K}(x)$ is identical with $\mathfrak{L}$.

As in the formation of the simplest *fields*, so also in the formation of the simplest *extensions* we have to distinguish two cases. The first corresponds to the assumption that two elements $\sum a_k x^k$, $\sum b_k x^k$ of $\mathfrak{K}[x]$ are equal only when $a_k = b_k$ for every k; the second is the contrary case. In the first case we call the element x *transcendental*, in the second *algebraic relative to* $\mathfrak{K}$. We shall also introduce here some further terminology that will be important later. If Ω is any extension field of $\mathfrak{K}$, then n elements $\alpha_1, \ldots, \alpha_n$ of Ω are called *linearly dependent* or *linearly independent relative to* $\mathfrak{K}$ according as a homogeneous linear relation

$$a_0 \alpha_0 + \cdots + a_n \alpha_n = 0,$$

in which the coefficients a are elements of $\mathfrak{K}$ and are not all equal to 0, exists or does not exist. The elements $\alpha_1, \ldots, \alpha_n$ need not all be distinct from one another; but it is clear that if equal ones occur among them, they are always linearly dependent. We also note that the elements $\alpha_1, \ldots, \alpha_n$ are linearly dependent if even a part of them is linearly dependent. On this basis we arrive at an extension of our definition to systems of infinitely many elements. A system $\mathfrak{S}$ of infinitely many elements from Ω is called linearly dependent or independent relative to $\mathfrak{K}$ according as there exists, or does not exist, a *finite* subsystem of $\mathfrak{S}$ which is linearly dependent relative to $\mathfrak{K}$. The distinction made above between (relatively) transcendental and algebraic elements can now also be formulated as follows: *The element x is transcendental or algebraic relative to $\mathfrak{K}$ according as its powers*

$$\varepsilon, x, x^2, \ldots, x^n, \ldots$$

are linearly independent or dependent relative to $\mathfrak{K}$.

We first consider the adjunction of a *transcendental* element, which alone will be treated in the present section. Here the following holds.

1. Every field $\mathfrak{K}$ can be extended by adjoining a transcendental element x.

Namely, using the elements of $\mathfrak{K}$ we form all expressions of the form (1.), stipulate that two such expressions $\sum a_k x^k$, $\sum b_k x^k$ are to count as equal only when $a_k = b_k$ for every k, and set equations (2.) as definitions for addition and multiplication. We have thereby created a system with double composition containing $\mathfrak{K}$, which, as direct calculation immediately gives, satisfies conditions 1.–6. of § 1. Further, if $\alpha = \sum a_k x^k$ and $\beta = \sum b_k x^k$ are two non-zero elements of this system, and if a_m, b_n are the last coefficients different from 0, then in the product $\alpha\beta = \sum c_k x^k$ the coefficient $c_{m+n} = a_m \cdot b_n \neq 0$, hence $\alpha\beta \neq 0$. Since the system also contains a unit, it is an integral domain $\mathfrak{J}$, and since all elements of $\mathfrak{J}$ are formed from x and the elements of $\mathfrak{K}$ by addition and multiplication, $\mathfrak{J} = \mathfrak{K}[x]$. From $\mathfrak{J}$, however, as from every integral domain, one obtains by formation of quotients a field, the field $\mathfrak{K}(x)$.

2. All extensions of a field which arise by adjoining a transcendental element are equivalent.

For if $\mathfrak{K}(x)$ and $\mathfrak{K}(y)$ are two such extensions, and to every element of $\mathfrak{K}(x)$, thought of as represented in the form of a quotient of two elements of the form (1.), one assigns that element of $\mathfrak{K}(y)$ which is obtained by replacing x by y, then it is easy to see that one has an isomorphic relation between $\mathfrak{K}(x)$ and $\mathfrak{K}(y)$ in which every element of $\mathfrak{K}$ corresponds to itself.

The elements of the field $\mathfrak{K}(x)$ are also called *rational functions*; the elements of the integral domain $\mathfrak{K}[x]$ are called *integral functions of the transcendental or indeterminate x in the field $\mathfrak{K}$.* For their notation we use, in the usual way, function symbols $f(x)$, $g(x)$, and so on. An integral function

$$f(x) = c_0 + c_1 x + \cdots + c_n x^n$$

is said to be of *degree n* if $c_n \neq 0$; thus the integral functions of degree 0 are represented by the non-zero elements of $\mathfrak{K}$. We assign no degree to the element 0; however, if it is asserted of an integral function that its degree lies below a certain bound, the case that the function is equal to 0 is always to be included.

If $f(x)$ and $g(x)$ are integral functions of the indeterminate x in the field $\mathfrak{K}$, with $g(x) \neq 0$ and n the degree of $g(x)$, then by the familiar division process in the field $\mathfrak{K}$ one can determine two integral functions $q(x)$ and $r(x)$ such that, first,

$$f(x) = q(x) \cdot g(x) + r(x) \tag{4.}$$

and, second, the degree of $r(x)$ is $< n$. The functions $q(x)$ and $r(x)$ are uniquely determined by these two requirements. The equation $r(x) = 0$ says that $f(x)$ is divisible by $g(x)$. It may also happen that each of the functions $f(x)$, $g(x)$ is divisible by the other; then they differ only by a factor of degree 0, hence by an element of $\mathfrak{K}$. In all questions concerning divisibility, two such functions are to be regarded as essentially equal. The unique determination of the functions $q(x), r(x)$ occurring in (4.) is the reason that, for these questions, unlike the questions of irreducibility treated later, it is immaterial in which field $\mathfrak{K}$ the functions are considered; it need only be a field to which the coefficients of the functions belong as elements, while x is transcendental relative to it. — This also applies to the question of the *greatest common divisor* $t(x)$ of two integral functions $f(x), g(x)$, in which every common divisor of $f(x)$ and $g(x)$ is contained. The function $t(x)$ is equal to 0 if $f(x) = 0$, $g(x) = 0$; otherwise it is different from 0 and determined only up to a factor of degree 0. To fix $t(x)$ uniquely, we stipulate that the coefficient of the highest power of x shall be the unit, and we denote the greatest common divisor $t(x)$ of $f(x)$ and $g(x)$ so determined by

$$(f(x), g(x)).$$

Euclid's procedure teaches how to find $t(x)$ and how to represent it in the form

$$t(x) = f(x) \cdot g_1(x) + g(x) \cdot f_1(x). \tag{5.}$$

The integral functions $f_1(x), g_1(x)$ occurring here are uniquely determined if it is required that their degrees be respectively smaller than the degrees of $f(x)$ and $g(x)$; like $t(x)$, they belong to every field to which the functions $f(x), g(x)$ belong. The equation

$$(f(x), g(x)) = \varepsilon$$

expresses that $f(x)$ and $g(x)$ are coprime.

From Euclid's algorithm one proceeds, in the usual way, to the further propositions:

3. If $f(x)$ is coprime to $g(x)$ and $f(x) \cdot h(x)$ is divisible by $g(x)$, then $h(x)$ is divisible by $g(x)$.

4. If $f(x)$ and $h(x)$ are coprime to $g(x)$, then $f(x) \cdot h(x)$ is also coprime to $g(x)$.

Let the field $\mathfrak{K}$ be extended by adjoining a transcendental element x to $\mathfrak{K}(x)$. Every element ξ of $\mathfrak{K}(x)$ can then be represented in infinitely many ways as a quotient of two elements $f(x), g(x)$ of $\mathfrak{K}[x]$. By cancelling by the greatest common divisor one can arrange that numerator and denominator are coprime; then by multiplying numerator and denominator by an element of $\mathfrak{K}$ one can arrange that in the denominator the coefficient of the highest power is equal to ε. In this way we obtain a "reduced" representation of ξ, and there is only one such representation. Indeed, suppose two reduced representations of ξ are given,

$$\xi = \frac{f(x)}{g(x)} = \frac{f_1(x)}{g_1(x)}.$$

Then it follows that

$$f(x) \cdot g_1(x) = f_1(x) \cdot g(x).$$

Since $g(x)$ is coprime to $f(x)$, but divides the product $f(x) \cdot g_1(x)$, it follows that $g(x)$ must divide $g_1(x)$. Likewise $g_1(x)$ must divide $g(x)$; and since both functions have ε as the coefficient of the highest power, we get $g(x) = g_1(x)$, $f(x) \cdot g(x) = f_1(x) \cdot g(x)$, hence also $f(x) = f_1(x)$. — Let again $\xi = \frac{f(x)}{g(x)}$ denote the reduced representation of the element ξ of $\mathfrak{K}(x)$. Assume that ξ is algebraic relative to $\mathfrak{K}$. Then there is a relation of the form

$$c_0 + c_1\xi + \cdots + c_n\xi^n = 0, \tag{6.}$$

where the coefficients c belong to the field $\mathfrak{K}$ and where we may suppose $c_n \neq 0$. If $\xi \neq 0$, we may also suppose $c_0 \neq 0$. For if c_μ is the first coefficient different from 0, then (6.) implies the equation

$$c_\mu + c_{\mu+1}\xi + \cdots + c_n\xi^{n-\mu} = 0,$$

in which again the term free of ξ is non-zero. Thus, assuming at once that $c_0 \neq 0$, and multiplying (6.) by $(g(x))^n$, we obtain an equation to which we can give the forms

$$g(x)\big[c_0(g(x))^{n-1} + \cdots + c_{n-1}(f(x))^{n-1}\big] = -c_n(f(x))^n,$$
$$-c_0(g(x))^n = f(x)\big[c_1(g(x))^{n-1} + \cdots + c_n(f(x))^{n-1}\big].$$

From these it is clear that $(f(x))^n$ is divisible by $g(x)$ and $(g(x))^n$ by $f(x)$. But since $f(x)$ and $g(x)$ are coprime, by 4. this is possible only when $f(x)$ and $g(x)$ have degree 0, hence when ξ is an element of the field $\mathfrak{K}$. Thus we have the theorem:

5. In the field $\mathfrak{K}(x)$, which arises from $\mathfrak{K}$ by adjoining a transcendental element x, all elements except those of $\mathfrak{K}$ itself are transcendental relative to $\mathfrak{K}$.

This theorem, which has its counterpart in Theorem 3 of § 6, shows that the extensions obtained by adjoining a transcendental element are quite different from those supplied by adjoining an algebraic element.

In the field $\mathfrak{K}(x)$, which arises from $\mathfrak{K}$ by adjoining the transcendental element x, all elements are rational functions of x, or of course also of any other primitive element; by contrast, the distinction of the elements of $\mathfrak{K}(x)$ into *integral* and *fractional* depends on a particular element x. A concept independent of the choice of primitive element is, as will still be shown, the *degree of a rational function* $R(x)$. If $R(x) = \frac{f(x)}{g(x)}$ is the representation of $R(x)$ in reduced form, and if m and n are the degrees of $f(x)$ and $g(x)$, then the degree of the rational function $R(x)$ is the larger of the numbers m and n, and, if they are equal, the common number $m = n$. We prove:

6. A rational function of degree n of a rational function of degree ν ($n > 0$, $\nu > 0$) is a function of degree $n \cdot \nu$.

Let, in reduced representation,

$$R(x) = \frac{f(x)}{g(x)}, \qquad P(x) = \frac{\varphi(x)}{\psi(x)} \tag{7.}$$

be rational functions of x in the field $\mathfrak{K}$, and let n and ν be their degrees,

$$\begin{cases} f(x) = a_0x^n + \cdots + a_n, & g(x) = b_0x^n + \cdots + b_n, \\ \varphi(x) = \alpha_0x^\nu + \cdots + \alpha_\nu, & \psi(x) = \beta_0x^\nu + \cdots + \beta_\nu. \end{cases} \tag{8.}$$

Then neither the elements a_0, b_0 nor the elements α_0, β_0 can vanish simultaneously. Since $f(x)$ and $g(x)$ are coprime, two integral functions $f_1(x), g_1(x)$ of degrees $< n$ can be chosen so that

$$f(x) \cdot g_1(x) + g(x) \cdot f_1(x) = \varepsilon. \tag{9.}$$

Put

$$\gamma = a_0\alpha_0^n + a_1\alpha_0^{n-1}\beta_0 + \cdots + a_n\beta_0^n, \qquad \delta = b_0\alpha_0^n + b_1\alpha_0^{n-1}\beta_0 + \cdots + b_n\beta_0^n. \tag{10.}$$

If $\beta_0 \neq 0$, then

$$\gamma = f\left(\frac{\alpha_0}{\beta_0}\right)\beta_0^n, \qquad \delta = g\left(\frac{\alpha_0}{\beta_0}\right)\beta_0^n,$$

and so by (9.)

$$\gamma g_1\left(\frac{\alpha_0}{\beta_0}\right) + \delta f_1\left(\frac{\alpha_0}{\beta_0}\right) = \beta_0^n \neq 0.$$

Thus γ and δ cannot both be 0. The same is also the case when $\beta_0 = 0$. For then $\alpha_0 \neq 0$, $\gamma = a_0\alpha_0^n$, $\delta = b_0\alpha_0^n$, and the assertion follows because a_0 and b_0 do not both vanish. But now γ and δ are the coefficients of the $n \cdot \nu$-th power of x in the integral functions

$$\begin{cases} \sigma(x) = a_0\varphi^n + a_1\varphi^{n-1}\psi + \cdots + a_n\psi^n, \\ \tau(x) = b_0\varphi^n + b_1\varphi^{n-1}\psi + \cdots + b_n\psi^n \end{cases} \tag{11.}$$

(where φ and ψ are written for $\varphi(x)$ and $\psi(x)$). Of the functions $\sigma(x)$ and $\tau(x)$, neither of which, as is clear from (11.) and (8.), exceeds degree $n \cdot \nu$, at least one must therefore attain this degree. By (8.)

$$\sigma(x) = f\left(\frac{\varphi}{\psi}\right)\psi^n, \qquad \tau(x) = g\left(\frac{\varphi}{\psi}\right)\psi^n, \tag{12.}$$

and hence, by (7.),

$$R(P(x)) = \frac{\sigma(x)}{\tau(x)}. \tag{13.}$$

From (12.) and (9.) it follows that

$$\sigma(x) \cdot g_1\left(\frac{\varphi}{\psi}\right)\psi^{n-1} + \tau(x) \cdot f_1\left(\frac{\varphi}{\psi}\right)\psi^{n-1} = \psi^{2n-1}, \tag{14.}$$

and here $g_1\left(\frac{\varphi}{\psi}\right)\psi^{n-1}$ and $f_1\left(\frac{\varphi}{\psi}\right)\psi^{n-1}$ are integral functions of x. Applying the notation introduced above for the greatest common divisor, put

$$(\sigma(x), \tau(x)) = t(x), \tag{15.}$$

$$(t(x), \psi(x)) = u(x). \tag{16.}$$

Then $u(x)$ divides $\sigma(x)$, $\tau(x)$, and $\psi(x)$, hence, by (11.), also divides φ^n. But since $\varphi(x)$ and $\psi(x)$ are coprime, we must have $u(x) = \varepsilon$. Thus $t(x)$ is coprime to $\psi(x)$, and hence also to ψ^{2n-1}. From (15.) and (14.), however, it follows that $t(x)$ divides ψ^{2n-1}. Therefore $t(x) = \varepsilon$, that is, the functions $\sigma(x)$ and $\tau(x)$ are coprime. Since at least one of them has degree $n \cdot \nu$ and neither has a higher degree, it follows from (13.) that the degree of the rational function $R(P(x))$ is $n \cdot \nu$, as was to be proved. In the preceding proof we used the theorem that equations holding between rational functions of a transcendental element x in a field $\mathfrak{K}$ remain correct when any element of the field $\mathfrak{K}$, or of any extension field of $\mathfrak{K}$, is substituted for x, provided only that this substitution introduces no vanishing denominators. This theorem is an immediate consequence of the fact that, for equality, sum, and product of two rational functions of a transcendental element, only such stipulations were made as hold for every element x on the basis of the general properties of fields, whether that element is transcendental or not.

If y is an element of the field $\mathfrak{K}(x)$, then $\mathfrak{K}(y)$ is a field between $\mathfrak{K}$ and $\mathfrak{K}(x)$. Every element of $\mathfrak{K}(y)$, viewed as a rational function of x in $\mathfrak{K}$, has a degree which by 6. is divisible by the degree n of the element y. If y is to be a primitive element of $\mathfrak{K}(x)$ relative to $\mathfrak{K}$, so that $\mathfrak{K}(y) = \mathfrak{K}(x)$, then the degree of the element x, namely 1, must also be divisible by n, hence $n = 1$. This condition is also sufficient. Indeed, if $y = \frac{ax+b}{cx+d}$ has degree 1, then, as is easy to see, $ad - bc \neq 0$, and $x = \frac{dy-b}{-cy+a}$; hence x belongs to the field $\mathfrak{K}(y)$, and $\mathfrak{K}(y) = \mathfrak{K}(x)$. Thus:

7. The primitive elements of the field $\mathfrak{K}(x)$ are the rational functions of degree 1.

If z is any element of the field $\mathfrak{K}(x) = \mathfrak{K}(y)$, which arises from $\mathfrak{K}$ by adjoining the transcendental element x or y, then z can be represented as a rational function of x or of y:

$$z = R(x) = R_1(y).$$

Since y, considered as a function of x, has degree 1, it follows from 6. that the degree of $R(x)$ in x is equal to the degree of $R_1(y)$ in y. Hence:

8. If the field Ω can be obtained from $\mathfrak{K}$ by adjoining a transcendental element, then every element of Ω has a definite degree relative to $\mathfrak{K}$, independent of the choice of primitive element.

We still prove the theorem: *9. Suppose that n elements $x_1, x_2, \ldots, x_n$ are successively adjoined to a field $\mathfrak{K}_0$, so that the fields*

$$\mathfrak{K}_0(x_1) = \mathfrak{K}_1, \quad \mathfrak{K}_1(x_2) = \mathfrak{K}_2, \ldots, \quad \mathfrak{K}_{n-1}(x_n) = \mathfrak{K}_n$$

arise, and that every element x_k $(k = 1, \ldots, n)$ is transcendental relative to $\mathfrak{K}_{k-1}$. Then every element x_k is transcendental relative to the field obtained from $\mathfrak{K}_0$ by adjoining the remaining $n-1$ elements.

Proof: We first show that x_1 is transcendental relative to the field $\mathfrak{K}_0(x_2)$. If this were not the case, there would be a relation

$$c_0 + c_1 x_1 + c_2 x_1^2 + \cdots + c_m x_1^m = 0, \tag{17.}$$

where the coefficients c are elements of $\mathfrak{K}_0(x_2)$ and are not all equal to 0. Think of the coefficients c as written as quotients of elements of $\mathfrak{K}_0[x_2]$ and multiply (17.) by the denominators; one obtains a relation of the same form (17.) in which the coefficients c are now elements of $\mathfrak{K}_0[x_2]$ and again are not all equal to 0. Let, for example,

$$c_k = c_{k0} + c_{k1} x_2 + \cdots + c_{kr} x_2^r \neq 0.$$

Then at least one of the coefficients $c_{k0}, c_{k1}, \ldots, c_{kr}$, all of which are elements of $\mathfrak{K}_0$, say c_{kl}, is different from 0. If the left hand side of (17.) is now ordered by powers of x_2, one obtains

$$C_0 + C_1 x_2 + C_2 x_2^2 + \cdots + C_s x_2^s = 0, \tag{18.}$$

where the coefficients C are elements of $\mathfrak{K}_0[x_1]$, hence also of $\mathfrak{K}_0(x_1)$, and where $C_l \neq 0$, since the coefficient of the power x_1^k occurring in C_l is $c_{kl} \neq 0$. But (18.) would imply that x_2 is algebraic relative to $\mathfrak{K}_0(x_1) = \mathfrak{K}_1$. Since this contradicts the hypothesis, x_1 is transcendental relative to $\mathfrak{K}_0(x_2)$. — Thus Theorem 9 is proved in the case $n = 2$. If $n > 2$, one can proceed inductively. The fields $\mathfrak{K}_2, \mathfrak{K}_3, \ldots, \mathfrak{K}_n$ arise from $\mathfrak{K}_1$ by successively adjoining the $n-1$ elements $x_2, x_3, \ldots, x_n$. Since x_k $(k \geq 2)$ is transcendental relative to $\mathfrak{K}_{k-1}$, it may be regarded as proved that each element x_k $(k \geq 2)$ is transcendental relative to the field $\mathfrak{K}'_k$ obtained from $\mathfrak{K}_1$ by adjoining the remaining elements of the row $x_2, x_3, \ldots, x_n$. This field $\mathfrak{K}'_k$ is identical with the field obtained from $\mathfrak{K}_0$ by adjoining the elements $x_1, \ldots, x_n$ with the exception of x_k. It remains only to show that x_1 is also transcendental relative to $\mathfrak{K}_0(x_2, x_3, \ldots, x_n)$. Put $\mathfrak{K}_0(x_2) = \mathfrak{K}'_1$. As has already been shown, x_1 is transcendental relative to $\mathfrak{K}'_1$. From $\mathfrak{K}'_1$ one obtains the fields $\mathfrak{K}_2, \mathfrak{K}_3, \ldots, \mathfrak{K}_n$ by adjoining successively $x_1, x_3, \ldots, x_n$. Since these are again only $n-1$ elements, it follows that x_1 is transcendental relative to $\mathfrak{K}'_1(x_3, x_4, \ldots, x_n)$; and since $\mathfrak{K}'_1(x_3, x_4, \ldots, x_n) = \mathfrak{K}_0(x_2, x_3, \ldots, x_n)$, the assertion is proved also in this case.

We say of the field $\mathfrak{K}_n$ that it arises from $\mathfrak{K}_0$ by adjoining n transcendentals or indeterminates $x_1, x_2, \ldots, x_n$, and we call the elements of $\mathfrak{K}_n$ *rational functions*, and those of the integral domain $\mathfrak{K}_0[x_1, x_2, \ldots, x_n]$ *integral functions* of these indeterminates. — Repeated application of Theorem 2 gives the following general fact: two extensions Ω and $\mathfrak{M}$ of a field $\mathfrak{K}$, each of which is derived by adjoining n transcendentals to $\mathfrak{K}$, are equivalent. If $\Omega = \mathfrak{K}(x_1, \ldots, x_n)$ and $\mathfrak{M} = \mathfrak{K}(y_1, \ldots, y_n)$, then there is a definite isomorphic relation between Ω and $\mathfrak{M}$ which assigns the elements $y_1, \ldots, y_n$ to the elements $x_1, \ldots, x_n$ and makes every element of $\mathfrak{K}$ correspond to itself.

§ 6.

Adjunction of an algebraic element.

Let $f(x)$ be an integral function of degree n in a field $\mathfrak{K}$, and let α be an element of $\mathfrak{K}$. By division by $x - \alpha$ one obtains

$$f(x) = (x - \alpha) q(x) + r, \tag{1.}$$

where $q(x)$ is a function of degree $n-1$ in $\mathfrak{K}$ and r is an element of $\mathfrak{K}$. From (1.) it follows that $f(\alpha) = r$; hence, if α is a zero of $f(x)$, that is, if $f(\alpha) = 0$, we have the decomposition of $f(x)$ into $(x - \alpha) \cdot q(x)$, from which one further concludes that an integral function of degree n

in a field cannot have more than n zeros. — From the representation of the greatest common divisor $t(x)$ of two integral functions $f(x), g(x)$ in § 5, (5.), it follows that every common zero of $f(x)$ and $g(x)$ must be a zero of $t(x)$, and therefore that coprime functions have no common zero. *Reducible and irreducible functions.* — An integral function $f(x)$ of degree $n \geq 1$, whose coefficients belong to a field $\mathfrak{K}$, is called *reducible in* $\mathfrak{K}$ if there are in $\mathfrak{K}$ two integral functions of smaller degree whose product is equal to $f(x)$; otherwise $f(x)$ is called *irreducible in* $\mathfrak{K}$. If $\mathfrak{K}_1$ is a subfield of $\mathfrak{K}$ and the coefficients of $f(x)$ already belong to $\mathfrak{K}_1$, then it is clear that if $f(x)$ is irreducible in $\mathfrak{K}$, it must also be irreducible in $\mathfrak{K}_1$; on the other hand, $f(x)$ can be irreducible in $\mathfrak{K}_1$ and reducible in $\mathfrak{K}$. — Since a function $\varphi(x)$ irreducible in the field $\mathfrak{K}$ is coprime to every other function over $\mathfrak{K}$ which it does not divide, by § 5, 4. the product of two integral functions $f(x), g(x)$ over $\mathfrak{K}$ is divisible by $\varphi(x)$ only if at least one of the two functions $f(x), g(x)$ is divisible by $\varphi(x)$. From this one concludes in the usual way that every integral function $f(x)$ over $\mathfrak{K}$ can be decomposed into irreducible factors in only one way. Of course factors of degree zero are to be disregarded[12].

Let $\varphi(x)$ be an integral function of the indeterminate x in the field $\mathfrak{K}$, with degree $n\,(\geq 1)$, and call two elements $g_1(x), g_2(x)$ of $\mathfrak{K}[x]$ congruent modulo $\varphi(x)$,

$$g_1(x) \equiv g_2(x) \quad \text{mod. } \varphi(x),$$

if $g_1(x) - g_2(x)$ is divisible by $\varphi(x)$. Then all functions that are congruent to a given one are also congruent to one another, and every function $f(x)$ is congruent to a *reduced* function, i.e. to a function whose degree is $< n$. From two congruences

$$g_1(x) \equiv g_2(x), \qquad h_1(x) \equiv h_2(x) \quad \text{mod. } \varphi(x)$$

we obtain the congruences

$$g_1(x) + h_1(x) \equiv g_2(x) + h_2(x), \qquad g_1(x) \cdot h_1(x) \equiv g_2(x) \cdot h_2(x) \quad \text{mod. } \varphi(x).$$

If, therefore, congruent functions are collected into one *residue class*, then by two such classes A, B two further classes C, D are uniquely determined in such a way that, whenever $g(x)$ is any element of A and $h(x)$ one of B, $g(x) + h(x)$ belongs to C and $g(x) \cdot h(x)$ belongs to D. Thus, putting

$$A + B = C, \qquad A \cdot B = D,$$

the residue classes form a system with double composition; and from the fact that the integral domain $\mathfrak{K}[x]$ satisfies conditions 1. through 6. of § 1, the same is easily inferred for the residue-class system. The *zero class*, which may be denoted by 0, consists here of the functions divisible by $\varphi(x)$. If $\varphi(x)$ is reducible and

$$g(x) \cdot h(x) = \varphi(x) \tag{2.}$$

is a decomposition, and if A and B denote the residue classes to which $g(x)$ and $h(x)$ respectively belong, then by (2.) we have

$$A \neq 0, \qquad B \neq 0, \qquad A \cdot B = 0.$$

On the other hand, assume that $\varphi(x)$ is irreducible. Then a product of two functions is divisible by $\varphi(x)$ only if at least one factor is divisible by $\varphi(x)$, and therefore a product of two residue classes is equal to 0 only if at least one factor is equal to 0. It further follows that from $A \neq 0$ and $AB = AC$ one always obtains $B = C$. Finally, let A and B be two residue classes, let $A \neq 0$, and let $f(x), g(x)$ be functions from A and B respectively. Then $f(x)$ is coprime to $\varphi(x)$; hence two elements $\psi(x), \chi(x)$ of $\mathfrak{K}[x]$ can be chosen such that

$$f(x) \cdot \psi(x) + \varphi(x) \cdot \chi(x) = \varepsilon,$$

[12] On the extension of these theorems to functions of several variables, see Weber, Math. Ann. 43, p. 521 ff., § 3.

and so

$$f(x) \cdot (\psi(x) \cdot g(x)) \equiv g(x) \quad \text{mod. } \varphi(x). \tag{3.}$$

If C denotes the residue class to which the function $\psi(x) \cdot g(x)$ belongs, then by (3.)

$$A \cdot C = B,$$

and thus the residue-class system also satisfies condition 7. of § 1. We therefore obtain the theorem: *1. The residue-class system belonging to the integral function $\varphi(x)$ over $\mathfrak{K}$ is a field precisely when $\varphi(x)$ is irreducible in $\mathfrak{K}$.*

Now let $\mathfrak{K}(j)$ be a field arising from $\mathfrak{K}$ by adjoining an algebraic element j. Then the powers

$$\varepsilon, j, j^2, \ldots \quad \text{in inf.}$$

are linearly dependent; and if n is the smallest number for which

$$\varepsilon, j, j^2, \ldots, j^n$$

are linearly dependent, then among these powers there exists a homogeneous linear relation with coefficients in $\mathfrak{K}$ whose last coefficient is non-zero and hence, since one can divide by it, may be taken to be ε. The relation so obtained,

$$a_0 + a_1 j + a_2 j^2 + \cdots + a_{n-1} j^{n-1} + j^n = 0, \tag{4.}$$

is uniquely determined. For if there were a second one, then by subtraction one would obtain a homogeneous linear relation among the elements $\varepsilon, j, \ldots, j^{n-1}$, which are supposed to be linearly independent. We now consider the integral domain $\mathfrak{K}[j]$ contained in $\mathfrak{K}(j)$, which will subsequently prove to be identical with $\mathfrak{K}(j)$, and compare it with an integral domain $\mathfrak{K}[x]$ arising from $\mathfrak{K}$ by adjoining a transcendental element x. The domain $\mathfrak{K}[x]$ contains the function

$$\varphi(x) = x^n + a_{n-1} x^{n-1} + \cdots + a_0,$$

which by (4.) has the zero j. If $g(x), h(x)$ are elements of $\mathfrak{K}[x]$ and their degrees are $< n$, then, because $\varepsilon, j, \ldots, j^{n-1}$ are linearly independent, $g(j) \neq 0$ and $h(j) \neq 0$; hence

$$\varphi(j) = 0 \neq g(j) \cdot h(j),$$

and therefore

$$\varphi(x) \neq g(x) \cdot h(x).$$

That is, $\varphi(x)$ is not the product of two functions of smaller degree in $\mathfrak{K}$, hence is irreducible in $\mathfrak{K}$. By the stated theorems on irreducible functions, $\varphi(x)$ is the only irreducible function over $\mathfrak{K}$ with j as a zero, apart from a factor from $\mathfrak{K}$; but here that factor also disappears because of the additional condition that the coefficient of the highest power shall be equal to ε. We call the degree n of $\varphi(x)$ also the *degree of the element j relative to* $\mathfrak{K}$, in symbols:

$$n = [j : \mathfrak{K}].$$

We now assign to every element $f(x)$ of $\mathfrak{K}[x]$ the element $f(j)$ of $\mathfrak{K}[j]$. If $f(x)$ has degree 0, hence is an element of $\mathfrak{K}$, then $f(j) = f(x)$. The elements of $\mathfrak{K}[j]$ corresponding to elements of $\mathfrak{K}[x]$ contain all elements of the form

$$c_0 + c_1 j + \cdots + c_k j^k$$

with arbitrary coefficients c from $\mathfrak{K}$, and hence exhaust the integral domain $\mathfrak{K}[j]$. However, infinitely many elements of $\mathfrak{K}[x]$ correspond to each element of $\mathfrak{K}[j]$. If $f_1(x), f_2(x)$ are any two elements of $\mathfrak{K}[x]$, one can write

$$f_1(x) - f_2(x) = q(x) \cdot \varphi(x) + r(x), \tag{5.}$$

where $q(x), r(x)$ are elements of $\mathfrak{K}[x]$ and the degree of $r(x)$ is $< n$. It follows that

$$f_1(j) = f_2(j) + r(j),$$

and since $r(j)$ can be equal to 0 only when $r(x) = 0$, that is, when by (5.) $f_1(x) \equiv f_2(x)$ mod. $\varphi(x)$, the elements corresponding to a single element of $\mathfrak{K}[j]$ are precisely the elements of a definite residue class mod. $\varphi(x)$. Thus we have a one-to-one correspondence between these residue classes and $\mathfrak{K}[j]$. Further, since from

$$f(x) + g(x) = h(x), \qquad f(x) \cdot g(x) = k(x)$$

it always follows that

$$f(j) + g(j) = h(j), \qquad f(j) \cdot g(j) = k(j),$$

the correspondence is isomorphic. Since $\varphi(x)$ is irreducible, the residue-class system is a field; hence the isomorphic system $\mathfrak{K}[j]$ must also be a field, and therefore $\mathfrak{K}[j] = \mathfrak{K}(j)$. Since every residue class contains exactly *one* function whose degree is $< n$, every element of $\mathfrak{K}(j)$ is represented exactly once in the form

$$c_0 + c_1 j + \cdots + c_{n-1} j^{n-1}, \tag{6.}$$

where the coefficients c are arbitrary elements of $\mathfrak{K}$.

Let $\mathfrak{K}(\bar{j})$ be a second extension of $\mathfrak{K}$, and let $\bar{j}$ be a zero of the same irreducible function $\varphi(x)$. Then, just as there is an isomorphic correspondence between $\mathfrak{K}(j)$ and the residue-class system mod. $\varphi(x)$, there is one between this system and $\mathfrak{K}(\bar{j})$. These yield an isomorphism between $\mathfrak{K}(j)$ and $\mathfrak{K}(\bar{j})$ in which, by the preceding, every element of $\mathfrak{K}$ corresponds to itself. The two extensions are therefore equivalent. — Conversely, it is easy to see that to every irreducible function $\varphi(x)$ over $\mathfrak{K}$ there really belongs an extension $\mathfrak{K}(j)$ in which j is a zero of $\varphi(x)$. Namely, let $\overline{\Omega}$ denote the residue-class field belonging to $\varphi(x)$. There is an isomorphic correspondence π_0 between $\mathfrak{K}$ and a subfield $\overline{\mathfrak{K}}$ of $\overline{\Omega}$, assigning to each element a of $\mathfrak{K}$ the residue class represented by a. We now introduce a system $\mathfrak{S}$ of new elements, which we assign one-to-one to the elements of $\overline{\Omega}$ that do not belong to $\overline{\mathfrak{K}}$. This assignment and π_0 give a mapping π_1 between $\overline{\Omega}$ and the system $\Omega = \mathfrak{K} + \mathfrak{S}$. If A, B, C, D are elements of $\overline{\Omega}$ and a, b, c, d are the corresponding elements of Ω, and if $A + B = C$, $A \cdot B = D$, then, if a and b belong to $\mathfrak{K}$, $a + b = c$ and $a \cdot b = d$. If we decree that these equations are to hold also in the other cases, then Ω becomes a field isomorphic to $\overline{\Omega}$, and it is an extension of $\mathfrak{K}$. Let j be the element of Ω which under the isomorphism π_1 corresponds to the residue class represented by x. It follows immediately from the definition of addition and multiplication of residue classes that: *An element $g(x)$ of $\mathfrak{K}[x]$ represents a residue class to which π_1 assigns the element $g(j)$ of Ω.* Consequently $\varphi(j) = 0$, and the elements of the form $g(j)$ exhaust the field Ω; hence $\Omega = \mathfrak{K}(j)$. Thus:

2. Every irreducible function $\varphi(x)$ over $\mathfrak{K}$ determines one and, apart from equivalent extensions, only one extension $\mathfrak{K}(j)$ in which j is a zero of $\varphi(x)$.

If $\varphi(x)$ has degree n and if $\gamma_0, \gamma_1, \ldots, \gamma_n$ are any $n+1$ elements of $\mathfrak{K}(j)$, we can represent them in the form

$$\gamma_i = c_{i0} + c_{i1} j + \cdots + c_{i,n-1} j^{n-1} \qquad (i = 0, \ldots, n)$$

with coefficients in $\mathfrak{K}$. From the theorem on linear equations mentioned in § 1, it follows that we can choose in $\mathfrak{K}$ $n+1$ elements $b_0, \ldots, b_n$, not all zero, such that $\sum_{i=0}^{n} b_i \gamma_i = 0$. Thus $n+1$ elements are always linearly dependent. Since this also holds for the powers $\varepsilon, \gamma, \gamma^2, \ldots, \gamma^n$ of any element of $\mathfrak{K}(j)$, we have the theorem:

3. If the field Ω arises from $\mathfrak{K}$ by adjoining an algebraic element j, then all elements of Ω are algebraic. If n is the degree of j, then every system of more than n elements is linearly dependent.

§ 7.
Foundations of algebraic extensions.

Algebraic extensions; algebraically closed fields. — After §§ 5 and 6 have investigated the simplest extensions, the primitive ones, and have led to a distinction between (relatively) algebraic and transcendental elements, we shall now first consider algebraic extensions more closely. A field Ω is called *algebraic relative to a subfield* $\mathfrak{K}$, or an *algebraic extension of* $\mathfrak{K}$, if every element of Ω is algebraic relative to $\mathfrak{K}$. Every field Ω contains elements that are algebraic relative to a given subfield $\mathfrak{K}$. For it is clear that every element of $\mathfrak{K}$ itself is algebraic relative to $\mathfrak{K}$, and in fact of degree 1. Likewise one sees immediately that an element of Ω not belonging to $\mathfrak{K}$, if it is algebraic relative to $\mathfrak{K}$, has degree > 1, hence is a zero of a function irreducible in $\mathfrak{K}$ and of degree greater than the first. Thus, if an algebraic extension Ω of $\mathfrak{K}$ is to exist, and specifically a proper one, i.e. one different from $\mathfrak{K}$, it is necessary, and by § 6, 2. also sufficient, that there be irreducible functions in $\mathfrak{K}$ whose degree is > 1. If none are present, so that every integral function of higher degree decomposes into linear factors, then the field $\mathfrak{K}$ is no longer capable of any algebraic extension and will therefore be called *algebraically closed.* The existence of such fields was first proved by the fundamental theorem of algebra, which asserts that the field of complex numbers is algebraically closed. *Absolutely algebraic elements and fields.* — If an element α of a field Ω is algebraic relative to a subfield $\mathfrak{K}_1$, then of course it is also algebraic relative to every field between $\mathfrak{K}_1$ and Ω. Therefore, if α is algebraic relative to the prime field contained in Ω, then α is algebraic relative to every subfield. Such an element α shall therefore be called *absolutely algebraic.* Likewise we call a field *absolutely algebraic* if it is already algebraic relative to its prime field, or, equivalently, if all of its elements are absolutely algebraic.

Finite extensions. — If $\mathfrak{K}$ is a subfield of Ω, we say that Ω is *finite relative to* $\mathfrak{K}$ *and of degree* n — in symbols

$$[\Omega : \mathfrak{K}] = n$$

— if one can specify in Ω n elements linearly independent relative to $\mathfrak{K}$, while every system of more than n elements from Ω is linearly dependent relative to $\mathfrak{K}$.

1. Every finite extension of a field is algebraic.

For if $[\Omega : \mathfrak{K}] = n$ and α is an element of Ω, then the elements $\varepsilon, \alpha, \alpha^2, \ldots, \alpha^n$ are linearly dependent relative to $\mathfrak{K}$, i.e. α is algebraic relative to $\mathfrak{K}$. But there are also infinite (non-finite) algebraic extensions, for example the extension of the field of rational numbers to the field of all algebraic numbers. — One sees at once that every field $\mathfrak{K}$ is finite relative to itself, namely of degree 1, while every proper finite extension has degree > 1.

Basis of a finite extension. — Suppose the field Ω is finite relative to its subfield $\mathfrak{K}$, $[\Omega : \mathfrak{K}] = n$, and that the elements $\alpha_1, \ldots, \alpha_n$ of Ω are linearly independent relative to $\mathfrak{K}$. If β is any element of Ω, then among $\alpha_1, \ldots, \alpha_n$ and β there is a homogeneous linear relation with coefficients in $\mathfrak{K}$, which, because the coefficient of β is non-zero, can be given the form

$$\beta = c_1\alpha_1 + \cdots + c_n\alpha_n. \tag{1.}$$

This representation is possible in only one way, because two equations for β of this form would by subtraction give a homogeneous linear relation among $\alpha_1, \ldots, \alpha_n$. — Conversely, let us now assume that $\alpha_1, \ldots, \alpha_n$ are elements of some extension field Ω of $\mathfrak{K}$. If it is known that every element β of Ω can be represented *at most* one way in the form (1.), then this also holds for $\beta = 0$; hence the expression $c_1\alpha_1 + \cdots + c_n\alpha_n$ is equal to 0 only when all coefficients are 0. Therefore the extension Ω, if it is finite at all, has degree *at least* n. If, on the other hand, it is known that every element β of Ω can be represented *at least* one way in the form (1.), then for any $n + 1$ elements $\beta_0, \beta_1, \ldots, \beta_n$ one obtains representations of the form

$$\beta_i = c_{i1}\alpha_1 + \cdots + c_{in}\alpha_n \qquad (i = 0, \ldots, n),$$

and hence, by the theorem on homogeneous linear equations cited in § 1, one can choose $n+1$ elements $d_0, \ldots, d_n$ in $\mathfrak{K}$, not all zero, such that $d_0\beta_0 + \cdots + d_n\beta_n = 0$. Thus the extension Ω is finite and has degree *at most* n. — Consequently, if the representation (1.) is to be possible in one and only one way for all elements of Ω, then one must have $[\Omega : \mathfrak{K}] = n$, and the system $\alpha_1, \ldots, \alpha_n$ must be linearly independent. For this reason, a system of n linearly independent elements of an extension field of degree n is called a *basis* of this field relative to the ground field. — If the elements $\alpha_1, \ldots, \alpha_n$ form a basis for the extension field Ω, and if $\beta_1, \ldots, \beta_n$ are any elements of Ω whose representation in the form (1.) is

$$\beta_i = c_{i1}\alpha_1 + \cdots + c_{in}\alpha_n \qquad (i = 1, \ldots, n),$$

then one shows in the usual way that the elements $\beta_1, \ldots, \beta_n$ form a basis for Ω if and only if the determinant $|c_{ik}|$ is different from zero.

In connection with this we derive some simple theorems on finite extensions:

2. If Ω is an extension of $\mathfrak{K}$ and $\mathfrak{M}$ an extension of Ω, then $\mathfrak{M}$ is finite relative to $\mathfrak{K}$ if and only if Ω is finite relative to $\mathfrak{K}$ and $\mathfrak{M}$ is finite relative to Ω; and in this case

$$[\mathfrak{M} : \mathfrak{K}] = [\Omega : \mathfrak{K}] \cdot [\mathfrak{M} : \Omega].$$

Indeed, if $[\mathfrak{M} : \mathfrak{K}] = \nu$, then among $\nu + 1$ elements of $\mathfrak{M}$ there is a homogeneous linear relation with coefficients belonging to $\mathfrak{K}$, hence also to Ω. Thus $\mathfrak{M}$ is finite relative to Ω. Likewise, among $\nu + 1$ elements of Ω, since they belong to $\mathfrak{M}$, there is a homogeneous linear relation with coefficients in $\mathfrak{K}$; that is, Ω is finite relative to $\mathfrak{K}$. Conversely, suppose that

$$[\Omega : \mathfrak{K}] = m, \qquad [\mathfrak{M} : \Omega] = n.$$

Choose a basis $\alpha_1, \ldots, \alpha_m$ of Ω relative to $\mathfrak{K}$ and a basis $\beta_1, \ldots, \beta_n$ of $\mathfrak{M}$ relative to Ω. Then the $m \cdot n$ elements $\alpha_i \cdot \beta_k$ $(i = 1, \ldots, m;\ k = 1, \ldots, n)$ are linearly independent relative to $\mathfrak{K}$. For from a relation $\sum_{i,k} c_{ik}\alpha_i\beta_k = 0$, in which the coefficients c_{ik} belong to the field $\mathfrak{K}$, it follows, if

$$\sum_{i=1}^{m} c_{ik}\alpha_i = \gamma_k \qquad (k = 1, \ldots, n) \tag{2.}$$

is put, that $\sum_{k=1}^{n} \gamma_k\beta_k = 0$. Since the coefficients γ_k belong to Ω and the elements β_k form a basis of $\mathfrak{M}$, one must have $\gamma_1 = 0, \ldots, \gamma_n = 0$. But then (2.) implies that all $m \cdot n$ coefficients c_{ik} must be equal to 0, proving the linear independence of the elements $\alpha_i\beta_k$. Further, if τ is any element of $\mathfrak{M}$, then one obtains

$$\tau = \sigma_1\beta_1 + \cdots + \sigma_n\beta_n,$$

where the coefficients σ_k belong to Ω, and

$$\sigma_k = a_{1k}\alpha_1 + \cdots + a_{mk}\alpha_m,$$

where the coefficients a_{ik} are elements of $\mathfrak{K}$. Hence

$$\tau = \sum_{i,k} a_{ik}\alpha_i\beta_k.$$

Thus every element of $\mathfrak{M}$ is expressible linearly and homogeneously, with coefficients from $\mathfrak{K}$, in terms of the $m \cdot n$ elements $\alpha_i\beta_k$, which are linearly independent relative to $\mathfrak{K}$; hence

$$[\mathfrak{M} : \mathfrak{K}] = m \cdot n.$$

3. If Ω and $\mathfrak{M}$ are finite extensions of equal degree over $\mathfrak{K}$ and Ω is contained in $\mathfrak{M}$, then Ω is identical with $\mathfrak{M}$. For by 2. $[\mathfrak{M} : \Omega] = 1$, hence $\Omega = \mathfrak{M}$.

4. If $\mathfrak{K}(j)$ is a simple algebraic extension of $\mathfrak{K}$, then

$$[\mathfrak{K}(j) : \mathfrak{K}] = [j : \mathfrak{K}].$$

Indeed, if $[j : \mathfrak{K}] = n$, then, as was shown in the preceding section, every element of $\mathfrak{K}(j)$ is representable in one and only one way in the form

$$c_0 + c_1 j + \cdots + c_{n-1} j^{n-1}$$

with coefficients from $\mathfrak{K}$; hence the elements $\varepsilon, j, \ldots, j^{n-1}$ form a basis of $\mathfrak{K}(j)$ relative to $\mathfrak{K}$.

5. If the field Ω is finite of degree n relative to its subfield $\mathfrak{K}$, then the degree of every element of Ω (relative to $\mathfrak{K}$) divides n; the extension Ω is simple if and only if elements of degree n occur, and in that case these elements are primitive and all others imprimitive.

For if j is any element of Ω, then by 2. and 4.

$$n = [\Omega : \mathfrak{K}] = [\Omega : \mathfrak{K}(j)] \cdot [j : \mathfrak{K}],$$

so $[j : \mathfrak{K}]$ divides n. Further, for j to be a primitive element of Ω, i.e. for $\Omega = \mathfrak{K}(j)$, it is necessary and sufficient that $[\Omega : \mathfrak{K}(j)] = 1$, hence that $[j : \mathfrak{K}] = n$.

6. An extension is finite if and only if it can be obtained by adjoining a finite number of (relatively) algebraic elements.

Indeed, if $\Omega = \mathfrak{K}_0(j_1, j_2, \ldots, j_\nu)$, where the elements $j_1, j_2, \ldots, j_\nu$ are algebraic relative to $\mathfrak{K}_0$, and if one sets

$$\mathfrak{K}_0(j_1) = \mathfrak{K}_1, \quad \mathfrak{K}_1(j_2) = \mathfrak{K}_2, \ldots, \quad \mathfrak{K}_{\nu-1}(j_\nu) = \mathfrak{K}_\nu = \Omega,$$

then every element j_k is algebraic relative to the field $\mathfrak{K}_{k-1}$ lying between $\mathfrak{K}_0$ and Ω; hence $\mathfrak{K}_k$ is a simple algebraic and therefore finite extension of $\mathfrak{K}_{k-1}$. It follows from 2. that $\mathfrak{K}_\nu = \Omega$ is also a finite extension of $\mathfrak{K}_0$. — Conversely, suppose Ω is a finite extension of $\mathfrak{K}_0$, $[\Omega : \mathfrak{K}_0] = n$, and let j_1 be an element of Ω not belonging to $\mathfrak{K}_0$; further, if $\mathfrak{K}_1 = \mathfrak{K}_0(j_1)$ is not yet equal to Ω, let j_2 be an element not belonging to $\mathfrak{K}_1$; if $\mathfrak{K}_2 = \mathfrak{K}_1(j_2)$ is not yet equal to Ω, let j_3 be an element of Ω not belonging to $\mathfrak{K}_2$, and so on. Then the degrees $[\mathfrak{K}_1 : \mathfrak{K}_0]$, $[\mathfrak{K}_2 : \mathfrak{K}_0], \ldots$ form a strictly increasing sequence not exceeding n, which can be continued as long as the number n has not yet been reached. Therefore, after a finite number ν of steps the number n is reached, and then $\mathfrak{K}_0(j_1, j_2, \ldots, j_\nu) = \mathfrak{K}_\nu = \Omega$.

To these theorems are added several more that concern *arbitrary* algebraic extensions:

7. If the field $\Omega = \mathfrak{K}(\mathfrak{S})$ is obtained from its subfield $\mathfrak{K}$ by adjoining a system $\mathfrak{S}$, all of whose elements are algebraic relative to $\mathfrak{K}$, then Ω is algebraic relative to $\mathfrak{K}$.

If $\mathfrak{S}$ is a finite system, then by 6. Ω is finite relative to $\mathfrak{K}$, and hence algebraic. If, however, $\mathfrak{S}$ is infinite, then nevertheless (§ 2, 3.) every element α of Ω is contained in a field arising from $\mathfrak{K}$ by adjoining a finite subsystem of $\mathfrak{S}$; hence, as in the preceding case, it is algebraic relative to $\mathfrak{K}$.

8. The elements of a field Ω that are algebraic relative to a subfield $\mathfrak{K}$ form a field between $\mathfrak{K}$ and Ω; in particular, the absolutely algebraic elements of Ω form a field.

Indeed, let $\mathfrak{S}$ be the system of all elements of Ω algebraic relative to $\mathfrak{K}$. Then the field $\mathfrak{K}(\mathfrak{S})$ lying between $\mathfrak{K}$ and Ω contains only elements algebraic relative to $\mathfrak{K}$ (Theorem 7), whence $\mathfrak{S} = \mathfrak{K}(\mathfrak{S})$.

9. If the field Ω lies between the fields $\mathfrak{K}$ and $\mathfrak{M}$, and if Ω is algebraic relative to $\mathfrak{K}$ and $\mathfrak{M}$ is algebraic relative to Ω, then $\mathfrak{M}$ is algebraic relative to $\mathfrak{K}$.

Let β be any element of $\mathfrak{M}$. If $[\beta : \Omega] = n$, then we have an equation

$$\alpha_0 + \alpha_1 \beta + \cdots + \alpha_{n-1} \beta^{n-1} + \beta^n = 0, \tag{3.}$$

where $\alpha_0, \alpha_1, \ldots, \alpha_{n-1}$ are elements of Ω. These elements are algebraic relative to $\mathfrak{K}$, and so by 6. the field $\mathfrak{K}' = \mathfrak{K}(\alpha_0, \alpha_1, \ldots, \alpha_{n-1})$ is finite relative to $\mathfrak{K}$. Further, by equation (3.), the element β is algebraic relative to $\mathfrak{K}'$, so $\mathfrak{K}'(\beta)$ is finite relative to $\mathfrak{K}'$. Thus by 2. $\mathfrak{K}'(\beta)$ is finite relative to $\mathfrak{K}$, and hence β is algebraic relative to $\mathfrak{K}$, which proves Theorem 9.

§ 8. Decomposition of an arbitrary polynomial into linear factors. Normal fields and normal functions.

In § 6 it was shown that every polynomial irreducible in a field has a root in a suitable extension field. Following this, one can now treat the further problem of extending a field in such a way that an arbitrarily given function from the original field decomposes completely, that is, into linear factors, in the extension field.

Let $f(x)$ now be an arbitrary polynomial in a field $\mathfrak{K}_0$, let n be the degree of $f(x)$, and let m_0 be the number of factors of $f(x)$ irreducible in $\mathfrak{K}_0$. Then $m_0 \leq n$, and it is equal to n if and only if $f(x)$ decomposes completely in $\mathfrak{K}_0$. Suppose that $m_0 < n$. Then $f(x)$ has an irreducible factor $g_1(x)$ whose degree is > 1. We extend $\mathfrak{K}_0$ by adjoining an element j_1 for which $g_1(j_1) = 0$ (§ 6). Since in the field $\mathfrak{K}_1 = \mathfrak{K}_0(j_1)$ the function $g_1(x)$ is no longer irreducible, the number m_1 of irreducible factors of $f(x)$ in $\mathfrak{K}_1$ is $> m_0$; hence $m_0 < m_1 \leq n$. If $m_1 < n$, let $g_2(x)$ be an irreducible non-linear factor of $f(x)$ within $\mathfrak{K}_1$. We adjoin a root j_2 of $g_2(x)$, put $\mathfrak{K}_1(j_2) = \mathfrak{K}_2$, and denote by m_2 the number of irreducible factors of $f(x)$ in $\mathfrak{K}_2$, so that $m_0 < m_1 < m_2 \leq n$. We continue in this way until we arrive at a field $\mathfrak{K}_\nu$ for which $m_\nu = n$, that is, in which $f(x)$ decomposes into linear factors. The field $\mathfrak{K}_\nu$ is then a finite extension of $\mathfrak{K}_0$. Let $j_1, \ldots, j_\nu$ be the elements successively adjoined, let $\mathfrak{K}_r = \mathfrak{K}_{r-1}(j_r)$ $(r = 1, \ldots, \nu)$ be the fields successively obtained, and let $g_r(x)$ be the factor of $f(x)$ irreducible in $\mathfrak{K}_{r-1}$ with root j_r. We may say that the extension $\mathfrak{K}_\nu$ just suffices for the complete decomposition of $f(x)$. This means that no field different from $\mathfrak{K}_\nu$ between $\mathfrak{K}_0$ and $\mathfrak{K}_\nu$ already suffices for this purpose; and this follows from the fact that the adjoined elements $j_1, \ldots, j_\nu$ are all roots of $f(x)$ and therefore must occur in every subfield of $\mathfrak{K}_\nu$ which effects the decomposition.

Let $\overline{\mathfrak{K}}_0$ now be a field isomorphic to $\mathfrak{K}_0$, and let π_0 denote an isomorphism between $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$. By π_0, every polynomial $g(x)$ from $\mathfrak{K}_0$ is assigned a definite polynomial $\overline{g}(x)$ from $\overline{\mathfrak{K}}_0$, obtained by replacing the coefficients of $g(x)$ by the corresponding elements in $\overline{\mathfrak{K}}_0$. The resulting relation between $\mathfrak{K}_0[x]$ and $\overline{\mathfrak{K}}_0[x]$ is plainly also an isomorphism. Therefore, if $\overline{f}(x)$ and $\overline{g}_1(x)$ are the functions from $\overline{\mathfrak{K}}_0$ assigned to the functions $f(x)$ and $g_1(x)$ occurring above, then $\overline{f}(x)$ decomposes in $\overline{\mathfrak{K}}_0$ into m_0 irreducible factors, and $\overline{g}_1(x)$ is one of them. Let Ω be any extension field of $\overline{\mathfrak{K}}_0$ in which $\overline{f}(x)$ decomposes completely. Then the extension Ω either just suffices for the complete decomposition of $\overline{f}(x)$ or it does not; in every case a definite field $\mathfrak{M}$ between $\overline{\mathfrak{K}}_0$ and Ω, namely the one obtained by adjoining to $\overline{\mathfrak{K}}_0$ the roots $\overline{j}$ of $\overline{f}(x)$ lying in Ω, has this property. Since $\overline{f}(x)$ decomposes completely in $\mathfrak{M}$, the factor $\overline{g}_1(x)$ of $\overline{f}(x)$ has roots in $\mathfrak{M}$. Let $\overline{j}_1$ be such a root, and let r denote the degree of $g_1(x)$ and $\overline{g}_1(x)$. By the isomorphism between $\mathfrak{K}_0[x]$ and $\overline{\mathfrak{K}}_0[x]$, the functions $g_1(x)$ and $\overline{g}_1(x)$ are assigned to one another; hence the residue-class fields belonging to these functions as moduli are isomorphic to one another. These residue-class fields, however, stand in isomorphic relation to the fields $\mathfrak{K}_0(j_1) = \mathfrak{K}_1$ and $\overline{\mathfrak{K}}_0(\overline{j}_1) = \overline{\mathfrak{K}}_1$, and there results an isomorphism π_1 between $\mathfrak{K}_1$ and $\overline{\mathfrak{K}}_1$ by which each element of $\mathfrak{K}_1$, written in the form

$$a_0 + a_1 j_1 + \cdots + a_{r-1} j_1^{r-1}$$

with coefficients a from $\mathfrak{K}_0$, is assigned the element

$$\overline{a}_0 + \overline{a}_1 \overline{j}_1 + \cdots + \overline{a}_{r-1} \overline{j}_1^{r-1}$$

of $\overline{\mathfrak{K}}_1$ obtained by replacing a_k $(k = 0, \ldots, r-1)$ by the corresponding element $\overline{a}_k$ of $\overline{\mathfrak{K}}_0$ under π_0. The relation π_0 is contained in π_1, since π_1 assigns to one another the elements of the fields

$\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$ in the same way as π_0. We may therefore say: there is an isomorphism π_1 between $\mathfrak{K}_1$ and $\overline{\mathfrak{K}}_1$ under which the function $f(x)$ from $\mathfrak{K}_1$ is assigned the function $\overline{f}(x)$ from $\overline{\mathfrak{K}}_1$; the field $\mathfrak{M}$ is an extension of $\overline{\mathfrak{K}}_1$ which just suffices for the complete decomposition of $\overline{f}(x)$. We can therefore apply to $\mathfrak{K}_1, \overline{\mathfrak{K}}_1, \pi_1, f(x), \overline{f}(x)$ the same argument just applied to $\mathfrak{K}_0, \overline{\mathfrak{K}}_0, \pi_0, f(x), \overline{f}(x)$. Continuing in this way, we arrive at a sequence of fields

$$\overline{\mathfrak{K}}_0, \overline{\mathfrak{K}}_1, \ldots, \overline{\mathfrak{K}}_\nu$$

with the following properties: each preceding field is contained in the following one, and all are contained in $\mathfrak{M}$. Between $\mathfrak{K}_r$ and $\overline{\mathfrak{K}}_r$ $(r = 0, \ldots, \nu)$ there is an isomorphism π_r which includes π_0 and therefore assigns the functions $f(x)$ and $\overline{f}(x)$ to one another. Since $f(x)$ decomposes into linear factors in $\mathfrak{K}_\nu$, the same holds for $\overline{f}(x)$ in $\overline{\mathfrak{K}}_\nu$; and since $\overline{\mathfrak{K}}_\nu$ lies between $\overline{\mathfrak{K}}_0$ and $\mathfrak{M}$, while $\mathfrak{M}$ was to be an extension of $\overline{\mathfrak{K}}_0$ just sufficient for the complete decomposition of $\overline{f}(x)$, we have $\overline{\mathfrak{K}}_\nu = \mathfrak{M}$. Thus we have the theorem:

1. Let $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$ be isomorphic fields, and let π_0 denote an isomorphism between them which assigns to the polynomial $f(x)$ from $\mathfrak{K}_0$ the polynomial $\overline{f}(x)$ from $\overline{\mathfrak{K}}_0$. If $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are extensions of $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$, respectively, which just suffice for the complete decomposition of $f(x)$ and $\overline{f}(x)$, respectively, then there is an isomorphism π' between $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ which includes the relation π_0.

The hypotheses of theorem 1 are fulfilled if $\overline{\mathfrak{K}}_0 = \mathfrak{K}_0$ is taken and π_0 is understood as the relation which lets every element of $\mathfrak{K}_0$ correspond to itself. Then $\overline{f}(x) = f(x)$, and the relation π' likewise leaves every element of $\mathfrak{K}_0$ unchanged; that is, $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are equivalent extensions. Thus we have the following result:

2. To every polynomial $f(x)$ of a field $\mathfrak{K}_0$ there belongs one, and essentially only one, extension of this field which just suffices for the complete decomposition of $f(x)$.

This is the reason why all theorems about roots of equations that matter for us are independent of the way in which we extend the ground field in order to solve the equations. To spell this out somewhat more closely, let

$$f(x) = (x - \omega_1)(x - \omega_2) \cdots (x - \omega_n)$$

be the decomposition of a function $f(x)$ of a field $\mathfrak{K}_0$ in an extension field Ω. Then the field $\mathfrak{N} = \mathfrak{K}_0(\omega_1, \ldots, \omega_n)$ lying between $\mathfrak{K}_0$ and Ω is an extension just sufficient for this decomposition. If one has a second decomposition of $f(x)$ in another extension field Ω', and if $\mathfrak{N}'$ is the field obtained from $\mathfrak{K}_0$ by adjoining the roots of $f(x)$ occurring in Ω', then there is an isomorphism π' between $\mathfrak{N}$ and $\mathfrak{N}'$ which leaves all elements of $\mathfrak{K}_0$ unchanged, and the decomposition of $f(x)$ in Ω' is

$$f(x) = (x - \omega_1')(x - \omega_2') \cdots (x - \omega_n'),$$

where ω_i and ω_i' $(i = 1, \ldots, n)$ correspond to one another under the isomorphism π'. Thus if we have a rational combination of $\omega_1, \ldots, \omega_n$ with coefficients from $\mathfrak{K}_0$, hence an element of $\mathfrak{N}$, we obtain the corresponding element of $\mathfrak{N}'$ by replacing each ω_i by ω_i'. In particular, if certain of the roots ω_i are equal, then the corresponding roots ω_i' must also be equal. Hence we can speak of the number of simple, double, triple, ... roots of an equation without specifying the extension field. The determination of these numbers will be carried out later (§ 10).

Normal extensions. — The extensions of a field $\mathfrak{K}_0$ which just suffice for the complete decomposition of a function $f(x)$ belong to the special kind of normal extensions. An extension field $\mathfrak{N}$ of $\mathfrak{K}_0$ is called normal with respect to $\mathfrak{K}_0$ if it is algebraic with respect to $\mathfrak{K}_0$ and if, moreover, every polynomial $f(x)$ from $\mathfrak{K}_0$ which is irreducible in $\mathfrak{K}_0$ and has a root in $\mathfrak{N}$ decomposes into linear factors in $\mathfrak{N}$. Plainly every extension equivalent to a normal extension is again normal. We derive a few simple theorems concerning normal fields.

3. If the field $\mathfrak{N}$ is normal with respect to its subfield $\mathfrak{K}_0$, then it is also normal with respect to every field $\mathfrak{K}_1$ between $\mathfrak{K}_0$ and $\mathfrak{N}$.

First, $\mathfrak{N}$ is algebraic with respect to $\mathfrak{K}_1$. Now let $\varphi(x)$ be a polynomial irreducible in $\mathfrak{K}_1$, and suppose that $\varphi(x)$ has a root j in $\mathfrak{N}$. Then j is also a root of an irreducible polynomial $f(x)$ of the field $\mathfrak{K}_0$, and since $\mathfrak{N}$ is normal with respect to $\mathfrak{K}_0$, $f(x)$ decomposes into linear factors in $\mathfrak{N}$. But $f(x)$ is also a function in the field $\mathfrak{K}_1$ and has the root j in common with the irreducible function $\varphi(x)$ of this field; therefore $\varphi(x)$ is a factor of $f(x)$. Hence $\varphi(x)$ also decomposes into linear factors in $\mathfrak{N}$. This proves the assertion.

Normal functions. — A polynomial $f(x)$ of a field $\mathfrak{K}_0$ is to be called normal in $\mathfrak{K}_0$ if it is irreducible in $\mathfrak{K}_0$ and, in addition, the extension $\mathfrak{K}_0(j)$ obtained by adjoining a root j of $f(x)$ is normal. If $f(x)$ is normal in $\mathfrak{K}_0$, then $f(x)$ decomposes into linear factors in the field $\mathfrak{K}_0(j)$ obtained by adjoining a root j of $f(x)$. This follows directly from the definition of normal extensions and functions. But the converse also holds.

4. If a function $f(x)$ irreducible in a field $\mathfrak{K}_0$ decomposes completely in the field $\mathfrak{K}_0(j)$ obtained by adjoining a root j, then it is normal with respect to $\mathfrak{K}_0$.

Indeed, if $g(x)$ is an irreducible function from $\mathfrak{K}_0$ and $g(x)$ has a root α in $\mathfrak{K}_0(j)$, then we can put $\alpha = \varphi(j)$, where $\varphi(x)$ denotes a polynomial from $\mathfrak{K}_0$. If

$$f(x) = (x - j)(x - j_1)\cdots(x - j_{n-1})$$

is the assumed decomposition of $f(x)$ in $\mathfrak{K}_0(j)$, then one concludes in the known way that the coefficients of the function

$$h(x) = (x - \varphi(j))(x - \varphi(j_1))\cdots(x - \varphi(j_{n-1}))$$

are symmetric functions of $j, j_1, \ldots, j_{n-1}$ and hence elements of $\mathfrak{K}_0$. Thus $h(x)$ is a function in $\mathfrak{K}_0$ which has with the function $g(x)$ irreducible in $\mathfrak{K}_0$ the common factor $x - \varphi(j)$, and therefore is divisible by $g(x)$. Since $h(x)$ decomposes completely in $\mathfrak{K}_0(j)$, the same holds for $g(x)$; that is, for every irreducible function in $\mathfrak{K}_0$ which has a root in $\mathfrak{K}_0(j)$. Thus $\mathfrak{K}_0(j)$ is a normal extension of $\mathfrak{K}_0$, and consequently $f(x)$ is normal in $\mathfrak{K}_0$.

5. If $f(x)$ is a normal function of the field $\mathfrak{K}_0$, and Ω is any extension of $\mathfrak{K}_0$, then every factor of $f(x)$ irreducible in Ω is normal in Ω.

Indeed, if $g(x)$ is such a factor and Ω is extended by adjoining a root j of $g(x)$, then $\Omega(j)$ contains the field $\mathfrak{K}_0(j)$, and j is a root of $f(x)$. Since the function $f(x)$ is normal in $\mathfrak{K}_0$, it decomposes into linear factors in $\mathfrak{K}_0(j)$ and therefore also in $\Omega(j)$. Since in the field Ω the function $g(x)$ is a factor of $f(x)$, $g(x)$ also decomposes into linear factors in $\Omega(j)$. Therefore by theorem 4, $g(x)$ is normal in Ω, as was to be shown. *6. An extension $\mathfrak{N}$ of a field $\mathfrak{K}_0$ which just suffices to decompose completely a function $f(x)$ from $\mathfrak{K}_0$ is normal.*

For if

$$f(x) = (x - j_1)(x - j_2)\cdots(x - j_n)$$

is the decomposition of $f(x)$ in $\mathfrak{N}$, and if $g(x)$ is an irreducible function of the field $\mathfrak{K}_0$ which has in $\mathfrak{N}$ a root β, then β can be represented in the form

$$\beta = \psi(j_1, j_2, \ldots, j_n),$$

where $\psi(x_1, x_2, \ldots, x_n)$ denotes a polynomial from $\mathfrak{K}_0$. The $n!$ permutations of the elements $j_1, j_2, \ldots, j_n$ give from β $n!$ elements of the field $\mathfrak{N}$, which we may call $\beta, \beta_1, \ldots, \beta_{n!-1}$. The function

$$\varphi(x) = (x - \beta)(x - \beta_1)\cdots(x - \beta_{n!-1})$$

lies in the field $\mathfrak{K}_0$, since its coefficients are symmetric functions of $j_1, j_2, \ldots, j_n$ with coefficients from $\mathfrak{K}_0$; it has with the function $g(x)$ irreducible in $\mathfrak{K}_0$ the common factor $x - \beta$, and so is divisible by $g(x)$. Since $\varphi(x)$ decomposes completely in $\mathfrak{N}$, $g(x)$ also decomposes completely in $\mathfrak{N}$. Thus the conditions for a normal extension are fulfilled.

In theorems 1, 2, and 6 of this paragraph, in place of a single function $f(x)$ of the field $\mathfrak{K}_0$ one may of course put a system S consisting of a finite number of functions of this field. For if the system S consists of the functions

$$\varphi_1(x),\ \varphi_2(x), \dots, \varphi_m(x)$$

and if

$$f(x) = \varphi_1(x) \cdot \varphi_2(x) \cdots \varphi_m(x)$$

is set, then it is clear that in an extension field Ω of $\mathfrak{K}_0$ all functions of the system S decompose completely if and only if the function $f(x)$ decomposes completely in Ω.

§ 9. Differentiation.

To determine the number of roots of specified multiplicity, we bring in the derivatives of functions. The derivative, or differential quotient, $\frac{d}{dx}f(x) = f'(x)$ of a polynomial

$$f(x) = a_0 + a_1 x + \cdots + a_n x^n$$

in an indeterminate x with respect to this indeterminate is defined by us purely formally as the expression

$$a_1 + 2a_2 x + \cdots + n a_n x^{n-1},$$

so that $f'(x)$ occurs as a function in every field $\mathfrak{K}$ in which $f(x)$ occurs. From this definition it follows immediately, if

$$g(x) = b_0 + b_1 x + \cdots + b_m x^m$$

is a second function from $\mathfrak{K}$, that

$$\frac{d}{dx}(f(x) \pm g(x)) = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x).$$

Furthermore,

$$\begin{aligned} \frac{d}{dx}(f(x)g(x)) &= \frac{d}{dx}\sum a_q b_r x^{q+r} = \sum (q+r) a_q b_r x^{q+r-1} \\ &= \sum a_q x^q \, r b_r x^{r-1} + \sum q a_q x^{q-1} b_r x^r = f(x)\frac{d}{dx}g(x) + \frac{d}{dx}f(x)\, g(x), \end{aligned}$$

hence

$$\frac{d}{dx}(f(x)g(x)) = f(x)g'(x) + f'(x)g(x).$$

Let $R(x)$ now be a rational function of x. We can represent it in different ways as a quotient of polynomials,

$$R(x) = \frac{u}{v} = \frac{u_1}{v_1}.$$

Then

$$uv_1 = vu_1. \tag{1}$$

Taking the derivative on both sides, one obtains

$$uv_1' + u'v_1 = vu_1' + v'u_1,$$

and hence

$$u'v_1 - vu_1' = v'u_1 - uv_1'. \tag{2}$$

Multiplying this equation by vv_1, one obtains

$$v_1^2 vu' - v^2 v_1 u_1' = vv_1(v'u_1 - uv_1'). \tag{3}$$

Multiplying equation (1) by $vv_1' + v'v_1$, one obtains

$$vv_1uv_1' + v_1^2 uv' = vv_1v'u_1 + v^2u_1v_1',$$

therefore

$$v_1^2 uv' - v^2 u_1 v_1' = vv_1(v'u_1 - uv_1'). \tag{4}$$

From (3) and (4) follows

$$(vu' - uv')v_1^2 = (v_1u_1' - u_1v_1')v^2,$$

so that

$$\frac{vu' - uv'}{v^2} = \frac{v_1u_1' - u_1v_1'}{v_1^2}. \tag{5}$$

We are therefore justified in understanding by the derivative of a rational function $R(x) = \frac{u}{v}$ the expression $\frac{vu'-uv'}{v^2}$, because this definition, as equation (5) shows, is independent of the particular representation of $R(x)$, and because, in the case in which $R(x)$ is equal to a polynomial u (which can be written in the form $\frac{u}{\varepsilon}$), it agrees with the earlier definition. Direct calculation then immediately gives, for any two rational functions u, v, the laws

$$\frac{d}{dx}(u \pm v) = u' \pm v', \tag{6}$$

$$\frac{d}{dx}(uv) = uv' + u'v, \tag{7}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}. \tag{8}$$

From (7) one further obtains

$$\frac{d}{dx}u^n = nu^{n-1}u',$$

and hence, if $\lambda(u) = a_0 + a_1u + \cdots + a_nu^n$ is a polynomial in u (in the field $\mathfrak{K}$),

$$\frac{d}{dx}\lambda(u) = (a_1 + 2a_2u + \cdots + na_nu^{n-1})u' = \frac{d}{du}\lambda(u) \cdot \frac{du}{dx}.$$

Application of this equation gives, if $\mu(x)$ likewise denotes a polynomial,

$$\frac{d}{dx}\frac{\lambda(u)}{\mu(u)} = \frac{\mu(u)\frac{d}{du}\lambda(u) - \lambda(u)\frac{d}{du}\mu(u)}{(\mu(u))^2}\frac{du}{dx} = \frac{d}{du}\frac{\lambda(u)}{\mu(u)} \cdot \frac{du}{dx}.$$

In other words: if v is a rational function of u, and u is a rational function of x, then

$$\frac{dv}{dx} = \frac{dv}{du} \cdot \frac{du}{dx}.$$

The concept of differential quotient can also be extended still further, but this will not be pursued here.

§ 10. The multiplicity of the roots of a polynomial.

In the questions treated so far, the characteristic of the field has been of no importance. In the following investigations, however, we encounter essential differences between fields of characteristic 0 and fields of prime characteristic. This first appears in the question under what circumstances the derivative of a rational function vanishes.

Let a field $\mathfrak{K}$ be given, and let

$$f(x) = a_0 + a_1 x + \cdots + a_n x^n$$

be a polynomial in the indeterminate x over $\mathfrak{K}$. Then

$$f'(x) = a_1 + 2a_2 x + \cdots + n a_n x^{n-1},$$

so that in the case of characteristic 0, $f'(x)$ vanishes only when $a_1 = 0$, $a_2 = 0$, ..., $a_n = 0$, that is, when $f(x)$ does not contain x. But if the characteristic of $\mathfrak{K}$ is a prime p, then the coefficient $q \cdot a_q$ occurring in $f'(x)$ is equal to 0 when $a_q = 0$ or when q is divisible by p (§ 4, 4). Hence $f'(x)$ vanishes if and only if in $f(x)$ only powers of x whose exponent is divisible by p actually occur — in other words, if $f(x)$ is a polynomial in x^p. If we have a rational function in $\mathfrak{K}$ which we write, in reduced form, as the quotient $\frac{u}{v}$ of two polynomials, then its derivative is $\frac{vu'-uv'}{v^2}$ and is therefore equal to 0 only when $vu' = uv'$. Since u and v are relatively prime, this equation can hold only when v divides v' and u divides u' (§ 5, 3). But since the degrees of u' and v' are respectively smaller than the degrees of u and v, this is possible only when $u' = 0$, $v' = 0$, that is, when u and v are free of x, or, in the case of characteristic p, functions of x^p. Thus we have the theorem:

1. For the derivative of a rational function $R(x)$ to vanish, it is necessary and sufficient that $R(x)$ be free of x, respectively a rational function of x^p, according as the characteristic is 0 or p.

Let $f(x)$ now be a polynomial in a field $\mathfrak{K}$. Within a suitable extension field $f(x)$ decomposes into linear factors, and if $x - \alpha$ is an r-fold factor ($r > 0$), then

$$f(x) = (x-\alpha)^r g(x), \tag{1}$$

where $g(x)$ no longer contains the factor $x - \alpha$. From (1) follows

$$f'(x) = (x-\alpha)^{r-1}(r\varepsilon g(x) + (x-\alpha)g'(x)), \tag{2}$$

from which one sees that the derivative $f'(x)$ contains the factor $x - \alpha$ at least $(r-1)$ times, and contains it exactly $(r-1)$ times when $r\varepsilon \neq 0$. If, however, $\mathfrak{K}$ has characteristic p and r is divisible by p, then $r\varepsilon = 0$, and $f'(x) = (x-\alpha)^r g'(x)$ is divisible by $(x-\alpha)^r$. Thus:

2. If $f(x)$ has an r-fold linear factor $(r \geq 1)$, then the derivative $f'(x)$ has this factor at least $(r-1)$ times; it has it at least r times if the field $\mathfrak{K}$ to which $f(x)$ belongs has characteristic p and r is divisible by the prime p, otherwise exactly $(r-1)$ times. (If the derivative vanishes, then it is divisible by every arbitrarily high power of $x - \alpha$, and in this sense contains the factor $x - \alpha$ infinitely often.)

From 2 one easily obtains, for fields of arbitrary characteristic: *3. If $f(x)$ has only simple factors, then the greatest common divisor*

$$(f(x), f'(x))$$

of $f(x)$ and $f'(x)$ is equal to ε; otherwise it is at least of first degree.

Let

$$f(x) = (x-\alpha_1)^{r_1}(x-\alpha_2)^{r_2}\cdots(x-\alpha_\nu)^{r_\nu}, \tag{3}$$

where the roots α_i are distinct and the exponents $r_i > 0$. If the characteristic is p and $f(x)$ is the p-th power of a polynomial (of the original field or of an extension field), then all exponents r_i are divisible by p. Therefore, by 2, $f'(x)$ contains each factor $x - \alpha_i$ at least as often as $f(x)$ and is thus divisible by $f(x)$. Since, however, the degree of $f'(x)$ is smaller than the degree of $f(x)$, this is possible only when $f'(x) = 0$. This also follows immediately from the fact that, by differentiating an equation of the form $f(x) = [\varphi(x)]^p$, one obtains $f'(x) = p[\varphi(x)]^{p-1}\varphi'(x)$, and hence, in characteristic p, $f'(x) = 0$. It follows, however, that $f(x)$ is a polynomial in x^p. Conversely, if $f(x)$ is assumed to be a polynomial in x^p, then $f'(x)$ vanishes, and by 2 it follows that all exponents r_i are divisible by p. Thus:

4. In characteristic p, $f(x)$ is a polynomial in x^p if and only if $f(x)$ is the p-th power of a polynomial.

We can derive this result in another way. If a and b are any two elements of a field $\mathfrak{K}$ of characteristic p, the binomial expansion gives

$$(a + b)^p = a^p + b^p,$$

because all other terms of the expansion have a numerical coefficient divisible by p and hence disappear. Raising this equation to the p-th power, one obtains

$$(a + b)^{p^2} = (a^p + b^p)^p = a^{p^2} + b^{p^2},$$

and, continuing in this way, in general

$$(a + b)^{p^f} = a^{p^f} + b^{p^f} \tag{4}$$

for every natural number f. Similarly,

$$(a - b)^{p^f} = a^{p^f} - b^{p^f}. \tag{5}$$

For odd p this follows from the same expansion. If $p = 2$, however, then $2b = 0$, $2b^{p^f} = 0$, so that $-b = b$, $-b^{p^f} = b^{p^f}$, and formulas (4) and (5) are identical. From (4) follows immediately

$$(a_1 + a_2 + \cdots + a_n)^{p^f} = a_1^{p^f} + a_2^{p^f} + \cdots + a_n^{p^f}. \tag{4'}$$

We can also apply this equation to polynomials in an indeterminate x over the field $\mathfrak{K}$, since these belong to the field $\mathfrak{K}(x)$, which also has characteristic p (§ 4, 3). Then one sees at once that the p^f-th power of such a function is a polynomial in x^{p^f}. Conversely, if $f(x)$ is a polynomial in x^{p^f},

$$f(x) = c_0 + c_1 x^{p^f} + c_2 x^{2p^f} + \cdots + c_n x^{np^f},$$

and if we extend the field $\mathfrak{K}$ to a field Ω in which the functions $x^{p^f} - c_0$, $x^{p^f} - c_1$, ..., $x^{p^f} - c_n$ decompose into linear factors, then one can determine in Ω elements $\gamma_0, \gamma_1, \ldots, \gamma_n$ such that $\gamma_k^{p^f} = c_k$. Then

$$f(x) = (\gamma_0 + \gamma_1 x + \gamma_2 x^2 + \cdots + \gamma_n x^n)^{p^f}.$$

Thus theorem 4 is again proved, in a somewhat more general form:

4a. In characteristic p, $f(x)$ is a polynomial in x^{p^f} if and only if $f(x)$ is the p^f-th power of a polynomial in x.

Let $F(x)$ be a polynomial in the indeterminate x over a field $\mathfrak{K}$ of arbitrary characteristic; let the coefficient of the highest power of x be ε, and let

$$F(x) = (x - \alpha_1)^{r_1} \cdots (x - \alpha_\nu)^{r_\nu} \tag{6}$$

be the decomposition of $F(x)$ in a suitable extension field, with $\alpha_1, \ldots, \alpha_\nu$ assumed distinct. For every natural number m we denote by $f_m(x)$ the product of those linear factors, taken simply, of $F(x)$ which occur in $F(x)$ with multiplicity m. If none occur, set $f_m(x) = \varepsilon$. Thus

$$F(x) = f_1(x) \cdot (f_2(x))^2 \cdot (f_3(x))^3 \cdots . \tag{7}$$

We now prove the following theorems: *5. If the characteristic of $\mathfrak{K}$ is 0, then each of the functions $f_m(x)$ belongs to the field $\mathfrak{K}$ and can be determined rationally.*

6. If the characteristic of $\mathfrak{K}$ is p, and m is not divisible by p, then $f_m(x)$ belongs to the field $\mathfrak{K}$ and can be determined rationally.

7. If the characteristic of $\mathfrak{K}$ is p, and $\pi = p^f$ is the highest power of p contained in m, then $(f_m(x))^\pi$ belongs to the field $\mathfrak{K}$ and can be determined rationally.

Indeed, if $F'(x)$ denotes the derivative of $F(x)$ with respect to x, then from 2 it follows that the function belonging to the field $\mathfrak{K}$,

$$\varphi(x) = \frac{F(x)}{(F(x), F'(x))},$$

is, in characteristic 0, equal to the product of all linear factors taken simply, and in characteristic p equal to the product of those linear factors $x - \alpha_k$, taken simply, for which the exponent r_k in equation (6) is not divisible by p. Consider the expression

$$\chi_m(x) = \frac{((\varphi(x))^m, F'(x))}{((\varphi(x))^{m-1}, F'(x))} \qquad (m \geq 1).$$

This is a polynomial composed of factors $x - \alpha_k$, and in characteristic p it can contain only those factors $x - \alpha_k$ for which r_k is not divisible by p. Assuming this, in characteristic p, for r_k, if $r_k < m$, both numerator and denominator of $\chi_m(x)$ contain the factor $x - \alpha_k$ exactly r_k times. If $r_k \geq m$, then $x - \alpha_k$ occurs in the numerator m times and in the denominator $(m-1)$ times. Therefore $\chi_m(x)$ is the product of those linear factors $x - \alpha_k$ for which $r_k \geq m$, and which in characteristic p are not divisible by p. Hence

$$\frac{\chi_m(x)}{\chi_{m+1}(x)} = f_m(x),$$

provided the characteristic is not p with m divisible by p.

This proves theorems 5 and 6. In the case $f = 0$, theorem 7 coincides with theorem 6. We prove theorem 7 by induction, assuming that it has been proved for every m which contains the prime p as a factor fewer than f times. Then those factors $(f_m(x))^m$ in which m is not divisible by $\pi = p^f$ are already known to belong to the field $\mathfrak{K}$ and to have been determined. Dividing both sides of (7) by these factors, we obtain an equation of the form

$$\Phi(x) = (f_\pi(x))^\pi \cdot (f_{2\pi}(x))^{2\pi} \cdots , \tag{8}$$

where $\Phi(x)$ is a known function from $\mathfrak{K}$. Let

$$\Phi(x) = (x - \beta_1)^{\varrho_1 \pi} (x - \beta_2)^{\varrho_2 \pi} \cdots (x - \beta_\lambda)^{\varrho_\lambda \pi} \tag{9}$$

be the decomposition of $\Phi(x)$ into powers of distinct linear factors, and let $m = \mu\pi$, where μ no longer contains the factor p. The functions $(f_{k\pi}(x))^\pi$ are polynomials in x^π (theorem 4a), and the same is therefore true of $\Phi(x)$. If we put

$$x^\pi = y, \quad \beta_k^\pi = \gamma_k, \quad (f_{k\pi}(x))^\pi = g_k(y), \quad \Phi(x) = G(y),$$

then equations (8) and (9) pass into

$$G(y) = g_1(y) \cdot (g_2(y))^2 \cdot (g_3(y))^3 \cdots , \tag{10}$$

$$G(y) = (y - \gamma_1)^{\varrho_1} (y - \gamma_2)^{\varrho_2} \cdots (y - \gamma_\lambda)^{\varrho_\lambda}, \tag{11}$$

where $G(y)$ is a known function from $\mathfrak{K}$. The roots $\gamma_1, \gamma_2, \dots, \gamma_\lambda$ are distinct; for from $\gamma_k = \gamma_l$ would follow

$$0 = \gamma_k - \gamma_l = \beta_k^\pi - \beta_l^\pi = (\beta_k - \beta_l)^\pi,$$
$$0 = \beta_k - \beta_l, \qquad \beta_k = \beta_l,$$

although $\beta_1, \beta_2, \dots, \beta_\lambda$ were assumed distinct. The function $g_\mu(y)$ is the product of those linear factors $y - \gamma_l$, taken simply, for which $\varrho_l = \mu$. Since by hypothesis μ is not divisible by p, theorem 6 gives that $g_\mu(y)$ is a function in $\mathfrak{K}$ and can be determined rationally. Putting x^π back for y in $g_\mu(y)$, we obtain, since $m = \mu\pi$,

$$g_\mu(y) = (f_m(x))^\pi,$$

which proves theorem 7.

Thus in all cases $(f_m(x))^m$ is a function in $\mathfrak{K}$ and can be determined rationally. Dividing the degree of this function by m, we obtain the number of m-fold roots of the function $F(x)$.

§ 11. Perfect and imperfect fields.

In § 7 it was shown that a number of theorems of *Galois* theory hold in every field. But if this theory is to hold in a field $\mathfrak{K}$ unchanged in all its basic outlines, one still needs the theorem that a polynomial irreducible in $\mathfrak{K}$ can have only simple roots. This theorem, however, does not hold in full generality, and we are thus compelled to make a distinction, expressed by the following definition:

A field $\mathfrak{K}$ is called perfect if every polynomial $f(x)$ from $\mathfrak{K}$ which has multiple roots (in a suitable extension field) is reducible in $\mathfrak{K}$; it is called imperfect if there are irreducible functions in $\mathfrak{K}$ with multiple roots.

We obtain the simple criteria:

1. Every field of characteristic 0 is perfect.

2. Among fields whose characteristic is a prime number p, there are perfect and imperfect fields; such a field is perfect if extraction of the p-th root (which here turns out to be a unique operation) can always be carried out in it, and otherwise imperfect.

For the proof we first observe that in a field $\mathfrak{K}$ of characteristic p there can never be more than one p-th root, and hence never more than one p^f-th root, of an element γ. For if, for two elements α, β, one has $\alpha^p = \beta^p = \gamma$, then (compare p. 217, lines 17–19) $0 = \alpha^p - \beta^p = (\alpha - \beta)^p$, hence $\alpha - \beta = 0$, $\alpha = \beta$. No extension of $\mathfrak{K}$ can alter this, since the characteristic remains p. The function $x^{p^f} - \gamma$ is therefore, in $\mathfrak{K}$ or in an extension field in which it decomposes completely, the p^f-th power of a linear factor:

$$x^{p^f} - \gamma = (x - \delta)^{p^f},$$

which also follows from § 10, 4a. The prime field $\mathfrak{P}_p$ (§ 4) gives the simplest example of a field in which extraction of p-th roots can always be carried out. For by Fermat's theorem, every element of $\mathfrak{P}_p$ satisfies $a^p = a$, hence also $\sqrt[p]{a} = a$. Now let $\mathfrak{K}$ be any field of characteristic p; we extend it by adjoining a transcendental element u to obtain $\Omega = \mathfrak{K}(u)$. If α is an element of Ω, then either α is an element of $\mathfrak{K}$, in which case the same holds for α^p, or α does not belong to $\mathfrak{K}$; then, considered as a rational function of u, α has a definite degree > 0, and the degree of α^p is p times this degree (§ 5, 6). In no case can $\alpha^p = u$. Thus the element u of the field Ω has no p-th root in Ω. Now let $\mathfrak{K}$ again be any field of characteristic p, and let γ be any element of $\mathfrak{K}$. Then for the function $x^{p^f} - \gamma$, either in $\mathfrak{K}$ itself or in an extension field, one has the decomposition

$$x^{p^f} - \gamma = (x - \delta)^{p^f}. \tag{1}$$

If one decomposes $x^{p^f} - \gamma$ in $\mathfrak{K}$ itself into irreducible factors, then all these factors are powers of $x - \delta$, and all must have the same exponent, since otherwise a factor with larger exponent would be divisible by one with smaller exponent and hence would not be irreducible. The factor irreducible in $\mathfrak{K}$ therefore has the form

$$(x - \delta)^{p^r} = x^{p^r} - \delta^{p^r}, \tag{2}$$

where $0 \leq r \leq f$. The case $r = 0$ occurs when δ, that is, the p^f-th root of γ, belongs to the field $\mathfrak{K}$. But if the p-th (and hence also the p^f-th) root of γ is not present in $\mathfrak{K}$, then $r > 0$, and then $x^{p^r} - \delta^{p^r}$ is an irreducible function in $\mathfrak{K}$ with $p^r > 1$ equal roots. The field $\mathfrak{K}$ is therefore imperfect.

Now suppose that $\mathfrak{K}$ either has characteristic 0, or has characteristic p but contains the p-th root of each of its elements. If $f(x)$ is then a polynomial of degree n in $\mathfrak{K}$ which has multiple roots, then

$$(f(x), f'(x))$$

is at least of degree 1. If $f'(x)$ is different from 0, then the degree of $(f(x), f'(x))$, like that of $f'(x)$, is smaller than n; hence $f(x)$ is reducible. If, however, $f'(x) = 0$, which can occur only when the characteristic is a prime p, then $f(x)$ is a polynomial in x^p:

$$f(x) = a_0 + a_1 x^p + \cdots + a_\nu x^{\nu p} \quad (\nu p = n),$$

and it follows that

$$f(x) = \left(\sqrt[p]{a_0} + \sqrt[p]{a_1}x + \cdots + \sqrt[p]{a_\nu}x^\nu\right)^p,$$

where by our hypothesis the elements $\sqrt[p]{a_0}, \sqrt[p]{a_1}, \ldots, \sqrt[p]{a_\nu}$ belong to the field $\mathfrak{K}$; thus in this case too $f(x)$ is reducible in $\mathfrak{K}$. This proves theorems 1 and 2.

Of the theorems of ordinary *Galois* theory which do not hold without exception in all fields, we single out two which, as will be shown later, are most closely connected with one another and may be called the theorem of primitive elements and the theorem of intermediate fields. The first says that every finite extension is simple; the second says that between a field $\mathfrak{K}$ and a finite extension Ω of it there are always only finitely many fields. The converse of the latter theorem holds without exception, as is easy to see:

3. If between a field $\mathfrak{K}$ and its extension field Ω there lie only finitely many fields, then Ω is finite with respect to $\mathfrak{K}$.

First, Ω cannot contain an element x that is transcendental over $\mathfrak{K}$; for otherwise one would have infinitely many fields

$$\mathfrak{K}(x),\ \mathfrak{K}(x^2),\ \mathfrak{K}(x^3), \ldots, \mathfrak{K}(x^n), \ldots,$$

which are all distinct, as one sees at once (for example from § 5, 6), and lie between $\mathfrak{K}$ and Ω. By § 2, 3, every element of Ω belongs to a field $\mathfrak{K}_i$ between $\mathfrak{K}$ and Ω which is obtained from $\mathfrak{K}$ by adjoining a system S_i of finitely many elements. By hypothesis, however, there are only finitely many such fields. Let them be denoted by $\mathfrak{K}_1, \mathfrak{K}_2, \ldots, \mathfrak{K}_n$; let S_i $(i = 1, \ldots, n)$ be a finite system of elements for which $\mathfrak{K}(S_i) = \mathfrak{K}_i$; and let S be the finite system containing all elements of the systems S_i. Then $\mathfrak{K}(S)$ contains all the fields $\mathfrak{K}_i$, and hence $\mathfrak{K}(S) = \Omega$. Since, moreover, the elements of S are finite in number and, as we saw, algebraic over $\mathfrak{K}$, it follows (§ 7, 6) that $\mathfrak{K}(S) = \Omega$ is finite over $\mathfrak{K}$.

The theorem of primitive elements in ordinary *Galois* theory rests on the following theorem, valid for all fields:

4. If the field Ω is obtained from its subfield $\mathfrak{K}$ by adjoining finitely many elements $\alpha, \beta, \gamma, \ldots$, each of which satisfies over $\mathfrak{K}$ an equation with only simple roots, then Ω is simple over $\mathfrak{K}$ and can also be obtained from $\mathfrak{K}$ by adjoining one element which likewise satisfies over $\mathfrak{K}$ an equation with only simple roots.

It is of course enough to prove the assertion for the case in which Ω is first derived from $\mathfrak{K}$ by adjoining two elements α, β; repeating the same argument then gives the same result for any finite number. We give the proof here under the hypothesis that $\mathfrak{K}$ contains infinitely many elements; for the case of a field with finitely many elements a different proof will appear below (§ 15, 9). Let $f(x)$ and $g(x)$ be polynomials from $\mathfrak{K}$ which vanish for $x = \alpha$ and $x = \beta$, respectively, and which decompose in a suitable extension field $\mathfrak{M}$ into pairwise distinct linear factors. Let the decompositions be

$$f(x) = (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_m),$$
$$g(x) = (x - \beta_1)(x - \beta_2) \cdots (x - \beta_n),$$

where α_1 and β_1 are put for α and β. By our hypothesis the elements $\alpha_1, \alpha_2, \ldots, \alpha_m$ are all distinct, and likewise $\beta_1, \beta_2, \ldots, \beta_n$. If μ is any element of $\mathfrak{K}$, then $\alpha + \mu\beta$ is an element of $\mathfrak{K}(\alpha, \beta) = \Omega$. Consider the $m \cdot n$ elements $\alpha_i + \mu\beta_k$. If two of them, $\alpha_i + \mu\beta_k$ and $\alpha_r + \mu\beta_s$, are equal without $i = r$ and $k = s$ both holding, then $\beta_k \neq \beta_s$, since otherwise $k = s$, $\alpha_i = \alpha_r$, and $i = r$ would follow. Thus in this case one obtains

$$\mu = \frac{\alpha_r - \alpha_i}{\beta_k - \beta_s}. \tag{3}$$

Since only finitely many elements can be represented in this form, while $\mathfrak{K}$ contains infinitely many, we can and shall choose an element μ from $\mathfrak{K}$ which is not of the form (3), and we fix it for what follows. Let now $\delta = \delta_1$ be an arbitrary element of $\mathfrak{K}(\alpha, \beta)$, hence a polynomial $\chi(\alpha, \beta) = \chi(\alpha_1, \beta_1)$ in α and β with coefficients from $\mathfrak{K}$. We denote by

$$\gamma_1, \gamma_2, \ldots, \gamma_{mn}, \qquad \delta_1, \delta_2, \ldots, \delta_{mn}$$

the elements of $\mathfrak{M}$ which arise from

$$\gamma = \gamma_1 = \alpha + \mu\beta, \qquad \delta = \delta_1 = \chi(\alpha, \beta)$$

when all possible combinations α_i, β_k are substituted for α, β, and form the function

$$h(x) = (x - \gamma_1)(x - \gamma_2) \cdots (x - \gamma_{mn}).$$

By the preceding, the elements $\gamma_1, \gamma_2, \ldots, \gamma_{mn}$ are all distinct, so

$$\frac{h(x)}{x - \gamma_1}, \quad \frac{h(x)}{x - \gamma_2}, \quad \ldots, \quad \frac{h(x)}{x - \gamma_{mn}}$$

are polynomials in x; the first takes at $x = \gamma_1$ the nonzero value $h'(\gamma_1)$ (where $h'(x)$ denotes the derivative of $h(x)$), while the others vanish at $x = \gamma_1$. The function $h(x)$ is symmetric in $\alpha_1, \ldots, \alpha_m$ and in $\beta_1, \ldots, \beta_n$, hence is a function in $\mathfrak{K}$; therefore $h'(x)$ is also a function in $\mathfrak{K}$. Finally, the function

$$\varphi(x) = \frac{h(x)}{x - \gamma_1}\delta_1 + \frac{h(x)}{x - \gamma_2}\delta_2 + \cdots + \frac{h(x)}{x - \gamma_{mn}}\delta_{mn} \tag{4}$$

is also symmetric in $\alpha_1, \ldots, \alpha_m$ and in $\beta_1, \ldots, \beta_n$, and hence a function in $\mathfrak{K}$. Setting $x = \gamma_1 = \gamma$ in (4), one obtains

$$\varphi(\gamma) = h'(\gamma) \cdot \delta, \qquad \delta = \frac{\varphi(\gamma)}{h'(\gamma)},$$

from which it follows that δ belongs to the field $\mathfrak{K}(\gamma)$. Since γ belongs to $\mathfrak{K}(\alpha, \beta)$ and δ is an arbitrary element of $\mathfrak{K}(\alpha, \beta)$, it follows that

$$\mathfrak{K}(\alpha, \beta) = \mathfrak{K}(\gamma).$$

Moreover, the equation $h(x) = 0$ satisfied by γ over the field $\mathfrak{K}$ has only simple roots. Thus theorem 4 is proved.

If $\mathfrak{K}$ is a perfect field, then the irreducible equations in $\mathfrak{K}$ satisfied by α and β have only simple roots; the hypotheses imposed on α and β in theorem 4 therefore always hold in this case. It follows from § 7, 6: *5. Every finite extension of a perfect field is simple.*

To show now that neither the theorem of primitive elements nor the theorem of intermediate fields is valid without restriction, we prove here the following theorem, which will be needed later in the more detailed investigations of this subject:

6. Let $\mathfrak{K}$ be a field of characteristic p, and let Ω be an extension of $\mathfrak{K}$. Suppose that the p-th power of every element of Ω is contained in $\mathfrak{K}$, but that $[\Omega : \mathfrak{K}] > p$ (provided Ω is finite over $\mathfrak{K}$ at all). Then Ω is not a simple extension, and there lie infinitely many fields between $\mathfrak{K}$ and Ω.

Proof. The first assertion is immediate: the degree of a primitive element would have to be $[\Omega : \mathfrak{K}] > p$, whereas the hypothesis implies that every element of Ω satisfies an equation of degree p over $\mathfrak{K}$. For the second assertion, it follows first from the hypothesis that $\mathfrak{K}$ is an imperfect field; otherwise the elements of Ω, which are p-th roots of elements of $\mathfrak{K}$, would all belong to $\mathfrak{K}$, so that $[\Omega : \mathfrak{K}] = 1$. It follows further that $\mathfrak{K}$ contains infinitely many elements, because all fields with only finitely many elements are perfect (see § 12, 2). Let α be any element contained in Ω but not in $\mathfrak{K}$. Then α is a root of the function $x^p - \alpha^p$ of the field $\mathfrak{K}$, which is irreducible in this field. $\mathfrak{K}(\alpha)$ is a field between $\mathfrak{K}$ and Ω and is different from Ω, since $[\mathfrak{K}(\alpha) : \mathfrak{K}] = p$. Let β be an element belonging to Ω but not to $\mathfrak{K}(\alpha)$. Then also $[\mathfrak{K}(\beta) : \mathfrak{K}] = p$, and $\mathfrak{K}(\alpha)$ and $\mathfrak{K}(\beta)$ are distinct. Let $\mu_1, \mu_2, \ldots, \mu_k, \ldots$ be infinitely many nonzero elements of $\mathfrak{K}$, all distinct from one another. Then all elements $\omega_k = \alpha + \mu_k\beta$ belong to Ω, but none belongs to $\mathfrak{K}$. For if ω_k belonged to $\mathfrak{K}$, then $\alpha = \omega_k - \mu_k\beta$ would belong to $\mathfrak{K}(\beta)$, which cannot be the case, since α and β have the same degree over $\mathfrak{K}$, and $\mathfrak{K}(\alpha)$ and $\mathfrak{K}(\beta)$ are distinct. The fields $\mathfrak{K}(\omega_k)$ all lie between $\mathfrak{K}$ and Ω, have degree p over $\mathfrak{K}$, and are all distinct. For from $\mathfrak{K}(\omega_k) = \mathfrak{K}(\omega_l)$ it would follow that

$$\alpha = \frac{\mu_l\omega_k - \mu_k\omega_l}{\mu_l - \mu_k} \quad \text{and} \quad \beta = \frac{\omega_l - \omega_k}{\mu_l - \mu_k}$$

belong to the field $\mathfrak{K}(\omega_k) = \mathfrak{K}(\omega_l)$, and hence are primitive elements of this field over $\mathfrak{K}$. But then $\mathfrak{K}(\alpha) = \mathfrak{K}(\omega_k) = \mathfrak{K}(\beta)$, whereas $\mathfrak{K}(\alpha)$ and $\mathfrak{K}(\beta)$ have already been shown to be distinct. Thus infinitely many fields $\mathfrak{K}(\omega_k)$ lie between $\mathfrak{K}$ and Ω. This completes the proof of 6.

It is now easy to show that the hypotheses of theorem 6 can be realized by finite extensions. For this purpose we adjoin two transcendental elements t, u to an arbitrary field $\mathfrak{M}$ of characteristic p and consider, besides the field $\Omega = \mathfrak{M}(t, u)$, the fields $\mathfrak{K} = \mathfrak{M}(t^p, u^p)$ and $\mathfrak{K}' = \mathfrak{M}(t, u^p)$. Then neither t occurs in $\mathfrak{K}$ nor u in $\mathfrak{K}'$, and therefore the function $x^p - t^p$ is irreducible in $\mathfrak{K}$ and the function $x^p - u^p$ is irreducible in $\mathfrak{K}'$, whence

$$[\Omega : \mathfrak{K}] = [\mathfrak{K}' : \mathfrak{K}] \cdot [\Omega : \mathfrak{K}'] = [t : \mathfrak{K}] \cdot [u : \mathfrak{K}'] = p^2.$$

On the other hand, if v is any element of $\Omega = \mathfrak{M}(t, u)$ and we write it in the form $v = \frac{v_1}{v_2}$, where v_1 and v_2 belong to the integral domain $\mathfrak{M}[t, u]$, then v_1 and v_2 are sums of terms of the form $ct^m u^n$, where c is an element of $\mathfrak{M}$. Thus v_1^p and v_2^p are sums of terms of the form $c^p t^{mp} u^{np}$ (§ 10, (4')), and so belong to the integral domain $\mathfrak{M}[t^p, u^p]$. Therefore $v^p = \frac{v_1^p}{v_2^p}$ is an element of the field $\mathfrak{K} = \mathfrak{M}(t^p, u^p)$. Hence the extension Ω of the field $\mathfrak{K}$, although finite, is not simple, and there are infinitely many fields between $\mathfrak{K}$ and Ω.

§ 12.

Imperfect fields: smallest containing and largest contained perfect field. – Root-fields.

Let $\mathfrak{K}$ be a field of characteristic p, and let a and b be elements of $\mathfrak{K}$. Then the equations

$$\begin{aligned} a^p + b^p &= (a+b)^p, \quad & a^p - b^p &= (a-b)^p, \\ a^p \cdot b^p &= (a \cdot b)^p, \quad & a^p : b^p &= (a : b)^p \end{aligned} \tag{1.}$$

hold. From these equations we read off the following result:

1. The p-th powers of all elements of a field $\mathfrak{K}$ of characteristic p form a field $\mathfrak{K}^p$. If the field $\mathfrak{K}$ is perfect, then $\mathfrak{K}^p$ is identical with $\mathfrak{K}$; otherwise it is only a part of $\mathfrak{K}$ (§ 11, 2.). One obtains an isomorphic relation between $\mathfrak{K}$ and $\mathfrak{K}^p$ by assigning to each element of $\mathfrak{K}$ its p-th power.

Since the isomorphic relation between $\mathfrak{K}$ and $\mathfrak{K}^p$ is one-to-one, if $\mathfrak{K}$ has only finitely many elements, then necessarily $\mathfrak{K}^p = \mathfrak{K}$. Thus:

2. Every field which has only finitely many elements is perfect.

Starting from a field $\mathfrak{K}$ of characteristic p, we form the infinite sequence of fields

$$\mathfrak{K} = \mathfrak{K}^{p^0},\ \mathfrak{K}^p,\ \mathfrak{K}^{p^2}, \ldots, \mathfrak{K}^{p^f}, \ldots,$$

in which $\mathfrak{K}^{p^f}$ denotes, in general, the field consisting of the p-th powers of the elements of $\mathfrak{K}^{p^{f-1}}$, hence of the p^f-th powers of the elements of $\mathfrak{K}$. All these fields are isomorphic, and therefore either all are perfect or all are imperfect. In the first case $\mathfrak{K} = \mathfrak{K}^p = \mathfrak{K}^{p^2} = \cdots$; in the second case, which alone interests us here, each following field is only a part of the preceding one. We can also continue this sequence in the opposite direction. To do so, we must first extend the field $\mathfrak{K}$ to a field $\mathfrak{K}^{p^{-1}}$ such that $\mathfrak{K}$ consists of the p-th powers of all elements of $\mathfrak{K}^{p^{-1}}$. For this purpose we first observe that every element of $\mathfrak{K}$ can be written in the form $\sqrt[p]{a}$, where a is an element of $\mathfrak{K}^p$. If $\sqrt[p]{a}$ and $\sqrt[p]{b}$ are any two elements of $\mathfrak{K}$, then, as is immediately seen from (1.),

1) $\sqrt[p]{a}$ is equal to $\sqrt[p]{b}$ if and only if $a = b$,

2) $\sqrt[p]{a} + \sqrt[p]{b} = \sqrt[p]{a+b}$,

3) $\sqrt[p]{a} \cdot \sqrt[p]{b} = \sqrt[p]{a \cdot b}$.

We now form a system $\mathfrak{K}^{p^{-1}}$ by adjoining to $\mathfrak{K}$ all symbols of the form $\sqrt[p]{a}$ in which a denotes an element belonging to $\mathfrak{K}$ but not to $\mathfrak{K}^p$, without at first assigning to such a symbol the meaning of a p-th root of a. We further prescribe that, even when the elements a, b of $\mathfrak{K}$ do not belong to $\mathfrak{K}^p$, or when one of these elements does not belong to $\mathfrak{K}^p$, the statements made under 1), 2), 3) are to hold for equality, sum, and product. In this way we have made $\mathfrak{K}^{p^{-1}}$ into a system with double composition. From 1), 2), 3) it further follows that we obtain an isomorphic relation between $\mathfrak{K}$ and $\mathfrak{K}^{p^{-1}}$ by assigning to each element a of $\mathfrak{K}$ the element $\sqrt[p]{a}$ of $\mathfrak{K}^{p^{-1}}$; hence $\mathfrak{K}^{p^{-1}}$, like $\mathfrak{K}$, is a field. If we now apply the rules of calculation valid in every field, then with reference to 3) we obtain

$$(\sqrt[p]{a})^p = \sqrt[p]{a^p}.$$

But since the element a^p belongs to $\mathfrak{K}^p$, the symbol $\sqrt[p]{a^p}$ already has the meaning of a p-th root of a^p, and therefore $\sqrt[p]{a^p} = a$; hence

$$(\sqrt[p]{a})^p = a.$$

Thus the meaning of $\sqrt[p]{a}$ as a p-th root of a is established generally, and at the same time it is established that $\mathfrak{K}$ consists of the p-th powers of the elements of $\mathfrak{K}^{p^{-1}}$. Repeating the step which led us from $\mathfrak{K}$ to $\mathfrak{K}^{p^{-1}}$, we obtain a sequence of fields $\mathfrak{K}^{p^{-1}}, \mathfrak{K}^{p^{-2}}, \ldots$, and hence $\mathfrak{K}^{p^n}$ is defined for every integer n.

3. All elements which occur in the fields $\mathfrak{K}^{p^n}$ belonging to a field $\mathfrak{K}$ of characteristic p form a field $\mathfrak{C}$; likewise, the fields $\mathfrak{K}^{p^n}$ have as their intersection a field $\mathfrak{D}$. Both fields $\mathfrak{C}$ and $\mathfrak{D}$ are perfect; more precisely, $\mathfrak{D}$ is the largest perfect field contained in $\mathfrak{K}$, while $\mathfrak{C}$ proves to be a smallest perfect extension field of $\mathfrak{K}$.

Indeed, of any two of the fields $\mathfrak{K}^{p^n}$ $(n \leq 0)$, one is always contained in the other. Thus the totality of all elements occurring in these fields gives a field (§ 2, 2.). If a is an element of this field $\mathfrak{C}$ and occurs, say, in $\mathfrak{K}^{p^n}$, then in $\mathfrak{K}^{p^{n-1}}$, and hence in $\mathfrak{C}$, there is a p-th root of a. The field $\mathfrak{C}$ is therefore perfect. When we call it the smallest perfect extension field of $\mathfrak{K}$, this is to mean that no field different from $\mathfrak{C}$ between $\mathfrak{K}$ and $\mathfrak{C}$ is perfect. For if $\mathfrak{C}'$ is a field between $\mathfrak{K}$ and $\mathfrak{C}$, if a is an element of $\mathfrak{C}$ not contained in $\mathfrak{C}'$, and if n is the least natural number such that a occurs in $\mathfrak{K}^{p^{-n}}$, then a^{p^n} is an element of $\mathfrak{K}$, hence also of $\mathfrak{C}'$. If now, among the elements $a^p, a^{p^2}, a^{p^3}, \ldots$, a^{p^m} is the first one contained in $\mathfrak{C}'$, then a^{p^m} has no p-th root in $\mathfrak{C}'$, and the field $\mathfrak{C}'$ is therefore imperfect. – Further, if a is an element of the intersection field $\mathfrak{D}$ and n is an arbitrary integer, then a occurs in $\mathfrak{K}^{p^{n+1}}$, and therefore $\sqrt[p]{a}$ occurs in $\mathfrak{K}^{p^n}$, that is, in every field of the sequence $\mathfrak{K}^{p^n}$, and hence in $\mathfrak{D}$. The field $\mathfrak{D}$ is therefore perfect. Finally, if $\mathfrak{D}'$ is any perfect field contained in $\mathfrak{K}$, if a is an arbitrary element of $\mathfrak{D}'$, and if n is an arbitrarily large natural number, then a has in $\mathfrak{D}'$ a p^n-th root b. Then b belongs to $\mathfrak{K}$, and hence $a = b^{p^n}$ belongs to $\mathfrak{K}^{p^n}$, that is, to all fields of our sequence, and therefore to $\mathfrak{D}$. Thus $\mathfrak{D}'$ is contained in $\mathfrak{D}$, and $\mathfrak{D}$ is the largest perfect field contained in $\mathfrak{K}$.

Let $\mathfrak{K}$ be an arbitrary field of characteristic p, and let a be an element of $\mathfrak{K}$. It determines a sequence of elements

$$a,\ a^p,\ a^{p^2}, \ldots,$$

in which each element is a p-th root of the following one. If a p-th root of a exists in $\mathfrak{K}$, we can continue this sequence in the opposite direction without extending the field. Either, for every natural number n, however large, there exists a p^n-th root of a, which we denote by $a^{p^{-n}}$, or there is a largest number n such that $a^{p^{-n}}$ still exists but has no further p-th root. In the first case we obtain a sequence of elements extending infinitely in both directions; in the second it has an initial element. Perfect fields are characterized by having only sequences of the first kind. The elements of a sequence a^{p^n} are either all distinct from one another, or they repeat periodically, in which case the sequence is of course of the first kind. Here the following simple theorem holds:

4. The sequence of elements a^{p^n} determined by an element a of a field $\mathfrak{K}$ of characteristic p is periodic if and only if the element a is absolutely algebraic. The number of elements in the period is equal to the absolute degree of a (= degree with respect to the prime field), and the elements of the period are all the roots of one and the same irreducible equation in the prime field $\mathfrak{P}_p$.

For if the sequence belonging to a is periodic and consists of n different elements, then every element is equal to its p^n-th power, and is therefore a solution of the equation $x^{p^n} - x = 0$, whose coefficients belong to the prime field $\mathfrak{P}_p$ contained in $\mathfrak{K}$; hence it is absolutely algebraic. Conversely, suppose that a is an absolutely algebraic element of $\mathfrak{K}$ and that

$$\varphi(x) = x^m + a_1 x^{m-1} + \cdots + a_m = 0 \tag{2.}$$

is the irreducible equation of the prime field which is satisfied by a. Then

$$a^m + a_1 a^{m-1} + \cdots + a_m = 0.$$

Raising this equation to the p-th power and taking into account that every element of the prime field is equal to its p-th power, we obtain

$$a^{pm} + a_1 a^{p(m-1)} + \cdots + a_m = 0,$$

that is, equation (2.) is also satisfied by a^p. In the same way one sees that the elements $a^{p^2}, a^{p^3}, \ldots$ must be solutions of equation (2.), and therefore there are at most m different elements among them. Suppose that there are n different ones, so that $a, a^p, a^{p^2}, \ldots, a^{p^{n-1}}$ are distinct while $a^{p^n} = a$. Then $\varphi(x)$ is divisible by

$$(x-a)(x-a^p)\cdots(x-a^{p^{n-1}}) = x^n + b_1x^{n-1} + \cdots + b_n = \psi(x). \qquad (3.)$$

Raising to the p-th power and then replacing x^p by x (which is permissible because x^p is transcendental with respect to $\mathfrak{K}$), the factors on the left side of (3.) undergo only a cyclic permutation. Thus one obtains

$$\psi(x) = x^n + b_1^p x^{n-1} + \cdots + b_n^p,$$

and hence $b_1^p = b_1, \ldots, b_n^p = b_n$. But the equation $x^p = x$ is satisfied by the p elements of the prime field, and since it can have no more than p roots, it follows that the coefficients of $\psi(x)$ belong to the prime field. Since $\varphi(x)$ is divisible by $\psi(x)$ and is irreducible in the prime field, we must have $\varphi(x) = \psi(x)$. This proves theorem 4 completely.

If the field $\mathfrak{K}$ is absolutely algebraic, then every element of $\mathfrak{K}$ belongs to a periodic sequence, and therefore has a p-th root. We thus obtain the following theorem, of which 2. is a special case:

5. Every absolutely algebraic field is perfect.

Root-elements and root-fields. – Let $\mathfrak{K}$ be a field of characteristic p, and let Ω be an extension of $\mathfrak{K}$. An element a of Ω is to be called a *root-element with respect to* $\mathfrak{K}$ if a^{p^n} belongs to $\mathfrak{K}$ for sufficiently large n. We then call the least number $n \geq 0$ for which a^{p^n} becomes an element of $\mathfrak{K}$ the *exponent of a with respect to* $\mathfrak{K}$. If this exponent is f, then, as is easily seen (see p. 219), $x^{p^f} - a^{p^f} = 0$ is the irreducible equation in $\mathfrak{K}$ which is satisfied by a. We call the field Ω a *root-field with respect to* $\mathfrak{K}$ if all its elements are root-elements. All elements of $\mathfrak{K}$ itself are root-elements; they are characterized by exponent 0. If a and b are root-elements and their exponents do not exceed the natural number f, then a^{p^f}, b^{p^f} are elements of $\mathfrak{K}$, and the same is therefore true of $(a \pm b)^{p^f} = a^{p^f} \pm b^{p^f}$, as well as of $(a \cdot b)^{p^f}$ and $\left(\frac{a}{b}\right)^{p^f}$. Hence the elements $a \pm b, a \cdot b, \frac{a}{b}$ are also root-elements, and their exponents are $\leq f$. Therefore the root-elements contained in Ω whose exponent does not exceed f form a field between $\mathfrak{K}$ and Ω; and the same is true, as follows at once, of the totality of all root-elements contained in Ω. Further, an extension Ω which arises by adjoining a finite or infinite system $\mathfrak{S}$ of root-elements is a root-field. For the field formed by the root-elements contains $\mathfrak{K}$ and the system $\mathfrak{S}$, and hence is identical with the field $\mathfrak{K}(\mathfrak{S}) = \Omega$. – From the definitions just given it also follows immediately that if the field Ω lies between $\mathfrak{K}$ and $\mathfrak{M}$, and if Ω is a root-field with respect to $\mathfrak{K}$ while $\mathfrak{M}$ is a root-field with respect to Ω, then $\mathfrak{M}$ is also a root-field with respect to $\mathfrak{K}$. – Now suppose that the extension field Ω itself, or more generally some field $\mathfrak{T}$ between $\mathfrak{K}$ and Ω, is perfect, and denote by $\mathfrak{C}$ the field of all root-elements contained in Ω with respect to $\mathfrak{K}$. If a is an element of $\mathfrak{C}$ and f is its exponent, then a^{p^f} belongs to $\mathfrak{K}$, hence also to $\mathfrak{T}$. But a itself must already belong to $\mathfrak{T}$; for otherwise, if n denoted the least number for which a^{p^n} belongs to $\mathfrak{T}$, then a^{p^n} would have no p-th root within $\mathfrak{T}$, whereas $\mathfrak{T}$ was supposed to be a perfect field. Hence a belongs to $\mathfrak{T}$, and so the field $\mathfrak{C}$ is contained in $\mathfrak{T}$. On the other hand, $\mathfrak{C}$ itself is a perfect field. For if a is again an element of $\mathfrak{C}$ and f denotes its exponent, then as an element of $\mathfrak{T}$ it has, within this perfect field, a p-th root $\sqrt[p]{a} = b$, and $b^{p^{f+1}} = a^{p^f}$ becomes an element of $\mathfrak{K}$. Thus the element $\sqrt[p]{a}$ is a root-element and therefore belongs to $\mathfrak{C}$. Hence every extension field Ω of $\mathfrak{K}$, if it is perfect, contains a smallest field of this kind, and this field is formed by all root-elements contained in Ω. – Now let $\mathfrak{C}$ be a smallest perfect extension field of $\mathfrak{K}$ (therefore a root-field), and let $\mathfrak{K}_f$ ($f \geq 0$) denote the field of those elements of $\mathfrak{C}$ whose exponent is $\leq f$. Then $\mathfrak{K}_0 = \mathfrak{K}$. If a is an element of $\mathfrak{K}_f$, then the exponent of $\sqrt[p]{a}$ cannot be $> f+1$, that is, $\sqrt[p]{a}$ belongs to $\mathfrak{K}_{f+1}$; conversely, if $\sqrt[p]{a}$ belongs to $\mathfrak{K}_{f+1}$, one may conclude that a is an element of $\mathfrak{K}_f$. Hence $\mathfrak{K}_{f+1}^p = \mathfrak{K}_f$. Let $\overline{\mathfrak{K}}$ be a field of the same type as $\mathfrak{K}$, let π_0 be an isomorphic relation between $\mathfrak{K}$ and $\overline{\mathfrak{K}}$, let $\overline{\mathfrak{C}}$ be a smallest perfect extension field of $\overline{\mathfrak{K}}$, and let $\overline{\mathfrak{K}}_f$ be the field of those elements of $\overline{\mathfrak{C}}$ whose exponent with respect to $\overline{\mathfrak{K}}$ is $\leq f$. Then on the one hand we have an isomorphic relation between $\mathfrak{K}_1$ and $\mathfrak{K} = \mathfrak{K}_0$, and on the other between $\overline{\mathfrak{K}}_1$ and $\overline{\mathfrak{K}}_0$, by assigning to each element of $\mathfrak{K}_1$ or $\overline{\mathfrak{K}}_1$ its p-th power. From these two isomorphisms, together with π_0, there results an isomorphic relation π_1 between $\mathfrak{K}_1$ and $\overline{\mathfrak{K}}_1$ which includes π_0. In the same way we obtain a relation π_2 between $\mathfrak{K}_2$ and $\overline{\mathfrak{K}}_2$ which includes π_1, and so on. The totality of these isomorphisms $\pi_0, \pi_1, \pi_2, \ldots$ establishes an isomorphism between $\mathfrak{C}$ and $\overline{\mathfrak{C}}$. Here we may again identify $\mathfrak{K}$ with $\overline{\mathfrak{K}}$ and suppose that π_0 carries every element of $\mathfrak{K}$ to itself. Then the following result is obtained:

6. Every imperfect field can essentially be extended in only one way to a smallest possible perfect field; and thus to every imperfect type there is assigned a definite perfect type.

§ 13.

Imperfect fields: algebraic extensions of the first kind.

Let $\mathfrak{K}$ be an arbitrary field. An irreducible function in $\mathfrak{K}$ shall be called *of the first kind* if it has only simple zeros. An element γ algebraic with respect to $\mathfrak{K}$ shall be called *of the first kind* if the irreducible function in $\mathfrak{K}$ which has γ as a zero is of the first kind. An algebraic extension shall be called of the first kind if all elements of the extension field are of the first kind.

For perfect fields, and therefore in particular for all fields of characteristic 0, there are only functions, algebraic elements, and algebraic extensions of the first kind.

Let the field $\mathfrak{K}$ have characteristic p, and let $g(x)$ be an irreducible function in $\mathfrak{K}$. We can then determine a non-negative integer f such that $g(x)$ is an integral rational function of x^{p^f}, but no longer of $x^{p^{f+1}}$. Then $g(x)$ has the form

$$g(x) = x^{np^f} + a_1 x^{(n-1)p^f} + \cdots + a_n,$$

and at least one term whose exponent is not divisible by p^{f+1} actually occurs. We call the number n the *reduced degree*, and f (generalizing a designation introduced in the preceding section) the *exponent* of the irreducible function $g(x)$; the same terminology is transferred to the zeros of the function.[13]

Two special cases are especially noteworthy. First, if the reduced degree has its smallest value, 1, then $g(x)$ assumes the form $x^{p^f} - \alpha$ and has all its zeros equal; the algebraic elements of reduced degree 1 are therefore the root elements considered in the preceding section. Secondly, if the exponent f has its smallest value, 0, then $g(x)$ is no longer an integral rational function of x^p, the derivative $g'(x)$ does not vanish and therefore cannot have a common factor with $g(x)$; hence the function $g(x)$ has only simple zeros and is of the first kind. By contrast, in the case $f > 0$, $g(x)$ is the p-th power of another integral function and therefore has only multiple zeros. We shall chiefly be concerned in this section with the case $f = 0$.

We now prove a series of propositions concerning a field $\mathfrak{K}$ of characteristic p; some of these propositions have significance only when $\mathfrak{K}$ is imperfect.

1. If γ is algebraic with respect to $\mathfrak{K}$ and has exponent f, $f > 0$, then γ^p has exponent $f - 1$ with respect to $\mathfrak{K}$, and $[\mathfrak{K}(\gamma) : \mathfrak{K}(\gamma^p)] = p$.

Indeed, if γ is a zero of the irreducible function

$$g(x) = x^{np^f} + a_1 x^{(n-1)p^f} + \cdots + a_n, \tag{1.}$$

then γ^p is a zero of

$$h(x) = x^{np^{f-1}} + a_1 x^{(n-1)p^{f-1}} + \cdots + a_n.$$

[13] The term "reduced degree" is justified by the fact that, as can easily be shown by means of the following propositions, the reduced degree of an element with respect to $\mathfrak{K}$ is equal to the degree which the element has with respect to the smallest perfect field containing $\mathfrak{K}$.

The function $h(x)$ is irreducible in $\mathfrak{K}$, since from $h(x) = \varphi(x) \cdot \psi(x)$ it would follow that

$$g(x) = h(x^p) = \varphi(x^p) \cdot \psi(x^p),$$

whereas $g(x)$ is irreducible. Thus $f - 1$ is the exponent of γ^p; and since $[\gamma : \mathfrak{K}] = np^f$ and $[\gamma^p : \mathfrak{K}] = np^{f-1}$, it follows that $[\mathfrak{K}(\gamma) : \mathfrak{K}(\gamma^p)] = p$.

2. If

$$\varphi(x) = x^n + a_1 x^{n-1} + \cdots + a_n \tag{2.}$$

is an irreducible function of the first kind in $\mathfrak{K}$, then

$$\psi(x) = x^n + a_1^p x^{n-1} + \cdots + a_n^p \tag{3.}$$

(more generally, $x^n + a_1^{p^f} x^{n-1} + \cdots + a_n^{p^f}$) is also irreducible and of the first kind.

Proof. Let $h(x)$ denote a factor of $\psi(x)$ belonging to $\mathfrak{K}$ and having degree greater than 0. Then $h(x^p)$ is a factor of $\psi(x^p) = (\varphi(x))^p$, and therefore, since $\varphi(x)$ is irreducible, it is a power of $\varphi(x)$. We thus obtain

$$h(x^p) = (\varphi(x))^r \qquad (0 < r \leq p). \tag{4.}$$

But by hypothesis $\varphi(x)$ has only simple zeros, and hence $(\varphi(x))^r$ has only r-fold zeros; while $h(x^p)$ is the p-th power of an integral function (of an extension field), and therefore every one of its zeros has multiplicity at least p. It follows from (4.) that $r = p$,

$$h(x^p) = (\varphi(x))^p = \psi(x^p),$$

and hence

$$h(x) = \psi(x).$$

Thus $\psi(x)$ is irreducible. Since $\varphi(x)$ actually contains at least one power of x whose exponent is not divisible by p, it follows from (2.) and (3.) that the same is true for $\psi(x)$. Therefore $\psi(x)$ is of the first kind.

3. If

$$\varphi(x) = x^n + a_1 x^{n-1} + \cdots + a_n \tag{2.}$$

is an irreducible function of the first kind in $\mathfrak{K}$, and if $\mathfrak{L}$ is an extension field of $\mathfrak{K}$ which contains the p-th roots of the coefficients of $\varphi(x)$ and one zero γ of $\varphi(x)$, then $\mathfrak{L}$ also contains the p-th root of γ.

Proof. By hypothesis $\mathfrak{L}$ contains the two functions

$$\chi(x) = x^n + \sqrt[p]{a_1} x^{n-1} + \cdots + \sqrt[p]{a_n}, \qquad x^p - \gamma = (x - \sqrt[p]{\gamma})^p,$$

and therefore also their greatest common divisor $t(x)$. Since γ is a zero of $\varphi(x)$, $\sqrt[p]{\gamma}$ is a zero of $\varphi(x^p) = (\chi(x))^p$, and therefore a zero of $\chi(x)$; moreover it is a simple zero, because the zeros of $\chi(x)$ are the p-th roots of the zeros of $\varphi(x)$ and therefore, like them, are distinct. Since $\sqrt[p]{\gamma}$ is the only zero of $x^p - \gamma$, it follows that $t(x) = x - \sqrt[p]{\gamma}$, and hence $\sqrt[p]{\gamma}$ is an element of $\mathfrak{L}$.

4. Every algebraic extension of a perfect field is a perfect field.

Proof. Let $\mathfrak{K}$ be a perfect field of characteristic p – for characteristic 0 the assertion is immediate – let $\mathfrak{L}$ be an algebraic extension of $\mathfrak{K}$, let γ be an element of $\mathfrak{L}$, and let $\varphi(x)$ be the irreducible function in $\mathfrak{K}$ having γ as a zero. Since $\mathfrak{K}$ is perfect, $\varphi(x)$ is of the first kind; and for the same reason $\mathfrak{K}(\gamma)$, just like $\mathfrak{K}$ itself, contains the p-th roots of the coefficients of $\varphi(x)$ and, in addition, contains γ. Therefore, by 3., it also contains $\sqrt[p]{\gamma}$. Since $\mathfrak{K}(\gamma)$ is a subfield of $\mathfrak{L}$, the p-th root of every element γ of $\mathfrak{L}$ is contained in $\mathfrak{L}$; hence $\mathfrak{L}$ is a perfect field. – The hypothesis of Proposition 4. is satisfied when $\mathfrak{K}$ is a prime field, so that $\mathfrak{L}$ is an absolutely algebraic field; this again proves the assertion of § 12, 5.

5. If γ is algebraic with respect to $\mathfrak{K}$ and is of the first kind, then $\mathfrak{K}(\gamma^p) = \mathfrak{K}(\gamma)$.

Indeed, if γ is a zero of the irreducible function $\varphi(x) = x^n + a_1 x^{n-1} + \cdots + a_n$ in $\mathfrak{K}$, then γ^p is a zero of the function $\psi(x) = x^n + a_1^p x^{n-1} + \cdots + a_n^p$, which is likewise irreducible by 2. Thus $n = [\gamma : \mathfrak{K}] = [\gamma^p : \mathfrak{K}]$, and the assertion follows. – Conversely, Proposition 1 already says that from $\mathfrak{K}(\gamma^p) = \mathfrak{K}(\gamma)$ one may conclude that γ is of the first kind.

6. If the element γ is algebraic with respect to $\mathfrak{K}$ and is of the first kind, then $\mathfrak{K}(\gamma)$ is an algebraic extension of the first kind.

Proof. Let α be an arbitrary element of $\mathfrak{K}(\gamma)$, and put

$$\begin{cases} [\mathfrak{K}(\alpha) : \mathfrak{K}] = m_1, & [\mathfrak{K}(\gamma) : \mathfrak{K}(\alpha)] = m_2, \\ [\mathfrak{K}(\alpha^p) : \mathfrak{K}] = n_1, & [\mathfrak{K}(\gamma) : \mathfrak{K}(\alpha^p)] = n_2. \end{cases} \tag{5.}$$

Then

$$m_1 \cdot m_2 = n_1 \cdot n_2 = [\mathfrak{K}(\gamma) : \mathfrak{K}], \tag{6.}$$

and, since $\mathfrak{K}(\alpha^p)$ is a subfield of $\mathfrak{K}(\alpha)$,

$$m_1 \geq n_1, \quad m_2 \leq n_2. \tag{7.}$$

Now let $\chi(x) = x^{m_2} + b_1 x^{m_2 - 1} + \cdots + b_{m_2}$ be the irreducible function of the field $\mathfrak{K}(\alpha)$ which has γ as a zero. Then γ^p is a zero of $x^{m_2} + b_1^p x^{m_2-1} + \cdots + b_{m_2}^p$, and the coefficients of this function belong, as is easily seen, to the field $\mathfrak{K}(\alpha^p)$. Hence

$$n_2 = [\mathfrak{K}(\gamma) : \mathfrak{K}(\alpha^p)] = [\mathfrak{K}(\gamma^p) : \mathfrak{K}(\alpha^p)] \leq m_2.$$

Consequently, by (7.) and (6.), $m_2 = n_2$ and $m_1 = n_1$. The elements α and α^p therefore have the same degree with respect to $\mathfrak{K}$. Hence $\mathfrak{K}(\alpha^p) = \mathfrak{K}(\alpha)$, so that α is an element of the first kind, as was to be shown.

7. Every finite extension of the first kind is simple.

Indeed, if $\mathfrak{L}$ is a finite extension of the first kind of $\mathfrak{K}$, then we can write $\mathfrak{L} = \mathfrak{K}(\mathfrak{S})$, where $\mathfrak{S}$ contains only finitely many elements (§ 7, 6.); and since all of these are of the first kind, $\mathfrak{L}$ is simple with respect to $\mathfrak{K}$ by § 11, 4.

8. A field $\mathfrak{K}(\mathfrak{S})$ which arises from $\mathfrak{K}$ by adjunction of a system $\mathfrak{S}$ of algebraic elements of the first kind is an extension of the first kind.

If $\mathfrak{S}$ is a finite system, this follows from § 11, 4. and Proposition 6 of this section; if $\mathfrak{S}$ is infinite, the same conclusion follows after adding § 2, 3.

9. If the field $\mathfrak{L}$ lies between $\mathfrak{K}$ and $\mathfrak{M}$, and if $\mathfrak{L}$ is of the first kind with respect to $\mathfrak{K}$ while $\mathfrak{M}$ is of the first kind with respect to $\mathfrak{L}$, then $\mathfrak{M}$ is of the first kind with respect to $\mathfrak{K}$.

Proof. Let δ be an arbitrary element of $\mathfrak{M}$. By § 7, 9., δ is algebraic with respect to $\mathfrak{K}$. To prove 9. it remains only to show that δ is of the first kind, i.e. has exponent 0, with respect to $\mathfrak{K}$. Denote the exponent in question first by f. Then by 1. δ^{p^f} has exponent 0 with respect to $\mathfrak{K}$. With respect to $\mathfrak{L}$, δ is of the first kind by hypothesis, whence by 5. we have $\mathfrak{L}(\delta) = \mathfrak{L}(\delta^p) = \mathfrak{L}(\delta^{p^2}) = \cdots$, and hence $\mathfrak{L}(\delta) = \mathfrak{L}(\delta^{p^f})$. The field $\mathfrak{L}(\delta^{p^f})$ is obtained from $\mathfrak{K}$ by adjoining all elements of $\mathfrak{L}$ as well as δ^{p^f}. But all these elements are of the first kind with respect to $\mathfrak{K}$; hence, by Proposition 8, the same holds for all elements of the field $\mathfrak{L}(\delta^{p^f}) = \mathfrak{L}(\delta)$, and in particular for δ.

Let $\mathfrak{K}$ be an arbitrary field, and let $\mathfrak{L}$ be a finite extension of the first kind of $\mathfrak{K}$. Then the extension is simple, and any primitive element α of $\mathfrak{L}$ is a zero of an irreducible function $\varphi(x)$ in $\mathfrak{K}$ of the first kind. It may happen that $\varphi(x)$ already splits completely in $\mathfrak{L}$; then we put $\mathfrak{N} = \mathfrak{L}$. Otherwise, let $\mathfrak{N}$ denote an extension of $\mathfrak{L}$ which is just sufficient to split $\varphi(x)$ completely.

Then $\mathfrak{N}$ arises from $\mathfrak{L}$, and, since $\mathfrak{L} = \mathfrak{K}(\alpha)$, also from $\mathfrak{K}$, solely by adjunction of zeros of the function $\varphi(x)$. It is therefore a finite extension of the first kind of $\mathfrak{K}$, and moreover is normal with respect to $\mathfrak{K}$ (§ 8, 6.). Thus, if j_1 is a primitive element of $\mathfrak{N}$ (with respect to $\mathfrak{K}$) and $g(x)$ is the irreducible function in $\mathfrak{K}$ having j_1 as a zero, then in $\mathfrak{N}$ one has a decomposition

$$g(x) = (x - j_1) \cdot (x - j_2) \cdots (x - j_n).$$

Here the elements $j_1, j_2, \ldots, j_n$ are distinct, and

$$\mathfrak{N} = \mathfrak{K}(j_1) = \mathfrak{K}(j_2) = \cdots = \mathfrak{K}(j_n), \qquad [\mathfrak{N} : \mathfrak{K}] = n.$$

Since $j_1, j_2, \ldots, j_n$ are zeros of the same irreducible function in $\mathfrak{K}$, we obtain an isomorphic relation π_k $(k = 1, \ldots, n)$ of $\mathfrak{N}$ with respect to itself, in which every element of $\mathfrak{K}$ remains unchanged, by replacing j_1 by j_k in each element of $\mathfrak{N}$, conceived as written in the form

$$c_0 + c_1 j_1 + \cdots + c_{n-1} j_1^{n-1} \tag{8.}$$

with coefficients in $\mathfrak{K}$ (§ 6). Let the system of these n isomorphisms (the *Galois group*) be denoted by $\mathfrak{G}$. Now let $\mathfrak{M}$ be any field between $\mathfrak{K}$ and $\mathfrak{N}$, let $[\mathfrak{M} : \mathfrak{K}] = m$, let $\beta = \psi(j_1)$ be a primitive element of $\mathfrak{M}$ represented in the form (8.), and let $h(x)$ be the irreducible function in $\mathfrak{K}$ with zero β. Since the isomorphisms π_k leave the element 0 unchanged and carry $\beta = \psi(j_1)$ into $\psi(j_k)$, it follows from $0 = h(\beta) = h(\psi(j_1))$ that also $h(\psi(j_k)) = 0$. The function

$$f(x) = (x - \psi(j_1)) \cdot (x - \psi(j_2)) \cdots (x - \psi(j_n))$$

belongs to the field $\mathfrak{K}$, since its coefficients are symmetric in the zeros j_k of $g(x)$; and, by the preceding argument, all its zeros are zeros of the irreducible function $h(x)$. Thus $f(x)$ can have no irreducible factor in $\mathfrak{K}$ other than $h(x)$, and, by considering the degrees of $f(x)$ and $h(x)$, one obtains

$$f(x) = h(x)^{n/m}.$$

Hence among the elements $\psi(j_k)$ exactly n/m are equal to $\beta = \psi(j_1)$. Thus among the isomorphisms π_k there are exactly n/m which leave the element β, and therefore every element of $\mathfrak{K}(\beta) = \mathfrak{M}$, unchanged. Let $\mathfrak{H}$ be the system of these n/m isomorphisms. – If $\mathfrak{M}'$ is a second field between $\mathfrak{K}$ and $\mathfrak{N}$, then to $\mathfrak{M}'$ there belongs a system $\mathfrak{H}'$ consisting of all isomorphisms π_k which leave all elements of $\mathfrak{M}'$ unchanged. Suppose that $\mathfrak{H}'$ is identical with $\mathfrak{H}$. If δ is then an element of $\mathfrak{M}'$, the isomorphisms belonging to $\mathfrak{H}$ leave all elements of $\mathfrak{M}$ and also δ unchanged, and hence also all elements of the field $\mathfrak{M}(\delta)$ unchanged. If

$$[\mathfrak{M}(\delta) : \mathfrak{M}] = \mu,$$

then $[\mathfrak{M}(\delta) : \mathfrak{K}] = m \cdot \mu$. By the preceding argument, therefore, the number of isomorphisms π_k which leave all elements of $\mathfrak{M}(\delta)$ unchanged is $n/(m \cdot \mu)$. Since all n/m isomorphisms of $\mathfrak{H}$ belong to them, it follows that $\mu = 1$, $\mathfrak{M}(\delta) = \mathfrak{M}$, and hence δ is an element of $\mathfrak{M}$. Since δ denoted an arbitrary element of $\mathfrak{M}'$, $\mathfrak{M}'$ is therefore a part of $\mathfrak{M}$; and since the converse can be proved in the same way, $\mathfrak{M} = \mathfrak{M}'$. Thus different fields $\mathfrak{M}$ between $\mathfrak{K}$ and $\mathfrak{N}$ correspond to different subsystems $\mathfrak{H}$ of the system $\mathfrak{G}$. Since the number of subsystems of $\mathfrak{G}$ is finite, only finitely many fields lie between $\mathfrak{K}$ and $\mathfrak{N}$, and hence also between $\mathfrak{K}$ and $\mathfrak{L}$. That is:

10. Between a field $\mathfrak{K}$ and a finite extension $\mathfrak{L}$ of the first kind there lie only finitely many fields.

Moreover, since for perfect fields all algebraic extensions are of the first kind, we obtain:

11. Between a perfect field and a finite extension of it there lie only finitely many fields.

§ 14.

Imperfect fields: arbitrary algebraic extensions, the theorem of primitive elements, and the theorem of intermediate fields.

According to § 11, 5. and § 13, 11., the theorems named in the heading hold for every perfect field. But there are certain imperfect fields in which these theorems still hold. For a closer investigation of the questions connected with this, after having studied in §§ 12 and 13 two special kinds of algebraic extensions, namely root fields and extensions of the first kind, we must now consider arbitrary algebraic extensions. First it appears that every such extension is composed in a definite way from two special ones.

1. If $\mathfrak{L}$ is an algebraic extension of the imperfect field $\mathfrak{K}$, then there is between $\mathfrak{K}$ and $\mathfrak{L}$ a definite field $\mathfrak{L}_0$ with the following property: $\mathfrak{L}_0$ is of the first kind with respect to $\mathfrak{K}$, and $\mathfrak{L}$ is a root field with respect to $\mathfrak{L}_0$. The field $\mathfrak{L}_0$ consists of all elements of $\mathfrak{L}$ which are of the first kind with respect to $\mathfrak{K}$.

Indeed, let $\mathfrak{L}_0$ denote the system of all elements of the first kind in $\mathfrak{L}$. Then, by § 13, 8., the field $\mathfrak{K}(\mathfrak{L}_0)$ contains only elements of the first kind, and is therefore identical with $\mathfrak{L}_0$; thus $\mathfrak{L}_0$ is a field between $\mathfrak{K}$ and $\mathfrak{L}$. Further, if δ is any element of $\mathfrak{L}$ and f is its exponent with respect to $\mathfrak{K}$, then δ^{p^f} is of the first kind, hence is an element of $\mathfrak{L}_0$, and consequently δ is a root element of $\mathfrak{L}_0$. This field $\mathfrak{L}_0$ is the only one having the properties required in 1. For if $\mathfrak{L}_0'$ has these properties, and if the element δ of $\mathfrak{L}$ is of the first kind with respect to $\mathfrak{K}$, then δ is also of the first kind with respect to $\mathfrak{L}_0'$; and since δ is at the same time to be a root element of $\mathfrak{L}_0'$, δ must itself belong to the field $\mathfrak{L}_0'$. Hence $\mathfrak{L}_0'$ must contain all elements of the first kind, and so is identical with $\mathfrak{L}_0$.

2. If $\mathfrak{L}$ is an algebraic extension of the imperfect field $\mathfrak{K}$, and f is an integer ≥ 0, then the system $\mathfrak{L}_f$ of those elements of $\mathfrak{L}$ whose exponent is $\leq f$ is a field (between $\mathfrak{K}$ and $\mathfrak{L}$).

This is first true for $\mathfrak{L}_0$ by Proposition 1. Next, the requirement that an element a of $\mathfrak{L}$ have exponent $\leq f$ is equivalent to the requirement that a^{p^f} have exponent 0, and therefore be an element of $\mathfrak{L}_0$. Thus, if a and b are elements of $\mathfrak{L}_f$, then a^{p^f} and b^{p^f}, and hence also the sum, difference, product, and quotient of a^{p^f} and b^{p^f}, are elements of $\mathfrak{L}_0$. But these combinations are the p^f-th powers of $a+b$, $a-b$, ab, and a/b. Therefore these four elements again belong to the system $\mathfrak{L}_f$, and this system is a field.

3. If to an imperfect field $\mathfrak{K}$ one adjoins a system $\mathfrak{S}$ of algebraic elements whose exponents do not exceed the number f, then the exponents of all elements of $\mathfrak{K}(\mathfrak{S})$ are $\leq f$.

For let $\mathfrak{S}'$ denote the system of all elements of $\mathfrak{K}(\mathfrak{S})$ whose exponents are $\leq f$. By 2. $\mathfrak{S}'$ is a field; and since all elements of $\mathfrak{K}$ and of $\mathfrak{S}$ occur in $\mathfrak{S}'$, we have $\mathfrak{S}' = \mathfrak{K}(\mathfrak{S})$.

Exponent and reduced degree of a finite extension. – Let $\mathfrak{L}$ be a finite extension of an imperfect field $\mathfrak{K}$. Then there is an integer $f \geq 0$ such that $\mathfrak{L}$ contains elements of exponent f, but none of larger exponent. We call the number f the *exponent of $\mathfrak{L}$ (with respect to $\mathfrak{K}$)*. Further, if $\mathfrak{L}_0$ is the subfield of $\mathfrak{L}$ formed by the elements of the first kind, and if $[\mathfrak{L}_0 : \mathfrak{K}] = n$, then we call the number n the *reduced degree of $\mathfrak{L}$ (with respect to $\mathfrak{K}$)*. – Now if δ is an element of $\mathfrak{L}$ of exponent f, then, among the powers

$$\delta,\ \delta^p,\ \delta^{p^2}, \ldots, \delta^{p^f},$$

only the last has exponent 0, and is therefore an element of $\mathfrak{L}_0$. Hence $[\mathfrak{L}_0(\delta) : \mathfrak{K}] = n \cdot p^f$, so that $n \cdot p^f$ divides $[\mathfrak{L} : \mathfrak{K}]$. Further, if η is an arbitrary element of $\mathfrak{L}$, with reduced degree n_1 and exponent f_1, then $f_1 \leq f$. The element $\eta^{p^{f_1}}$ belongs to $\mathfrak{L}_0$, and n_1 is equal to the degree of $\eta^{p^{f_1}}$, hence is a divisor of n. Since $n_1 \cdot p^{f_1}$ is the degree of η, it follows that:

4. If $\mathfrak{L}$ is a finite extension of the imperfect field $\mathfrak{K}$, with reduced degree n and exponent f, then the degree of $\mathfrak{L}$ (with respect to $\mathfrak{K}$) is a multiple, and the degree of each element of $\mathfrak{L}$ is a divisor, of the number $n \cdot p^f$.

If $\mathfrak{L}$ is a simple extension, then the primitive elements have the same degree as $\mathfrak{L}$ itself; by 4. this degree must therefore be equal to $n \cdot p^f$, and in that case f is the exponent and n the reduced degree of every primitive element.

Let $\mathfrak{L}$ again denote a finite, not necessarily simple extension of the imperfect field $\mathfrak{K}$, and let $\mathfrak{L}_0$, n, and f retain their earlier meanings. In $\mathfrak{L}$ there are elements of exponent f, and likewise elements of reduced degree n: for the field $\mathfrak{L}_0$ is of the first kind and therefore has primitive elements. The degree of such an element is n, and since it is of the first kind, n is also its reduced degree. The question is whether one can always find in $\mathfrak{L}$ an element which is simultaneously of exponent f and of reduced degree n. For understanding the structure of finite extensions it is important to show that such elements always exist. Thus, in supplement to 4., we shall now prove:

5. If $\mathfrak{L}$ is a finite extension of the imperfect field $\mathfrak{K}$, with reduced degree n and exponent f, then there exist in $\mathfrak{L}$ elements of degree $n \cdot p^f$.

We first observe that the field $\mathfrak{K}$, since it is imperfect, has infinitely many elements. In such a field $\mathfrak{K}$ the following proposition holds: in a non-zero integral function F of n indeterminates $t_0, t_1, \dots, t_{n-1}$ one can substitute elements of $\mathfrak{K}$ for these indeterminates in such a way that the resulting element of $\mathfrak{K}$ is also $\neq 0$. This is shown in the usual way, first for $n = 1$, using the fact that infinitely many values are available while the number of zeros is at most equal to the degree of the function, and then in general by induction from n to $n+1$. – We now make use of this fact. Returning to the previous notation, choose from the field $\mathfrak{L}_0$ a primitive element γ, from the field $\mathfrak{L}$ an element δ of exponent f (if $f = 0$, any non-zero element δ), and put $\delta^{p^f} = \eta$.

Of the powers

$$\delta,\ \delta^p, \dots, \delta^{p^f} = \eta \tag{1.}$$

the last is then the only one which belongs to $\mathfrak{L}_0$. The elements

$$\varepsilon,\ \gamma,\ \gamma^2, \dots, \gamma^{n-1} \tag{2.}$$

form a basis of $\mathfrak{L}_0$; and since γ is of the first kind and

$$\mathfrak{L}_0 = \mathfrak{K}(\gamma) = \mathfrak{K}(\gamma^p) = \mathfrak{K}(\gamma^{p^2}) = \cdots = \mathfrak{K}(\gamma^{p^f}),$$

so that γ^{p^f} is a primitive element of $\mathfrak{L}_0$, and since $\eta \neq 0$, the elements

$$\eta,\ \eta\gamma^{p^f},\ \eta\gamma^{2p^f}, \dots, \eta\gamma^{(n-1)p^f} \tag{3.}$$

also form a basis of $\mathfrak{L}_0$. We now introduce $2n$ indeterminates $t_0, \dots, t_{n-1}; u_0, \dots, u_{n-1}$ and form the two linear forms

$$\xi = t_0 + \gamma t_1 + \gamma^2 t_2 + \cdots + \gamma^{n-1} t_{n-1}, \tag{4.}$$

$$\zeta = \eta u_0 + \eta\gamma^{p^f} u_1 + \eta\gamma^{2p^f} u_2 + \cdots + \eta\gamma^{(n-1)p^f} u_{n-1}. \tag{5.}$$

Both ξ and ζ represent every element of $\mathfrak{L}_0$ once, as the t's respectively the u's range over all elements of $\mathfrak{K}$. For every non-negative integer k, ζ^k represents an integral homogeneous function of the indeterminates u with coefficients from $\mathfrak{L}_0$. Express these coefficients in the basis (3.) and then arrange the expression according to the elements of this basis. We obtain

$$\zeta^k = \eta A_{k0} + \eta\gamma^{p^f} A_{k1} + \cdots + \eta\gamma^{(n-1)p^f} A_{k,n-1}. \tag{6.}$$

Here the A's are integral homogeneous functions of the u's with coefficients in $\mathfrak{K}$, and the same is true of the determinant

$$|A_{kl}| \qquad (k, l = 0, \dots, n-1).$$

If arbitrary elements of $\mathfrak{K}$ are substituted for the indeterminates u, then ζ becomes an element of $\mathfrak{L}_0$ and $|A_{kl}|$ an element of $\mathfrak{K}$; it is equal to 0 precisely when the elements

$$\varepsilon,\ \zeta,\ \zeta^2,\ldots,\zeta^{n-1}$$

become linearly dependent (with respect to $\mathfrak{K}$). But since the u's can be chosen so that ζ is primitive, $|A_{kl}|$, considered as a function of the indeterminates u, is not zero. Therefore $|A_{kl}|$ passes into a non-zero function Δ of the indeterminates $t_0,\ldots,t_{n-1}$ when $u_0,\ldots,u_{n-1}$ are replaced by $t_0^{p^f},\ldots,t_{n-1}^{p^f}$. After this substitution, as is clear from (4.) and (5.),

$$\zeta = \eta\xi^{p^f}. \tag{7.}$$

Now substitute for $t_0,\ldots,t_{n-1}$ elements of $\mathfrak{K}$ for which $\Delta \neq 0$ (and in the case $n = 1$ assume $t_0 \neq 0$). Then $\zeta = \eta\xi^{p^f}$ is a primitive element, and ξ a non-zero element, of $\mathfrak{L}_0$. The element $\delta\xi$ of $\mathfrak{L}$ has exponent f: for, since among the elements (1.) only the last belongs to the field $\mathfrak{L}_0$, while ξ is an element of $\mathfrak{L}_0$, among the powers

$$\delta\xi,\ \delta^p\xi^p,\ldots,\delta^{p^f}\xi^{p^f} = \eta\xi^{p^f} \tag{8.}$$

again only the last is an element of $\mathfrak{L}_0$, and hence of the first kind. Moreover, the elements (8.) all have the same reduced degree; and since $\eta\xi^{p^f}$ is a primitive element of $\mathfrak{L}_0$, the reduced degree of $\delta\xi$ is equal to n. Therefore the degree of $\delta\xi$ is $n \cdot p^f$, and 5. is proved.

From 5. and § 7, 5. we immediately obtain:

I. A finite extension $\mathfrak{L}$ of an imperfect field $\mathfrak{K}$ is simple if and only if $[\mathfrak{L} : \mathfrak{K}] = n \cdot p^f$, where n denotes the reduced degree and f the exponent of $\mathfrak{L}$.

Let $\mathfrak{L}$ again be an arbitrary finite extension of the imperfect field $\mathfrak{K}$, and let $\mathfrak{M}$ be a field between $\mathfrak{K}$ and $\mathfrak{L}$. Let n and m denote the reduced degrees of the fields $\mathfrak{L}$ and $\mathfrak{M}$, and let f and g denote their exponents. Then $g \leq f$ and

$$[\mathfrak{M} : \mathfrak{K}] \geq mp^g. \tag{9.}$$

Further, for each non-negative integer r, let $\mathfrak{L}_r$ respectively $\mathfrak{M}_r$ denote the field of those elements of $\mathfrak{L}$ respectively $\mathfrak{M}$ whose exponents are $\leq r$, so that

$$\mathfrak{L}_f = \mathfrak{L}, \qquad \mathfrak{M}_g = \mathfrak{M},$$

$$[\mathfrak{L}_0 : \mathfrak{K}] = n, \qquad [\mathfrak{M}_0 : \mathfrak{K}] = m,$$

and every field $\mathfrak{M}_r$ is a subfield of $\mathfrak{L}_r$. The field $\mathfrak{L}_0$ is of the first kind with respect to $\mathfrak{K}$, hence also with respect to $\mathfrak{M}_0$, and therefore is simple. A primitive element γ of $\mathfrak{L}_0$ is thus a zero of an irreducible function $\varphi(x)$ in $\mathfrak{M}_0$ of degree n/m and of the first kind. But the function $\varphi(x)$ is also irreducible in the field $\mathfrak{M} = \mathfrak{M}_g$. For if in $\mathfrak{M}$ we had a decomposition

$$\varphi(x) = \chi(x) \cdot \psi(x), \tag{10.}$$

and if $\overline{\varphi}(x)$, $\overline{\chi}(x)$, $\overline{\psi}(x)$ denote the functions obtained from $\varphi(x)$, $\chi(x)$, $\psi(x)$ by replacing each coefficient by its p^g-th power, then (10.) would imply

$$\overline{\varphi}(x) = \overline{\chi}(x) \cdot \overline{\psi}(x). \tag{11.}$$

Since the exponent of every element of $\mathfrak{M}$ is $\leq g$, its p^g-th power is contained in $\mathfrak{M}_0$; hence by (11.) $\overline{\varphi}(x)$ would be reducible in the field $\mathfrak{M}_0$. This, however, contradicts § 13, 2. Thus $\varphi(x)$ is also irreducible in $\mathfrak{M}$, and consequently $[\gamma : \mathfrak{M}] = n/m$; with (9.) this gives

$$[\mathfrak{M}(\gamma) : \mathfrak{K}] = \frac{n}{m}[\mathfrak{M} : \mathfrak{K}] \geq n \cdot p^g. \tag{12.}$$

Since $\mathfrak{M}$ has exponent g and γ exponent 0, it follows by 3. that $\mathfrak{M}(\gamma)$ has exponent g, and is therefore a part of the field $\mathfrak{L}_g$, which contains all elements of $\mathfrak{L}$ whose exponents are $\leq g$. – The field $\mathfrak{L}$ contains elements of exponent f, hence elements of every exponent $\leq f$ (§ 13, 1.). Therefore, among the fields

$$\mathfrak{L}_0, \mathfrak{L}_1, \ldots, \mathfrak{L}_f = \mathfrak{L},$$

each preceding one is only a part of the following one.

Assume now that $\mathfrak{L}$ is simple with respect to $\mathfrak{K}$. Then by I, $[\mathfrak{L} : \mathfrak{K}] = n \cdot p^f$ and $[\mathfrak{L} : \mathfrak{L}_0] = p^f$; hence

$$[\mathfrak{L}_{r+1} : \mathfrak{L}_r] = p, \qquad (r = 0, \ldots, f-1)$$

and

$$[\mathfrak{L}_g : \mathfrak{K}] = n \cdot p^g. \tag{13.}$$

But since $\mathfrak{M}(\gamma)$ is a subfield of $\mathfrak{L}_g$, (12.) and (13.) imply

$$\mathfrak{M}(\gamma) = \mathfrak{L}_g, \qquad \frac{n}{m}[\mathfrak{M} : \mathfrak{K}] = n \cdot p^g, \qquad [\mathfrak{M} : \mathfrak{K}] = m \cdot p^g.$$

By I this says that $\mathfrak{M}$ is a simple extension of $\mathfrak{K}$. Thus we have the theorem:

II. If $\mathfrak{L}$ is a simple algebraic[14] *extension of $\mathfrak{K}$, then every field between $\mathfrak{K}$ and $\mathfrak{L}$ is also simple with respect to $\mathfrak{K}$.*

The field $\mathfrak{M}$ contains only those elements of $\mathfrak{L}$ whose p^g-th power belongs to $\mathfrak{M}_0$, but it must also contain all these elements. For these elements first form a field $\mathfrak{M}'$ (see Proposition 2.) between $\mathfrak{M}$ and $\mathfrak{L}$, and by II $\mathfrak{M}'$ must contain a primitive element δ. Since then δ^{p^g} belongs to $\mathfrak{M}_0$, we have $[\mathfrak{M}' : \mathfrak{K}] = [\delta : \mathfrak{M}_0] \cdot [\mathfrak{M}_0 : \mathfrak{K}] \leq p^g \cdot m$, hence $\mathfrak{M}' = \mathfrak{M}$. Every field $\mathfrak{M}$ between $\mathfrak{K}$ and $\mathfrak{L}$ is therefore uniquely determined by the field $\mathfrak{M}_0$ of its elements of the first kind and by its exponent g. Now $g \leq f$, and for $\mathfrak{M}_0$, since it lies between $\mathfrak{K}$ and $\mathfrak{L}_0$ and $\mathfrak{L}_0$ is an extension of the first kind, only finitely many fields can occur by § 13, 10. Hence only finitely many fields can lie between $\mathfrak{K}$ and $\mathfrak{L}$.

We now drop the hypothesis that the finite extension $\mathfrak{L}$ is simple, but instead assume that the number of fields between $\mathfrak{K}$ and $\mathfrak{L}$ is finite. Since the p-th power of every element of $\mathfrak{L}_r$ $(0 < r \leq f)$ belongs to $\mathfrak{L}_{r-1}$, $[\mathfrak{L}_r : \mathfrak{L}_{r-1}]$ cannot be $> p$; otherwise, contrary to our hypothesis, there would already be infinitely many fields between $\mathfrak{L}_{r-1}$ and $\mathfrak{L}_r$ (§ 11, 6.). Since, on the other hand, $\mathfrak{L}_{r-1}$ is only a part of $\mathfrak{L}_r$, we have $[\mathfrak{L}_r : \mathfrak{L}_{r-1}] = p$, and therefore

$$[\mathfrak{L} : \mathfrak{K}] = [\mathfrak{L}_f : \mathfrak{L}_0] \cdot [\mathfrak{L}_0 : \mathfrak{K}] = n \cdot p^f,$$

which says that $\mathfrak{L}$ is simple with respect to $\mathfrak{K}$. Taking account of the earlier results (§ 11, 3., § 11, 5., § 13, 11.), this proves:

III. In order that only finitely many fields lie between an arbitrary field $\mathfrak{K}$ and an extension $\mathfrak{L}$ of it, it is necessary and sufficient that $\mathfrak{L}$ be algebraic and simple with respect to $\mathfrak{K}$.

Finally we prove:

IV. The theorem of primitive elements and the theorem of intermediate fields (§ 11) hold, besides for perfect fields, also for those imperfect fields $\mathfrak{K}$ of characteristic p which are finite over $\mathfrak{K}^p$ and of degree p. They do not hold for the other imperfect fields.

First suppose that the stated theorems hold for the imperfect field $\mathfrak{K}$ of characteristic p, and form, as in § 12, the extension field $\mathfrak{L} = \mathfrak{K}^{p^{-1}}$ so that $\mathfrak{L}^p = \mathfrak{K}$. If a is then an element not belonging to $\mathfrak{K}$, and b an arbitrary element of $\mathfrak{L}$, then $[\mathfrak{K}(a) : \mathfrak{K}] = p$, but also $[\mathfrak{K}(a, b) : \mathfrak{K}] = p$. For if $[\mathfrak{K}(a, b) : \mathfrak{K}] > p$, then, since the p-th powers of all elements of $\mathfrak{K}(a, b)$ belong to $\mathfrak{K}$, infinitely many fields would lie between $\mathfrak{K}$ and $\mathfrak{K}(a, b)$ by § 11, 6., and the extension would not be simple. Since

[14] Exactly the same is also true, however, for simple transcendental extensions (Lüroth's theorem; see § 24).

this contradicts our assumption, $[\mathfrak{K}(a,b):\mathfrak{K}] = p = [\mathfrak{K}(a):\mathfrak{K}]$; the element b therefore belongs to $\mathfrak{K}(a)$. Since b was an arbitrary element of $\mathfrak{L}$, we have $\mathfrak{L} = \mathfrak{K}(a)$; hence $[\mathfrak{L}:\mathfrak{K}] = [\mathfrak{K}(a):\mathfrak{K}] = p$, and, because $\mathfrak{L}$ is isomorphic to $\mathfrak{K}$, also $[\mathfrak{K}:\mathfrak{K}^p] = p$.

Conversely, suppose that $[\mathfrak{K}:\mathfrak{K}^p] = p$, and let $\mathfrak{L}$ be any finite extension of $\mathfrak{K}$, with exponent f and reduced degree n. If $\mathfrak{L}_r$ has the earlier meaning, and if to each element of $\mathfrak{L}_r$ in $\mathfrak{L}_r^p$ we assign its p-th power, we obtain an isomorphic relation between $\mathfrak{L}_r$ and $\mathfrak{L}_r^p$, in which $\mathfrak{K}$ is mapped isomorphically onto $\mathfrak{K}^p$. From this it follows at once that

$$[\mathfrak{L}_r:\mathfrak{K}] = [\mathfrak{L}_r^p:\mathfrak{K}^p],$$

and hence

$$[\mathfrak{L}_r:\mathfrak{L}_r^p]\cdot[\mathfrak{L}_r^p:\mathfrak{K}^p] = [\mathfrak{L}_r:\mathfrak{K}^p] = [\mathfrak{L}_r:\mathfrak{K}]\cdot[\mathfrak{K}:\mathfrak{K}^p] = p[\mathfrak{L}_r:\mathfrak{K}],$$

so that

$$[\mathfrak{L}_r:\mathfrak{L}_r^p] = p. \tag{14.}$$

Let now $0 < r \le f$. Then $\mathfrak{L}_{r-1}$ is only a part of $\mathfrak{L}_r$; on the other hand, the p-th power of every element of $\mathfrak{L}_r$ belongs to $\mathfrak{L}_{r-1}$, that is, $\mathfrak{L}_{r-1}$ lies between $\mathfrak{L}_r^p$ and $\mathfrak{L}_r$. From (14.) it follows that $\mathfrak{L}_{r-1} = \mathfrak{L}_r^p$, and therefore

$$[\mathfrak{L}_r:\mathfrak{L}_{r-1}] = p.$$

Consequently

$$[\mathfrak{L}:\mathfrak{K}] = [\mathfrak{L}_f:\mathfrak{L}_0]\cdot[\mathfrak{L}_0:\mathfrak{K}] = np^f.$$

This equality expresses precisely that $\mathfrak{L}$ is simple with respect to $\mathfrak{K}$, as was to be shown.

§ 15.
Finite fields.

By a *finite field* we mean a field with only finitely many elements. If $\mathfrak{K}$ is a finite field, then the prime field $\mathfrak{P}$ contained in $\mathfrak{K}$ is also finite; hence the characteristic of $\mathfrak{K}$ is a prime number. Moreover $\mathfrak{K}$ must be finite with respect to its prime field. Conversely, suppose that $\mathfrak{K}$ has characteristic p and is finite with respect to its prime field $\mathfrak{P}_p = \mathfrak{P}$, say $[\mathfrak{K} : \mathfrak{P}] = n$, and that the elements $a_1, \dots, a_n$ form a basis of $\mathfrak{K}$ with respect to $\mathfrak{P}$. Then every element of $\mathfrak{K}$ is represented once in the form

$$c_1 a_1 + c_2 a_2 + \cdots + c_n a_n,$$

when $c_1, \dots, c_n$ are allowed to range over all elements of the prime field. Thus $\mathfrak{K}$ consists of p^n elements. Consequently the number of elements of a finite field is always a prime power. Conversely, let an arbitrary prime power p^n be given, and form, from the prime field $\mathfrak{P}_p$, an extension just sufficient to split the function $x^{p^n} - x$ completely. In the extension field $\mathfrak{K}$ the function $x^{p^n} - x$ has p^n distinct zeros, because its derivative is $p^n x^{p^n - 1} - \varepsilon = -\varepsilon$, and hence is relatively prime to $x^{p^n} - x$. If, moreover, a and b are any two zeros of $x^{p^n} - x$, then from $a^{p^n} = a$, $b^{p^n} = b$ it follows that

$$(a \pm b)^{p^n} = a^{p^n} \pm b^{p^n} = a \pm b,$$

and likewise $(a \cdot b)^{p^n} = a \cdot b$ and $\left(\frac{a}{b}\right)^{p^n} = \frac{a}{b}$; this says that the zeros of $x^{p^n} - x$ form a field with p^n elements. Conversely, if $\mathfrak{K}$ is a finite field with p^n elements, then p is its characteristic and n its degree with respect to the prime field. The field $\mathfrak{K}$ is absolutely algebraic, and likewise every element a of $\mathfrak{K}$ is absolutely algebraic; its absolute degree m is a divisor of n. By § 12, 4., the sequence

$$a,\ a^p,\ a^{p^2}, \dots$$

consists of a period of m distinct elements; hence for every exponent f divisible by m one has $a^{p^f} = a$, and therefore also $a^{p^n} = a$. Thus the p^n elements of $\mathfrak{K}$ are the zeros of the function $x^{p^n} - x$, and $\mathfrak{K}$ is an extension of the prime field which is just sufficient to split the function $x^{p^n} - x$ completely. But since all extensions of a field which are just sufficient for the complete decomposition of a given function are equivalent, all fields with p^n elements are isomorphic. We have therefore obtained:

1. The number of elements of a finite field is a prime power, and every prime power p^n gives the number of elements for one and only one[1] *field type; p is its characteristic and n its absolute degree.*

We shall denote this type by $\mathfrak{C}_{p,n}$.

In order that a field $\mathfrak{K}$ of characteristic p should contain a field $\mathfrak{C}_{p,n}$, it is, by the foregoing, necessary and sufficient that the function $x^{p^n} - x$ decompose in $\mathfrak{K}$ into linear factors. The p^n zeros of $x^{p^n} - x$ then form the field. From this it follows first that:

2. A field can contain no more than one subfield of type $\mathfrak{C}_{p,n}$ for a given n.

If d is a divisor of n, then $p^d - 1$ is a divisor of $p^n - 1$; hence

$$x^{p^d} - x = x\bigl(x^{p^d - 1} - \varepsilon\bigr)$$

is a divisor of

$$x^{p^n} - x = x\bigl(x^{p^n - 1} - \varepsilon\bigr).$$

[1]This was probably first stated and proved by *Moore* (Math. Papers read at the intern. math. congress, Chicago 1893, (New York 1896), p. 210 ff.).

Thus if the field $\mathfrak{K}$ contains a field $\mathfrak{C}_{p,n}$, the function $x^{p^n} - x$ decomposes in $\mathfrak{K}$, and therefore so too does the function $x^{p^d} - x$, into linear factors. Hence the following theorem holds:

3. If a field $\mathfrak{K}$ contains a field $\mathfrak{C}_{p,n}$, then it also contains fields of all types $\mathfrak{C}_{p,d}$, where d denotes a divisor of n.

A further consequence of 2. and 3. is:

4. A field $\mathfrak{C}_{p,n}$ has as many subfields as the number n has divisors; each divisor of n denotes the degree of a subfield; moreover:

5. If the field $\mathfrak{K}$ contains a field $\mathfrak{C}_{p,n}$, then this field consists of all absolutely algebraic elements of $\mathfrak{K}$ whose degree is a divisor of n.

For first, every element of $\mathfrak{C}_{p,n}$ is absolutely algebraic and its degree is a divisor of n. Conversely, if a is an absolutely algebraic element of $\mathfrak{K}$ and its degree d is a divisor of n, then the field $\mathfrak{P}_p(a)$ contained in $\mathfrak{K}$ is of type $\mathfrak{C}_{p,d}$ and, since $\mathfrak{K}$ contains only *one* field of this type, it is identical with the one which is also contained in $\mathfrak{C}_{p,n}$. Thus a is an element of $\mathfrak{C}_{p,n}$.

Let $\mathfrak{K}$ be an arbitrary field of characteristic p; let a and b be absolutely algebraic elements of $\mathfrak{K}$, let m and n be their degrees, let m be relatively prime to n, and let t be the degree of $a + b$. Then (§ 12, 4.)

$$a + b = (a + b)^{p^{tn}} = a^{p^{tn}} + b^{p^{tn}} = a^{p^{tn}} + b,$$

so that

$$a = a^{p^{tn}}.$$

It follows that m divides tn, and hence also divides t. Similarly one shows that n divides t, whence t is divisible by $m \cdot n$. On the other hand,

$$(a + b)^{p^{mn}} = a^{p^{mn}} + b^{p^{mn}} = a + b,$$

so that mn is divisible by t, and hence $t = mn$. This result extends at once to sums of more than two elements, and gives:

6. If $a_1, a_2, \ldots, a_\nu$ are absolutely algebraic elements of a field of prime characteristic, and if their degrees are pairwise relatively prime, then the degree of $a_1 + a_2 + \cdots + a_\nu$ is equal to the product of these degrees.

From this we draw a few simple consequences. Suppose

$$n = q_1^{r_1} \cdot q_2^{r_2} \cdots q_\nu^{r_\nu} \tag{1.}$$

is the decomposition of the natural number n into powers of distinct primes. Then a field of type $\mathfrak{C}_{p,n}$ contains a subfield $\mathfrak{A}$ of absolute degree $q_i^{r_i}$ $(i = 1, \ldots, \nu)$, and this contains a subfield $\mathfrak{B}$ of degree $q_i^{r_i - 1}$. If a_i is an element contained in $\mathfrak{A}$ but not in $\mathfrak{B}$, then the degree of a_i is a divisor of $q_i^{r_i}$ but, by 5., not a divisor of $q_i^{r_i - 1}$, and hence is equal to $q_i^{r_i}$. We can therefore take from the field $\mathfrak{C}_{p,n}$ ν elements $a_1, a_2, \ldots, a_\nu$ of degrees $q_1^{r_1}, q_2^{r_2}, \ldots, q_\nu^{r_\nu}$. The element $a_1 + a_2 + \cdots + a_\nu$ is then of degree n, hence primitive, and so we have proved:

7. Every finite field is simple with respect to its prime field, and therefore over that prime field there are irreducible functions of every degree n.

Suppose that the field $\mathfrak{K}$ of characteristic p contains the fields $\mathfrak{C}_{p,n_1}$ and $\mathfrak{C}_{p,n_2}$; let n be the least common multiple of n_1 and n_2, and let equation (1.) give the decomposition of n into prime powers. Then $q_i^{r_i}$ $(i = 1, \ldots, \nu)$ divides at least one of the numbers n_1, n_2; hence $\mathfrak{K}$ contains a finite field of degree $q_i^{r_i}$, and therefore also an element a_i of this degree. If for $i = 1, \ldots, \nu$ we choose one such element each, then $a = a_1 + a_2 + \cdots + a_\nu$ is an element and $\mathfrak{P}_p(a)$ is a field of absolute degree n. Thus:

8. If the field $\mathfrak{K}$ contains finite fields of (absolute) degrees n_1 and n_2, and if n is the least common multiple of n_1 and n_2, then $\mathfrak{K}$ also contains a field of degree n.

By theorem 7. we can fill the gap which remained in the proof of § 11, 4., where the case of a finite field had been excluded. Namely, if we extend a finite field $\mathfrak{K}$, by adjoining finitely many algebraic elements $\alpha, \beta, \gamma, \ldots$, to a field Ω, then Ω is finite with respect to $\mathfrak{K}$, and hence also with respect to the prime field; therefore Ω is a finite field. By 7. there are elements ξ in Ω such that $\mathfrak{P}_p(\xi) = \Omega$, and consequently also $\mathfrak{K}(\xi) = \Omega$. Thus Ω is simple with respect to $\mathfrak{K}$; and since $\mathfrak{K}$ is a perfect field, ξ is a zero of an irreducible function over $\mathfrak{K}$ with simple zeros. Hence:

9. The theorem § 11, 4. also holds for finite fields.

Every element of the field $\mathfrak{C}_{p,n}$ satisfies, in the prime field $\mathfrak{P}_p = \mathfrak{C}_{p,1}$, an irreducible equation $g(x) = 0$ whose degree is a divisor of n. Therefore, if the function $x^{p^n} - x$, whose zeros are all elements of $\mathfrak{C}_{p,n}$, is decomposed in $\mathfrak{P}_p$ into irreducible factors, these factors are all distinct and their degrees are divisors of n. Conversely, however, every irreducible function $g(x)$ over $\mathfrak{P}_p$ whose degree d divides n must occur among the factors of $x^{p^n} - x$. For let Ω be an extension of $\mathfrak{P}_p$ just sufficient for the decomposition of $g(x)$, and let a be a zero of $g(x)$ in Ω. Then $\mathfrak{P}_p(a)$ is a field $\mathfrak{C}_{p,d}$ and, by 5., contains all elements of Ω whose degree is a divisor of d, hence in particular all zeros of $g(x)$. It follows that $\mathfrak{P}_p(a) = \Omega$, and the function $x^{p^d} - x$, which has all elements of $\mathfrak{P}_p(a) = \mathfrak{C}_{p,d} = \Omega$ as zeros, must be divisible by $g(x)$. Therefore $g(x)$ also divides $x^{p^n} - x$, and we obtain the theorem:

10. The function $x^{p^n} - x$ is the product of all irreducible functions of the prime field $\mathfrak{P}_p$ whose degree is a divisor of n.

§ 16.
The absolutely algebraic fields of prime characteristic.

Let p be an arbitrary prime number, but one fixed once and for all for the following considerations. We now wish to consider the *absolutely algebraic fields of characteristic* p. Among these are all finite fields of characteristic p, which are characterized by being finite with respect to the prime field $\mathfrak{P}$. In the more comprehensive domain of all absolutely algebraic fields of characteristic p one now finds, in the main, the same laws as in the narrower domain of finite fields. In order to make this stand out more clearly, denote all prime numbers in their natural order by $q_1, q_2, q_3, \ldots$, so that

$$q_1 = 2, \quad q_2 = 3, \quad q_3 = 5, \ldots$$

Then every natural number is represented in one way in the form

$$\prod_{i=1}^{\infty} q_i^{x_i}, \tag{1.}$$

where the exponents x_i are non-negative integers, only finitely many of which differ from 0. We shall now more generally consider all symbolic expressions of the form (1.), in which for each exponent x_i one may put, at will, 0 or a natural number or also the sign ∞; for these expressions, which we call *G-numbers* (G = degree), and which include the natural numbers, we stipulate, in agreement with the laws valid for natural numbers, the following. Two G-numbers

$$m = \prod q_i^{x_i}, \qquad n = \prod q_i^{\lambda_i}$$

are called equal if and only if, for every i, $x_i = \lambda_i$. The number m is called divisible by n, a multiple of n, and so on, if for every i one has $\lambda_i \leq x_i$. Here ∞ counts as greater than every other exponent. If m is divisible by n, we also form the quotient by stipulating that

$$\frac{m}{n} = \prod q_i^{x_i - \lambda_i}$$

where $x_i - \lambda_i$ is to be set equal to 0 when $x_i = \infty$, $\lambda_i = \infty$, and equal to ∞ when $x_i = \infty$ but λ_i is finite. It then follows at once: all G-numbers are divisible by 1, and all divide the G-number whose exponents x_i are all equal to ∞. Given any system of finitely or infinitely many G-numbers, these always possess a greatest common divisor t, into which all common divisors divide, and a least common multiple v, which divides all common multiples. The exponent of q_i in t is equal to the smallest exponent which q_i has in the given G-numbers; the exponent of q_i in v is equal to the largest of these exponents, or, if no such largest exponent exists — namely, if no exponent of q_i is equal to ∞, but every finite number is exceeded by some of these exponents —, equal to ∞.

If m is any G-number, then the natural numbers which divide m form a system $\mathfrak{S}$ with the following properties:

1) If n is a number in $\mathfrak{S}$, then every divisor of n also belongs to $\mathfrak{S}$.

2) If n_1, n_2 are numbers in $\mathfrak{S}$, then their least common multiple is also a number in $\mathfrak{S}$.

The number m is then in every case the least common multiple of all the numbers in $\mathfrak{S}$, and hence is itself completely determined by the system $\mathfrak{S}$. Conversely, if one has any system $\mathfrak{S}$ of natural numbers which has the properties 1) and 2), and if m is the least common multiple of all numbers in $\mathfrak{S}$, then $\mathfrak{S}$ consists of all natural numbers which divide m.

Now let $\mathfrak{K}$ be any absolutely algebraic field of characteristic p. The degrees of the finite fields contained in $\mathfrak{K}$ then form a system $\mathfrak{S}$ of natural numbers which, by § 15, 3. and 8., has the properties 1) and 2). Let m be the least common multiple of the numbers in $\mathfrak{S}$. If m is a natural number, then $\mathfrak{K}$ is a finite field of degree m. Conversely, if $\mathfrak{K}$ is a finite field, then m represents the degree of this field. We shall now also, however, in the cases where $\mathfrak{K}$ is not finite and accordingly m is not a natural number, call m the absolute degree of $\mathfrak{K}$. Thus:

1. Every absolutely algebraic field of characteristic p has a definite degree, which is a G-number.

Moreover, the following theorems hold, as we still have to show:

2. There is an absolutely algebraic field of characteristic p whose degree is equal to any prescribed G-number.

3. All absolutely algebraic fields of the same characteristic p and the same degree are isomorphic.

To arrive at a proof of 2., we assign a numbering to the integral functions reduced modulo p. By the number ν of such a function

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0, \tag{2.}$$

in which the coefficients are therefore numbers from the series $0, 1, \ldots, p-1$, we mean the number $f(p)$, so that

$$\nu = a_n p^n + a_{n-1}p^{n-1} + \cdots + a_0. \tag{3.}$$

Then not only does every function considered here have a number ≥ 0, but every integer ≥ 0 also represents the number of one and only one of these functions. For equation (3.) is nothing other than the representation of the number ν in the system with base p. We then transfer this numbering to the integral functions of a prime field $\mathfrak{P} = \mathfrak{P}_p$, that is, to all elements of $\mathfrak{P}[x]$, which are also represented uniquely in the form (2.). As is seen at once, of two functions of different degree the one of smaller degree also has the smaller number; and among functions of the same degree, those in which the coefficient of the highest power is the unit have smaller numbers than the others.

If n is any natural number, then, as we know, there are irreducible functions of degree n in $\mathfrak{P}$. Among these let the one with the smallest number be denoted by $\varphi_n(x)$. If d is a divisor of n, then in a finite extension field $\mathfrak{C}_{p,n}$ there are elements $j, j_1, \ldots, j_{n-1}$ which satisfy the equation $\varphi_n(x) = 0$, and d elements which satisfy the equation $\varphi_d(x) = 0$. The latter can be represented as integral functions of j, of degree $< n$, with coefficients belonging to $\mathfrak{P}$. In the language of congruences this can be expressed as follows: there are in $\mathfrak{P}$ d functions $\chi_1(x), \ldots, \chi_d(x)$, of degree $< n$, for which

$$\varphi_d(\chi_i(x)) \equiv 0 \pmod{\varphi_n(x)} \qquad (i = 1, \ldots, d).$$

Among these we denote the one with the smallest number by $\psi_{d,n}(x)$. We can then say: if in an extension field of $\mathfrak{P}$ the element j is a zero of $\varphi_n(x)$, then $\psi_{d,n}(j)$ is a zero of $\varphi_d(x)$. The functions $\varphi_n(x)$ and $\psi_{d,n}(x)$ are well-defined functions from $\mathfrak{P}$ for every natural number n and every divisor d of n, and they are easily computed for each given n, d. Let it also be noted that $\varphi_1(x) = x$, $\psi_{1,n}(x) = 0$, while $\psi_{d,n}(x)$, as soon as $d > 1$, is at least of first degree. For $n > 1$, $\psi_{n,n}(x)$ is equal to x.

Now let any G-number

$$m = \prod q_i^{x_i}$$

be given. For each natural number n, denote by r_n the greatest common divisor of $n!$ and m; then

$$r_1, r_2, \ldots \tag{4.}$$

form a sequence of natural numbers, each of which divides the following ones. A natural number n which divides m divides r_n as well, because it divides $n!$. But a number which does not divide m cannot divide any of the numbers (4.). Hence the system $\mathfrak{S}$ of natural divisors of m is identical with the system of numbers which divide numbers of the sequence (4.). Apart from the already settled case in which m is finite, that is, a natural number, the numbers (4.) will grow beyond all bounds. By omitting elements from (4.) we can arrange that an infinite sequence of numbers remains in which the very first element is already > 1 and no two are equal. This sequence of numbers still has the properties on which alone we rely: namely, every number divides the following ones, and the natural numbers which divide numbers of the sequence are precisely the divisors of m.

We now introduce a system of elements

$$j_1, j_2, \ldots \text{ in inf.}$$

by stipulating that j_n is to be a zero of the function $\varphi_{r_n}(x)$ — for which we shall also write $\overline{\varphi}_n(x)$. Then $\mathfrak{P}(j_n) = \mathfrak{K}_n$ is a finite field of degree r_n. Thus we obtain a system of fields

$$\mathfrak{K}_0 = \mathfrak{P}, \quad \mathfrak{K}_1, \quad \mathfrak{K}_2, \ldots \tag{5.}$$

which at first have nothing to do with one another. If we replace the function symbol $\psi_{r_n, r_{n+1}}$ by $\overline{\psi}_{n,n+1}$, then $\overline{\psi}_{n,n+1}(j_{n+1})$ is a zero of $\overline{\varphi}_n(x)$, and therefore there is a uniquely determined isomorphic relation between $\mathfrak{K}_n = \mathfrak{P}(j_n)$ and the field $\mathfrak{P}(\overline{\psi}_{n,n+1}(j_{n+1}))$ contained in $\mathfrak{K}_{n+1} = \mathfrak{P}(j_{n+1})$, under which the element j_n corresponds to the element $\overline{\psi}_{n,n+1}(j_{n+1})$. We may therefore, without thereby invalidating the relations so far existing between the elements, stipulate that $j_n = \overline{\psi}_{n,n+1}(j_{n+1})$. By this stipulation, which is to hold for every n, each field of the series (5.) becomes part of the following one. The totality of all elements occurring in these fields forms a field Ω, and it is easy to show that this field has the desired properties. First it is clear that Ω is an absolutely algebraic field of characteristic p, since this holds for all the fields in (5.). If, moreover, n is a natural number which divides m, then n divides r_ν, and the field $\mathfrak{K}_\nu$, whose degree is r_ν, has a subfield of degree n, which is also a subfield of Ω. But if n does not divide m, and hence does not divide any number r_ν, then none of the fields $\mathfrak{K}_\nu$ contains an element of degree n; thus Ω contains no such element either, and hence no field of degree n. Therefore m is the degree of Ω.

We now come to the proof of 3. and suppose that, besides the field Ω just constructed, a second field Ω' is given which, like Ω, is absolutely algebraic, has characteristic p, and has degree m. We must show that Ω and Ω' are isomorphic. In this proof we may, as is easily seen, suppose that the prime field of Ω' is identical with that of Ω. Since each of the numbers r_n divides the degree m of Ω', by hypothesis Ω' has a finite subfield $\mathfrak{K}'_n$ of degree r_n, and therefore the function $\overline{\varphi}_n(x)$ irreducible in $\mathfrak{P}$, whose degree is r_n, must decompose into linear factors in Ω'. Let us now imagine that from Ω' some system of elements

$$i_1, i_2, \ldots \text{ in inf.}$$

has been selected in such a way that i_n becomes a zero of $\overline{\varphi}_n(x)$.[1] From this system we shall now derive, in a uniquely determined way, a second such system

$$j''_1, j''_2, \ldots \text{ in inf.}$$

For this purpose we stipulate that $j''_1 = i_1$, and give a method by which j''_{n+1} is determined once the elements $j''_1, \ldots, j''_n$ have already been determined. Now i_{n+1} is a zero of the function $\overline{\varphi}_{n+1}(x)$ irreducible in $\mathfrak{P}$; hence every power $i_{n+1}^{p^f}$ is also a zero of $\overline{\varphi}_{n+1}(x)$, and each zero is obtained once when the exponent f is given the values $0, 1, \ldots, (r_{n+1} - 1)$. Furthermore, from $\overline{\varphi}_{n+1}(i_{n+1}) = 0$ it follows that $\overline{\psi}_{n,n+1}(i_{n+1})$ is a zero of $\overline{\varphi}_n(x)$. By

$$\left(\overline{\psi}_{n,n+1}(i_{n+1})\right)^{p^f} \tag{6.}$$

[1] At this point the *principle of choice* is used here for the first time. It cannot in any way be avoided in the proof of the present theorem.

each zero of $\overline{\varphi}_n(x)$, and therefore also j''_n, is represented once when f is set equal to $0, 1, \dots, (r_n-1)$. Let ν be that one of these numbers for which the expression (6.) is equal to j''_n. We then stipulate that $j''_{n+1} = i^{p^\nu}_{n+1}$. Since $\overline{\psi}_{n,n+1}(x)$ is a function in the prime field, we have

$$j''_n = \big(\overline{\psi}_{n,n+1}(i_{n+1})\big)^{p^\nu} = \overline{\psi}_{n,n+1}(i^{p^\nu}_{n+1}) = \overline{\psi}_{n,n+1}(j''_{n+1}),$$

and hence for every n we have the equations

$$\overline{\varphi}_n(j''_n) = 0, \qquad j''_n = \overline{\psi}_{n,n+1}(j''_{n+1}). \tag{7.}$$

Of the fields in the series

$$\mathfrak{P}, \quad \mathfrak{K}'_1 = \mathfrak{P}(j''_1), \quad \mathfrak{K}'_2 = \mathfrak{P}(j''_2), \dots \tag{8.}$$

each is contained in the next, and all are subfields of Ω'. But every element of Ω' also occurs in fields of this series. For the degree of an element of Ω' is a divisor of m, hence a divisor of some number r_n, and the field $\mathfrak{K}'_n$, whose degree is r_n, contains (§ 15, 5.) all elements of Ω' whose degree divides r_n. Now the element j_n of $\mathfrak{K}_n$ and the element j''_n of $\mathfrak{K}'_n$ are zeros of the same irreducible function $\overline{\varphi}_n(x)$ from $\mathfrak{P}$; therefore there exists a definite isomorphic relation π_n between $\mathfrak{K}_n = \mathfrak{P}(j_n)$ and $\mathfrak{K}'_n = \mathfrak{P}(j''_n)$ in which the elements j_n and j''_n correspond to one another. From

$$j_n = \overline{\psi}_{n,n+1}(j_{n+1}), \qquad j''_n = \overline{\psi}_{n,n+1}(j''_{n+1})$$

it follows further that the relation π_{n+1} includes the relation π_n. It follows at once that the isomorphisms

$$\pi_1, \pi_2, \dots$$

produce a one-to-one, and indeed isomorphic, relation between Ω and Ω'. Theorem 3 is thereby proved.

We shall now, even when m is an infinite G-number, denote by $\mathfrak{C}_{p,m}$ an absolutely algebraic field of characteristic p and degree m. The degree m was defined as the least common multiple of the degrees of the finite fields contained in $\mathfrak{C}_{p,m}$, or, equivalently, of the degrees of the elements of $\mathfrak{C}_{p,m}$; and the system of these degrees is identical with the system of natural numbers which divide m. On this basis one easily proves:

4. If one adjoins to the prime field $\mathfrak{P}_p$ a system $\mathfrak{S}$ of algebraic elements, then there arises an absolutely algebraic field $\mathfrak{C}_{p,m}$ whose degree m is equal to the least common multiple of the degrees of the elements of $\mathfrak{S}$.

Let m' be this least common multiple. Then m, as least common multiple of the degrees of all elements of $\mathfrak{C}_{p,m}$, is divisible by m'. Suppose first that $\mathfrak{S}$ is a finite system; then $\mathfrak{C}_{p,m} = \mathfrak{P}_p(\mathfrak{S})$ is a finite field and contains a field $\mathfrak{C}_{p,m'}$. To this latter field, however, belong all elements of $\mathfrak{C}_{p,m}$ whose degree divides m'. Since all elements of $\mathfrak{S}$ belong to it, it follows that $\mathfrak{C}_{p,m'} = \mathfrak{C}_{p,m}$, and hence $m' = m$. If the system $\mathfrak{S}$ is infinite and α is an element of $\mathfrak{C}_{p,m}$, of degree n, then one can determine a finite subsystem $\mathfrak{S}'$ of $\mathfrak{S}$ such that α is contained in the finite field $\mathfrak{P}_p(\mathfrak{S}')$. Then n divides the degree of $\mathfrak{P}_p(\mathfrak{S}')$, and by what precedes this degree is equal to the least common multiple of the degrees of the elements of $\mathfrak{S}'$, hence is a divisor of m'. Thus n divides m', and m' is therefore the least common multiple not only of the degrees of the elements of $\mathfrak{S}$, but of all elements of $\mathfrak{C}_{p,m}$. Again, then, $m' = m$.

If a field Ω contains a field $\mathfrak{C}_{p,m}$, and the natural number n divides m, then $\mathfrak{C}_{p,m}$ contains a finite field $\mathfrak{C}_{p,n}$, and this contains all elements of degree n occurring in Ω (§ 15, 5.). On the other hand, the degree of every element of $\mathfrak{C}_{p,m}$ must divide m. Thus $\mathfrak{C}_{p,m}$ consists of all absolutely algebraic elements of Ω whose degree divides m. This is the extension of the theorem § 15, 5. to all absolutely algebraic fields of prime characteristic. It follows at once from it that the theorems 2., 3., 4., and 8. of § 15 also hold generally in the domain of absolutely algebraic fields of prime characteristic.

Let m and m' be two G-numbers, with m divisible by m'. Then a field $\mathfrak{C}_{p,m}$ contains a field $\mathfrak{C}_{p,m'}$. Now let α be an element of $\mathfrak{C}_{p,m}$, and let n be its absolute degree. By 4., the absolute degree of the field $\mathfrak{C}_{p,m'}(\alpha)$ is then equal to the least common multiple ν of m' and n. If m' is finite, then $\nu = [\mathfrak{C}_{p,m'}(\alpha) : \mathfrak{P}] = [\alpha : \mathfrak{C}_{p,m'}]\, m'$, whence

$$[\alpha : \mathfrak{C}_{p,m'}] = \frac{\nu}{m'},$$

or finally

$$[\alpha : \mathfrak{C}_{p,m'}] = \frac{n}{(n, m')}, \tag{9.}$$

where again (n, m') denotes the greatest common divisor of n and m'. The same equation also holds when m' is not a natural number. For if, in this case, ν denotes any natural number dividing m', then $\mathfrak{C}_{p,m'}$ contains a subfield $\mathfrak{C}_{p,\nu}$, and by (9.) one has

$$[\alpha : \mathfrak{C}_{p,m'}] \leq [\alpha : \mathfrak{C}_{p,\nu}] = \frac{n}{(n, \nu)}.$$

Conversely, ν can be chosen so that $[\alpha : \mathfrak{C}_{p,m'}] = \frac{n}{(n,\nu)}$. For the irreducible function $f(x)$ from $\mathfrak{C}_{p,m'}$ which has α as a zero belongs to a *finite* field, namely to the one obtained from $\mathfrak{P}$ by adjoining the coefficients of $f(x)$. If ν is the degree of this field, then $[\alpha : \mathfrak{C}_{p,m'}] = [\alpha : \mathfrak{C}_{p,\nu}] = \frac{n}{(n,\nu)}$. Hence $[\alpha : \mathfrak{C}_{p,m'}]$ is equal to the minimum of the expression $\frac{n}{(n,\nu)}$, and this occurs when (n, ν) reaches its maximum. Since ν is to be a divisor of m', this maximum cannot be $> (n, m')$, but this value is also reached, namely for $\nu = (n, m')$. Hence equation (9.) holds in general.

We now consider the fields $\mathfrak{C}_{p,k}$ between $\mathfrak{C}_{p,m'}$ and $\mathfrak{C}_{p,m}$. Every G-number k which is a multiple of m' and a divisor of m represents the degree of such a field. If this field is finite with respect to $\mathfrak{C}_{p,m'}$, then it is also simple (§ 11, 5.; § 12, 5.). If then α is a primitive element of $\mathfrak{C}_{p,k}$ with respect to $\mathfrak{C}_{p,m'}$, and n is its absolute degree, then by the preceding

$$[\mathfrak{C}_{p,k} : \mathfrak{C}_{p,m'}] = \frac{n}{(n, m')}.$$

If all fields $\mathfrak{C}_{p,k}$ finite with respect to $\mathfrak{C}_{p,m'}$ are considered, then n may be any natural number dividing m. The totality of all numbers of the form $\frac{n}{(n,m')}$ is, however, as follows at once from the definition given above of the quotient of two G-numbers, identical with the totality of all natural numbers dividing $\frac{m}{m'}$, and therefore has $\frac{m}{m'}$ as least common multiple. Thus if $\mathfrak{C}_{p,m}$ is finite with respect to $\mathfrak{C}_{p,m'}$, then $\frac{m}{m'}$ is a natural number, and conversely; and in this case

$$[\mathfrak{C}_{p,m} : \mathfrak{C}_{p,m'}] = \frac{m}{m'}.$$

It is therefore advisable to take this equation as the definition of the relative degree in the cases in which $\frac{m}{m'}$ is not finite. Now, if k is made to run through all multiples of m' which divide m, then $\frac{k}{m'}$ gives all divisors of $\frac{m}{m'}$, each exactly once. Thus:

5. If $\mathfrak{C}_{p,m'}$ is a subfield of $\mathfrak{C}_{p,m}$, then the degrees, taken with respect to $\mathfrak{C}_{p,m'}$, of the fields between $\mathfrak{C}_{p,m'}$ and $\mathfrak{C}_{p,m}$ are divisors of $\frac{m}{m'}$; every divisor of $\frac{m}{m'}$ gives the relative degree of one and only one such field.

Let

$$m = \prod q_i^{x_i},$$

let $\varphi(x)$ be an irreducible function in a field $\mathfrak{C}_{p,m}$, and let n be its degree. By adjoining a zero of $\varphi(x)$, $\mathfrak{C}_{p,m}$ is extended to a field $\mathfrak{C}_{p,\mu}$, and $n = \frac{\mu}{m}$. It follows that n can contain no prime factor q_i whose exponent x_i in the decomposition of m is equal to ∞. Conversely, if n is a natural number which contains no such prime factor, then μ can be chosen so that $\frac{\mu}{m} = n$. A field $\mathfrak{C}_{p,\mu}$ is then simple with respect to its subfield $\mathfrak{C}_{p,m}$, and every primitive element is a zero of a function irreducible in $\mathfrak{C}_{p,m}$ of degree n. Hence:

6. In a field $\mathfrak{C}_{p,m}$ there are irreducible functions of all and only the finite degrees n which contain only such prime factors q_i whose exponent x_i in

$$m = \prod q_i^{x_i}$$

is finite.

One further remark may be made here. The preceding investigations, as well as those which follow, can also be presented in the *Kroneckerian* form of expressions by means of systems of moduli; but one must then enlarge this concept by also bringing integral functions of infinitely many indeterminates into consideration. As such indeterminates we must then interpret the elements $j_1, j_2, \ldots$ introduced above, while $\varphi_n(x)$ and $\psi_{d,n}(x)$ are to be regarded as integral functions reduced modulo p of a further indeterminate x. Let m again be an arbitrary G-number, let $r_1, r_2, \ldots$ be an associated sequence of natural numbers of the kind described earlier, and write $\overline{\varphi}_n$, $\overline{\psi}_{n,n+1}$ for φ_{r_n}, $\psi_{r_n,r_{n+1}}$. Further let S denote the system of all integral functions of the indeterminates j (this is of course to be understood so that each individual element of S is always to contain only a finite number of these indeterminates), and let T be the subsystem of S consisting of the functions

$$p; \quad \overline{\varphi}_1(j_1), \quad \overline{\varphi}_2(j_2), \ldots; \quad \overline{\psi}_{1,2}(j_2) - j_1, \quad \overline{\psi}_{2,3}(j_3) - j_2, \ldots$$

The system T defines a *system of moduli* M. Two elements of S are called congruent modulo M if their difference can be represented in the form

$$A_1 B_1 + A_2 B_2 + \cdots + A_\nu B_\nu,$$

where $A_1, A_2, \ldots, A_\nu$ are elements of S and $B_1, B_2, \ldots, B_\nu$ are elements of T. Now every integral function of x, reduced modulo p, whose degree divides m, decomposes modulo M in a definite way into linear factors whose coefficients are elements of S, and the residue classes of the system of moduli M within S form a field of type $\mathfrak{C}_{p,m}$. If one takes

$$m = \prod q_i^{\infty},$$

then every integral function of x whose coefficients belong to S decomposes modulo M into linear factors (see § 17).

§ 17.
Algebraically closed fields.

In § 8 we treated the problem: in a field $\mathfrak{K}$ a polynomial $f(x)$ is given; sought is an extension field Ω in which $f(x)$ decomposes completely. But we can now also pose the substantially more far-reaching problem of finding, for the field $\mathfrak{K}$, an extension Ω in which *every* function from $\mathfrak{K}$ decomposes. The preceding paragraph contains the solution of this problem for the case of the prime field $\mathfrak{P}_p$. Namely, if we put

$$m = \prod q_i^\infty, \qquad \mathfrak{C}_{p,m} = \mathfrak{A}_p,$$

then m is divisible by every natural number n, and therefore in $\mathfrak{A}_p$ every irreducible, and hence also every reducible, function from $\mathfrak{P}_p$ must decompose completely.[1] From § 16, 6. it is further seen that in $\mathfrak{A}_p$ only irreducible functions of degree 1 exist. Thus $\mathfrak{A}_p$ belongs to that special kind of fields which we designated in § 7 as closed. These are of course of the greatest importance for the question touched on here, and we therefore collect here some simple propositions concerning them.

1. If Ω is an algebraic extension of a field $\mathfrak{K}$, and every polynomial from $\mathfrak{K}$ decomposes in Ω into linear factors, then Ω is algebraically closed.

Indeed, let $\varphi(x)$ be an irreducible function from Ω, and let j be a zero which $\varphi(x)$ has in Ω or in an extension field of Ω. Then $\Omega(j)$ is algebraic with respect to Ω, and hence also with respect to $\mathfrak{K}$, and j is a zero of a polynomial $f(x)$ from $\mathfrak{K}$. Since this polynomial decomposes in Ω into linear factors, while in $\Omega(j)$ it has the linear factor $x - j$, the element j belongs to Ω. Thus $\varphi(x)$ has degree 1, and Proposition 1 is proved.

Let Ω be an algebraically closed field, let $\mathfrak{K}$ be a subfield of Ω, and let $\mathfrak{A}$ be the field between $\mathfrak{K}$ and Ω which comprises the elements algebraic with respect to $\mathfrak{K}$. An arbitrary function $\varphi(x)$ from $\mathfrak{A}$ then decomposes in Ω into linear factors; its zeros are algebraic with respect to $\mathfrak{A}$, and therefore also with respect to $\mathfrak{K}$, and hence are elements of $\mathfrak{A}$. Thus $\varphi(x)$ already decomposes completely in $\mathfrak{A}$, so that the field $\mathfrak{A}$ is algebraically closed. — If $\mathfrak{T}$ is a field between $\mathfrak{K}$ and Ω, if α is an element of $\mathfrak{A}$, and if $\psi(x)$ is the irreducible function from $\mathfrak{K}$ with zero α, then $\psi(x)$ also belongs to the field $\mathfrak{T}$; and if $\mathfrak{T}$ is algebraically closed, then $\psi(x)$ decomposes in $\mathfrak{T}$ into linear factors, so that α is an element of $\mathfrak{T}$, and hence $\mathfrak{A}$ is a subfield of $\mathfrak{T}$. Thus $\mathfrak{A}$ proves to be the smallest algebraically closed field between $\mathfrak{K}$ and Ω; it is contained in every other such field. If Ω itself is to be as small as possible among algebraically closed extensions of $\mathfrak{K}$, then one must have $\Omega = \mathfrak{A}$. Therefore:

2. In every extension Ω of a field $\mathfrak{K}$ which represents an algebraically closed field, there is a smallest extension of this kind; it is distinguished from all the others by being algebraic.

In the preceding considerations take, for $\mathfrak{K}$, the prime field $\mathfrak{P}$ contained in Ω. Then $\mathfrak{A}$ becomes an absolutely algebraic field and consists of all absolutely algebraic elements of Ω. If the characteristic of Ω is a prime number p, then $\mathfrak{A}$ also has this characteristic and is therefore a field $\mathfrak{C}_{p,m}$. Since every irreducible function from $\mathfrak{P}_p$ decomposes in $\mathfrak{C}_{p,m}$, the number m must be divisible by every natural number, whence

$$m = \prod q_i^\infty, \qquad \mathfrak{C}_{p,m} = \mathfrak{A}_p.$$

Thus:

3. In every algebraically closed field Ω of characteristic p there is contained a field of type $\mathfrak{A}_p$; it consists of all absolutely algebraic elements of Ω.

[1] L. E. Dickson speaks of such a field (Transactions of the American Math. Soc. 8 (1907), p. 389). He explains it incorrectly, however, as the totality of all Galois fields belonging to the prime number p (Galois field = finite field, or the type of such a field); no proof of existence is given.

The analogous statement holds for fields of characteristic 0. Appealing first to known facts, we note that the field of all algebraic numbers is algebraically closed and that it is an algebraic extension of the field of rational numbers. For the latter type we use, as before, the symbol $\mathfrak{P}_0$; for the former we introduce the symbol $\mathfrak{A}_0$. One ordinarily arrives at the field of all algebraic numbers by regarding them as special complex numbers. But even without calling upon the general complex numbers, one can pass, just as from $\mathfrak{P}_p$ to $\mathfrak{A}_p$, from $\mathfrak{P}_0$ to an absolutely algebraic and algebraically closed field by defining an infinite sequence of fields

$$\mathfrak{K}_0 = \mathfrak{P}_0, \quad \mathfrak{K}_1, \quad \mathfrak{K}_2, \ldots,$$

each of which contains the preceding ones and is still finite with respect to $\mathfrak{P}_0$, and whose elements, taken together, represent an algebraically closed field. Here too, by using the principle of choice, one succeeds in proving that $\mathfrak{A}_0$ is the unique type of the required kind. Hence Proposition 3 also holds for characteristic 0.

We shall not enter further into these special investigations. Rather, we shall prove quite generally that *every field $\mathfrak{K}$ can be extended to an algebraically closed field, and that the problem of finding such an extension has, in essence, only one solution if one further requires that the extension be as small as possible, or — what by 2. amounts to the same thing — that it be algebraic.*

§ 18.
Relations between systems of polynomials and extension fields.

In the investigations of this paragraph we take as fixed an arbitrarily chosen field $\mathfrak{K}$. We consider systems of polynomials in one indeterminate x with coefficients from $\mathfrak{K}$. Such functions are always meant when function systems are mentioned without qualification; likewise expressions such as "extension field", "normal function", and so forth always refer to the ground field $\mathfrak{K}$.

Let Ω be an extension field, and let S be a system of finitely or infinitely many functions which decompose completely in Ω. Then Ω contains a definite extension Ω' which just suffices for the complete decomposition of the functions from S. Namely, one has $\Omega' = \mathfrak{K}(\mathfrak{T})$, where $\mathfrak{T}$ denotes the system of zeros which the functions of the system S possess inside Ω.

If S denotes any finite system of functions, then there always exist (see the end of § 8) extension fields in which the functions from S decompose completely. We now set up the definition:

Definition 1. A function $\varphi(x)$ is called subordinate to the finite function system S if it likewise decomposes completely in every extension field in which the functions from S decompose completely.

It follows immediately:

1. The function $\varphi(x)$ is subordinate to the finite system S as soon as it also decomposes in some extension field Ω_1 which just suffices for the complete decomposition of the system S.

For if Ω_2 is any second extension inside which S decomposes completely, then Ω_2 contains an extension Ω_2' which just suffices for the complete decomposition of S. By § 8, 2., the extensions Ω_1 and Ω_2' are equivalent; and since $\varphi(x)$ decomposes completely in Ω_1, it also decomposes completely in Ω_2' and hence in Ω_2.

There would now be nothing to prevent Definition 1 from being used also in the case where the system S consists of infinitely many functions, if it had already been proved that extension fields in which the functions of S decompose completely exist in this case too. Since this has not yet been done, we must rely on another definition. We arrive at one by starting from the observation that, in the case of a finite system S, a function $\varphi(x)$ which is subordinate to a subsystem of S is also subordinate to S itself.

Definition 2. A function $\varphi(x)$ is called subordinate to the infinite system S if it is subordinate to some finite subsystem of S. — If S_1, S_2 are any two function systems, we say that S_1 is subordinate to S_2 if every function from S_1 is subordinate to the system S_2.

Then the following proposition holds:

2. If the system S_1 is subordinate to the system S_2, and the latter is subordinate to the system S_3, then S_1 is subordinate to the system S_3.

Indeed, an arbitrary function $\varphi(x)$ from S_1 is subordinate to a finite subsystem S_2' of S_2, and this is subordinate to a finite subsystem S_3' of S_3 (Definition 2). Since S_3' is finite, there exist extensions inside which all functions of S_3' decompose completely. In every such extension the functions of S_2' decompose, and therefore $\varphi(x)$ also decomposes completely (Definition 1). Thus $\varphi(x)$ is subordinate to the system S_3', and hence also to the system S_3, proving 2.

Definition 3. Two function systems are called equivalent if each is subordinate to the other.

The preceding definitions and propositions show at once that every function system is equivalent to itself; that two function systems equivalent to a third are equivalent to one another; and that every finite system of functions is equivalent to the product of these functions.

Definition 4. A function system S will be called complete if every function subordinate to S belongs to S.

If S is any system, then a system S', if it is to be complete and equivalent to S, must contain only such functions as are subordinate to S, but it must contain all of them. Conversely, one sees at once that the system of all functions subordinate to S is in fact equivalent to S and complete. Thus every system S is equivalent to exactly one complete system. The smallest complete system is evidently formed by the functions which decompose completely in $\mathfrak{K}$ itself, and the largest by the totality of all functions from $\mathfrak{K}$. It is also evident that the intersection of two complete systems, or of any system, finite or infinite, of complete systems, is again a complete system.

To these propositions are appended some others which express relations between function systems and the associated decomposition fields. First, the preceding definitions and propositions immediately give:

3. If Ω is an extension of $\mathfrak{K}$, and S is a function system which decomposes completely in Ω, then every system S' subordinate to S also decomposes completely in Ω; and if S and S' are equivalent and the extension Ω just suffices for the complete decomposition of S, then it also just suffices for the complete decomposition of S'.

This proposition shows that in questions concerning the existence and the properties of extensions in which a given function system decomposes, that system may be replaced by any equivalent one.

We prove further:

4. If the extension $\mathfrak{N}$ just suffices to decompose the function system S completely, then it is normal, and every function which decomposes completely in $\mathfrak{N}$ is subordinate to the system S.

Proof. Let $\mathfrak{T}$ be the system of those elements of $\mathfrak{N}$ which are zeros of functions from S. Then our assumption implies that $\mathfrak{N} = \mathfrak{K}(\mathfrak{T})$. Now let $\varphi(x)$ denote any irreducible function from $\mathfrak{K}$ which has a zero α in $\mathfrak{N}$. Then one can specify a finite subsystem $\mathfrak{T}'$ of $\mathfrak{T}$ such that α already belongs to the field $\mathfrak{K}(\mathfrak{T}')$; and, because $\mathfrak{T}'$ is finite, a finite subsystem S' of S can be specified such that the elements of $\mathfrak{T}'$ are zeros of functions from S'. If $\mathfrak{T}''$ denotes the system of all zeros of the functions from S', then the extension field $\mathfrak{N}' = \mathfrak{K}(\mathfrak{T}'')$ also contains the element α and just suffices to decompose completely the functions from S'. Since S' is finite, $\mathfrak{N}'$ is therefore normal with respect to $\mathfrak{K}$ by § 8, 6.; and consequently the function $\varphi(x)$ must decompose completely in $\mathfrak{N}'$ and hence in $\mathfrak{N}$. This shows that the algebraic extension $\mathfrak{N} = \mathfrak{K}(\mathfrak{T})$ is likewise normal. Next, from the fact that $\varphi(x)$ also decomposes in the field $\mathfrak{N}'$, which just suffices for the complete decomposition of the finite system S', it follows that $\varphi(x)$ is subordinate to the system S' (Proposition 1). Hence $\varphi(x)$ is also subordinate to S. Finally, if $f(x)$ is any function which decomposes completely in $\mathfrak{N}$, then by what precedes every irreducible factor of $f(x)$ is subordinate to the system S; hence the same is true of $f(x)$ itself. Proposition 4 is proved. — It also follows from it that, once the existence of an extension field in which S decomposes completely is established for a system S, Definitions 1 and 2 amount to the same thing for this system S.

5. The totality S of all functions which decompose completely in an extension field $\mathfrak{N}$ forms a complete system, and if the extension is normal, then it just suffices for the complete decomposition of S.

Indeed, a function $\varphi(x)$ which is subordinate to S must decompose completely in $\mathfrak{N}$, and therefore belongs to S; that is, the system S is complete. Further, if $\mathfrak{T}$ denotes the system of elements of $\mathfrak{N}$ which are zeros of functions from S, then the extension $\mathfrak{K}(\mathfrak{T})$ contained in $\mathfrak{N}$ just suffices for the decomposition of S. If now $\mathfrak{N}$ is normal, hence also algebraic, then every element α of $\mathfrak{N}$ is a zero of a function $\varphi(x)$ irreducible in $\mathfrak{K}$ which decomposes completely in $\mathfrak{N}$, and therefore belongs to S. Hence α is an element of $\mathfrak{T}$, and so of $\mathfrak{K}(\mathfrak{T})$; thus $\mathfrak{N} = \mathfrak{K}(\mathfrak{T})$, proving 5.

We conclude these investigations with the proof of the theorem:

6. Every function system S is equivalent to a system of normal functions.

Proof. (Compare § 13, 3.) Suppose first that S consists of a single irreducible function $\varphi(x)$, and let $\mathfrak{N}$ denote an extension which just suffices for the complete decomposition of $\varphi(x)$. Then $\mathfrak{N}$ is a normal field. If $\mathfrak{N}$ has a primitive element α, then the function $\psi(x)$ irreducible in $\mathfrak{K}$ which has α as a zero is normal and equivalent to $\varphi(x)$. If $\mathfrak{N}$ has no primitive element, then $\mathfrak{K}$ must be an imperfect field and $\varphi(x)$ must have an exponent $f > 0$ (§ 13, 7.). Let

$$\varphi(x) = x^{np^f} + a_1 x^{(n-1)p^f} + \cdots + a_n.$$

Then, in the field $\mathfrak{N}$ in which it decomposes, the function $\varphi(x)$ is the p^f-th power of another function

$$\chi(x) = (x - \gamma_1) \cdots (x - \gamma_n) = x^n + b_1 x^{n-1} + \cdots + b_n,$$

and from $\varphi(x) = (\chi(x))^{p^f}$ it follows that

$$a_k = b_k^{p^f} \qquad (k = 1, \ldots, n).$$

The function $\varphi(x)$ has the zeros $\gamma_1, \gamma_2, \ldots, \gamma_n$, while the function

$$\overline{\varphi}(x) = x^n + a_1 x^{n-1} + \cdots + a_n,$$

which is an irreducible function of the first kind, has the zeros $\gamma_1^{p^f}, \gamma_2^{p^f}, \ldots, \gamma_n^{p^f}$. These are distinct from one another, and therefore the same is true of $\gamma_1, \gamma_2, \ldots, \gamma_n$. From $a_k = b_k^{p^f}$ it follows that $x^{p^f} - a_k = (x - b_k)^{p^f}$. Therefore the irreducible function from $\mathfrak{K}$ with zero b_k has the form

$$(x - b_k)^{p^{t_k}} \qquad (0 \leq t_k \leq f;\ k = 1, \ldots, n)$$

and is a normal function. The field $\mathfrak{N}' = \mathfrak{K}(\gamma_1^{p^f}, \gamma_2^{p^f}, \ldots, \gamma_n^{p^f})$, which just suffices for the complete decomposition of $\overline{\varphi}(x)$, is normal, is contained in $\mathfrak{N}$, and is a finite extension of the first kind of $\mathfrak{K}$. Hence there are primitive elements in $\mathfrak{N}'$. Let α be one such element, and let $\overline{\psi}(x)$ be the irreducible function from $\mathfrak{K}$ with zero α. Then $\overline{\psi}(x)$ is a normal function equivalent to $\overline{\varphi}(x)$. The system T of normal functions

$$\overline{\psi}(x), \quad (x - b_1)^{p^{t_1}}, \quad (x - b_2)^{p^{t_2}}, \ldots, \quad (x - b_n)^{p^{t_n}}$$

passes to an equivalent system T' if $\overline{\psi}(x)$ is replaced by $\overline{\varphi}(x)$. All these functions decompose completely in $\mathfrak{N}$, and hence are subordinate to the function $\varphi(x)$. A field $\mathfrak{M}$ between $\mathfrak{K}$ and $\mathfrak{N}$ which suffices for the complete decomposition of the system T' must contain the elements $b_1, b_2, \dots, b_n$, and hence the function $\chi(x)$; it must also contain the zeros $\gamma_k^{p^f}$ of $\overline{\varphi}(x)$ and therefore the functions $x^{p^f} - \gamma_k^{p^f} = (x - \gamma_k)^{p^f}$. Hence it also contains the functions

$$(\chi(x), (x - \gamma_k)^{p^f}) = x - \gamma_k$$

and therefore the elements γ_k as well. Since $\mathfrak{N} = \mathfrak{K}(\gamma_1, \gamma_2, \dots, \gamma_n)$, it follows that $\mathfrak{M} = \mathfrak{N}$; from this follows the equivalence of $\varphi(x)$ and T', and finally the equivalence of $\varphi(x)$ and the normal-function system T.

This proves Proposition 6 for the case where S consists of a single irreducible function. Now let S be an arbitrary function system; let S' be the system of all irreducible factors of the functions occurring in S, and let S'' be the system of all normal functions subordinate to S'. Then first S is equivalent to S', and S'' is subordinate to the system S'. Since, furthermore, every function from S' is, by what precedes, equivalent to a system of normal functions, and since this is a subsystem of S'', it follows that S' and S'' are equivalent; and S is equivalent to the normal-function system S''.

§ 19.
Auxiliary considerations from set theory.

By Proposition 5 of the preceding paragraph, every normal extension of a field $\mathfrak{K}$ corresponds to a class of equivalent function systems for whose decomposition the extension just suffices; and, as is immediately seen, the same system class corresponds to every equivalent extension. To prove conversely that to every system class there corresponds one, and, apart from equivalent extensions, only one, normal extension, it becomes necessary to bring in considerations from set theory; in particular, the proof cannot be carried out completely without using the principle of choice. We therefore give here a brief compilation of some concepts and propositions from that discipline which will be needed in what follows.

1. Ordered sets. — A set, or a system of elements, is called *ordered*, according to G. Cantor, if a stipulation has been made by which, of any two distinct elements a, b, one, say a, counts as the earlier (smaller), and the other b as the later (larger); in symbols:

$$a < b \quad \text{or} \quad b > a.$$

The stipulation must, however, be such that from $a < b$, $b < c$, it always follows that $a < c$.

2. Cut. — If the elements of an ordered set M are divided into two subsets A, B in such a way that every element of M belongs to one and only one of these subsets, and that all elements of A are earlier than all elements of B, such a division is called a *cut* (Dedekind); the subset A is called the *initial segment*, and B the *remainder* (Cantor).

3. Order types. — By a *similar mapping* of two ordered sets M and N one understands a one-to-one relation between their elements in which, whenever one of two elements of M is the earlier, the corresponding one of the two assigned elements in N is again the earlier. Ordered sets which admit such a mapping are called *similar*, or *of the same order type* (Cantor).

4. Well-ordered sets. — A set M is called *well ordered* (Cantor) if a stipulation has been made according to which, of any two elements a, b, one counts as the earlier and the other as the later, and if this stipulation is such that both M and every subset of M contain a *first* element, that is, one which is earlier than every other element of the set or subset. — *Every well-ordered* *set is also ordered;* for if $a < b$ and $b < c$, then it cannot be that $a = c$, since the cases $a < b$ and $b < a$ exclude one another; nor can it be that $c < a$, for otherwise the set consisting of the elements a, b, c would contain no first element; hence $a < c$. — The ordered finite sets also belong to the well-ordered sets; they are characterized by the fact that every subset of them also has a last element.

5. Ordinal numbers. — The order types of well-ordered sets are called *ordinal numbers*; the ordinal numbers of finite sets are denoted by $1, 2, 3, \ldots$. — *If M and N are well-ordered sets, then one of the following three mutually exclusive cases always occurs:*

a) M and N are similar; then there is one and only one similar relation between M and N;

b) M is similar to a definite initial segment of N;

c) N is similar to a definite initial segment of M.

In the first case M and N have the same ordinal number; in the second, the ordinal number of M is called smaller, and in the third, larger, than that of N. It is useful also to introduce an ordinal number 0, which counts as the smallest of all.

Every system of ordinal numbers, when they are ordered by increasing size, forms a well-ordered set; in particular, all ordinal numbers, including 0, which are smaller than an ordinal number σ ($\sigma > 0$), form a well-ordered set to which the ordinal number σ itself belongs. Thus if one has a well-ordered set of ordinal number σ, it is convenient to represent its elements by a symbol with an index:

$$a_0, a_1, a_2, \ldots, a_\nu, \ldots,$$

where the index ν runs through all ordinal numbers $0 \leq \nu < \sigma$. Each ordinal number ν (for $0 < \nu < \sigma$) then corresponds to a definite initial segment of the well-ordered system, which has precisely the ordinal number ν. It is formed by all elements preceding a_ν, and is therefore also called the *initial segment determined by a_ν*. — For simplicity of notation we make the following conventions: a well-ordered system of elements a_ν is denoted by

$$\{a_\nu\} \quad \text{or, more precisely,} \quad \{a_\nu\} \quad (0 \leq \nu < \sigma),$$

where σ indicates the ordinal number of the system, or else by a single letter, say A. In the last case A_ν ($0 < \nu < \sigma$) is always to be understood as the initial segment of A determined by a_ν, while $A_\sigma = A$ is to be put.

Among the ordinal numbers distinct from 0, those of the first and second kind are to be distinguished. An ordinal number α is said to be of the first or of the second kind according as a well-ordered set of ordinal number α does or does not contain a last element. Since the system of all ordinal numbers below α itself has ordinal number α, if α is of the first kind, then there is an ordinal number β which immediately precedes α. We put $\beta = \alpha - 1$, $\alpha = \beta + 1$. If α is of the second kind, then there is no immediately preceding ordinal number. These ordinal numbers are also called limit numbers. The reason for this designation is clear from the following consideration. Let S be a system of ordinal numbers with the property that there is no greatest among them, so that each is exceeded by other numbers of the same system (for example, the system of all finite ordinal numbers, or of all even finite ordinal numbers). Let T further denote the system of all those ordinal numbers which are exceeded by some ordinal number from S. It is then evident that S is contained in T; that T too contains no greatest number; and that if σ is any ordinal number from T, then every ordinal number $< \sigma$ also belongs to T. The system T itself has an ordinal number τ, which is a limit number. If now $\sigma > 0$ is any element of T, then the ordinal numbers which are $< \sigma$ form an initial segment of T whose ordinal number is smaller than the ordinal number of T. That is, $\sigma < \tau$. Conversely, every ordinal number $< \tau$ must belong to T. For if the system T were only a part of the system T' of all ordinal numbers $< \tau$, then the numbers belonging to T in T' would precede those not belonging to T; that is, T would be an initial segment of T', and the ordinal number τ of T would be smaller than that of

T', whereas T', as the system of all ordinal numbers $< \tau$, must have the ordinal number τ. Thus T consists of all and only the ordinal numbers $< \tau$, and by the definition of T, τ now represents the smallest ordinal number which is not exceeded by any ordinal number of the system S. Such a number therefore exists for every system of ordinal numbers which has no maximum, and by analogy with ordinary usage it would be called the upper bound of S. Every limit number τ is, among other things, the upper bound of all ordinal numbers $< \tau$.

6. Complete induction and the well-ordering theorem. — If a certain assertion concerning ordinal numbers holds for an ordinal number α, does not hold for another $\beta > \alpha$, and T denotes the system of those ordinal numbers between α and β (including α and β) for which the assertion does not hold, then T, as a well-ordered set, has a first element γ; and γ is then the smallest ordinal number $> \alpha$ for which the assertion does not hold. Thus in the domain of ordinal numbers there is the *principle of complete induction*: *if a proposition holds for the ordinal number α, and one can prove that it must hold for the ordinal number $\beta > \alpha$ whenever it is correct for all ordinal numbers which are $< \beta$ and $\geq \alpha$, then the proposition holds for all ordinal numbers $\geq \alpha$.* The importance of ordinal numbers for applications rests on the principle of complete induction. *Their extensive applicability, however, is opened up by Zermelo's theorem,*[2] *according to which every set is capable of being well ordered.* The theorem concerns, of course, only the existence of the well-ordering; it gives no method for actually carrying it out for an arbitrarily given set.

[2]E. Zermelo, Math. Ann. vol. 59 (1904), p. 514.

§ 20. The Next Applications.

Definition 1. Progressive systems of functions. – Let $\mathfrak{K}_0$ be an arbitrary field. A well-ordered system $S = \{f_\nu(x)\}$ of integral functions from $\mathfrak{K}_0$ shall be called *progressive* (with respect to $\mathfrak{K}_0$) if $f_0(x)$ does not split completely in $\mathfrak{K}_0$ and if, among the remaining functions $f_\nu(x)$, none is subordinated to the segment of S determined by it (§ 18).

1. *If S is a well-ordered system of integral functions from $\mathfrak{K}_0$, not all of which split completely in $\mathfrak{K}_0$, then S contains a progressive subsystem S' which is equivalent to S.*

Indeed, form a subsystem S' of S as follows: include the first function $f_0(x)$ in S' if it does not split completely in $\mathfrak{K}_0$, and include any other function if it is not subordinated to the segment of S determined by it. It is then clear, first, that the first function in S which does not split completely in $\mathfrak{K}_0$ belongs to S', and also that S' is progressive and subordinate to S. It remains only to show that every function in $S = \{f_\nu(x)\}$ is subordinate to S'. This is immediate for $f_0(x)$, since $f_0(x)$ either splits completely in $\mathfrak{K}_0$ or belongs to S'. To prove the same for every other function $f_\nu(x)$, we apply complete induction. We may then regard it as proved that the segment of S determined by $f_\nu(x)$ is subordinate to S'. If $f_\nu(x)$ is subordinate to this segment, then $f_\nu(x)$ is also subordinate to S'; but if $f_\nu(x)$ is not subordinate to the segment just indicated, then $f_\nu(x)$ belongs to S', and is therefore again subordinate to that system.

Definition 2. Progressive systems of elements of an extension field. – Let $\mathfrak{K}_0$ be an arbitrary field, Ω any extension of $\mathfrak{K}_0$, and $\mathfrak{S} = \{a_\nu\}$ a well-ordered system of elements of Ω. The system $\mathfrak{S}$ shall be called *progressive* if neither a_0 belongs to the field $\mathfrak{K}_0$, nor does any of the remaining elements a_ν belong to the field obtained from $\mathfrak{K}_0$ by adjoining the segment determined by a_ν. – The following theorem holds:

2. *If $\mathfrak{S} = \{a_\nu\}$ is a well-ordered system of elements of an extension field Ω of $\mathfrak{K}_0$, and not all elements of $\mathfrak{S}$ belong to $\mathfrak{K}_0$, then $\mathfrak{S}$ contains a progressive subsystem $\mathfrak{S}'$ for which $\mathfrak{K}(\mathfrak{S}') = \mathfrak{K}(\mathfrak{S})$. One can form $\mathfrak{S}'$ by including the element a_0 in $\mathfrak{S}'$ if it does not belong to $\mathfrak{K}_0$, and including any other element a_ν if it does not belong to the field obtained from $\mathfrak{K}_0$ by adjoining the segment of $\mathfrak{S}$ determined by a_ν.*

The proof is entirely analogous to that of Theorem 1 and may therefore be omitted. Since Ω itself can be obtained by adjoining a system $\mathfrak{S}$, and since this system can be well-ordered, it also follows that:

3. *Every extension of a field can be obtained by adjoining a progressive system.*

Definition 3. Well-ordered fields. – We may suppose that the elements of a field $\mathfrak{K}_0$ have been put into a well-ordered sequence $\{c_\nu\}$ $(0 \le \nu < \sigma)$. Under this ordering the field $\mathfrak{K}_0$ is assigned a definite ordinal number σ. We shall, however, explicitly make the convention that, whenever a well-ordered field $\mathfrak{K}_0 = \{c_\nu\}$ is spoken of, one always has $c_0 = 0$, $c_1 = \varepsilon$. If two well-ordered fields have the same ordinal number, then there is between them a unique well-defined similar relation (§ 19, 5.). If that relation is also an isomorphism, we call the fields *similar-isomorphic.*

4. *Let $\{\mathfrak{K}_\mu\}$ $(0 \le \mu < \sigma)$ be a well-ordered sequence of well-ordered fields, let σ be a limit number, and suppose that, for any two fields in this sequence, the earlier one is always similar-isomorphic to a segment of the later one. Then there are well-ordered fields Ω with the following property: each field $\mathfrak{K}_\mu$ is similar-isomorphic to a segment of Ω, and for each element of Ω a segment can be specified which contains this element and is similar-isomorphic to some field $\mathfrak{K}_\mu$. All fields Ω having the indicated properties are similar-isomorphic to one another.*

Proof. Let the field $\mathfrak{K}_\mu$ of the sequence $\{\mathfrak{K}_\mu\}$ have ordinal number τ_μ, and let its elements in their well-ordered sequence be denoted by $c_{\mu\nu}$ $(0 \le \nu < \tau_\mu)$. From our hypotheses it follows that the numbers τ_μ increase with μ and that there is no largest among them. We denote by τ the upper bound of the numbers τ_μ (§ 19, 5.):

$$\tau = \lim_{\mu\to\sigma} \tau_\mu$$

and take a well-ordered system $\Omega = \{c_\lambda\}$ $(0 \le \lambda < \tau)$, for which we define addition and multiplication as follows. If c_λ, c_ρ $(\lambda \le \rho)$ are any two elements of Ω, then from $\rho < \tau$ it follows that among the ordinal numbers τ_μ of the fields $\mathfrak{K}_\mu$ there are some which are $> \rho$. Let $\tau_{\mu'}$ be the smallest of these. The field $\mathfrak{K}_{\mu'} = \{c_{\mu'\nu}\}$ $(0 \le \nu < \tau_{\mu'})$ then contains the two elements $c_{\mu'\lambda}$, $c_{\mu'\rho}$, as well as their sum and their product. If

$$c_{\mu'\lambda} + c_{\mu'\rho} = c_{\mu'\alpha}, \qquad c_{\mu'\lambda} \cdot c_{\mu'\rho} = c_{\mu'\beta}, \tag{1}$$

then $\alpha < \tau_{\mu'} < \tau$, $\beta < \tau_{\mu'} < \tau$; hence the system Ω contains two elements c_α, c_β. We stipulate that

$$c_\lambda + c_\rho = c_\rho + c_\lambda = c_\alpha, \qquad c_\lambda \cdot c_\rho = c_\rho \cdot c_\lambda = c_\beta \tag{2}$$

shall hold. This makes Ω into a system with double composition.

Now let $\mathfrak{K}_\mu$ be any field in the sequence $\{\mathfrak{K}_\mu\}$. Since $\tau_\mu < \tau$, Ω has a segment Ω_μ of ordinal number τ_μ. In the similar relation π_μ between $\mathfrak{K}_\mu$ and Ω_μ, the elements $c_{\mu\lambda}$ and c_λ $(0 \le \lambda < \tau_\mu)$ correspond to one another. Suppose now that c_λ and c_ρ $(\lambda \le \rho)$ are any two elements of Ω_μ, and that their sum and product are given by equations (2.). Then $\rho < \tau_\mu$, and hence, if $\tau_{\mu'}$ is again the smallest number in the sequence $\{\tau_\mu\}$ exceeding ρ, one has

$$\tau_{\mu'} \le \tau_\mu, \qquad \mu' \le \mu. \tag{3}$$

By the stipulations made for the compositions in Ω, equations (2.) entail equations (1.). From (3.) it follows that $\mathfrak{K}_{\mu'}$ is either identical with $\mathfrak{K}_\mu$ or similar-isomorphic to a segment of $\mathfrak{K}_\mu$. In both cases (1.) implies

$$c_{\mu\lambda} + c_{\mu\rho} = c_{\mu\alpha}, \qquad c_{\mu\lambda} \cdot c_{\mu\rho} = c_{\mu\beta}. \tag{4}$$

Since the elements $c_{\mu\alpha}$, $c_{\mu\beta}$ again belong to the field $\mathfrak{K}_\mu$, one has $\alpha < \tau_\mu$, $\beta < \tau_\mu$, and therefore the elements c_α, c_β again belong to the segment Ω_μ of Ω. Comparison of (2.) and (4.) also shows that the similar relation π_μ between $\mathfrak{K}_\mu$ and Ω_μ is an isomorphism. Thus Ω_μ represents a field similar-isomorphic to $\mathfrak{K}_\mu$. Since for every element c_λ of Ω one can choose $\tau_\mu > \lambda$, every element of Ω occurs in some field Ω_μ; and since of any two of these fields one contains the other, Ω itself is a field. As shown, it satisfies the conditions of Theorem 4.

Let $\Omega' = \{c'_\lambda\}$ be another field satisfying these conditions, let τ' be its ordinal number, and let Ω'_μ be the segment similar-isomorphic to $\mathfrak{K}_\mu$. Then the ordinal number of Ω'_μ is τ_μ, whence $\tau_\mu < \tau'$. Conversely, if λ is any ordinal number $< \tau'$, then Ω' contains an element c'_λ; this occurs in some segment Ω'_μ, so that $\lambda < \tau_\mu$. Hence $\tau' = \lim_{\mu\to\sigma} \tau_\mu = \tau$, and there is a similar relation π between Ω and Ω' which assigns to each element c_λ of Ω the element c'_λ of Ω'. If the composition equations (2.) hold for any two elements c_λ, c_ρ $(\lambda \le \rho)$ of Ω, and if μ is chosen so large that $\tau_\mu > \rho$, then the field Ω_μ contains the elements c_λ, c_ρ, and hence also c_α and c_β. The fields Ω_μ and Ω'_μ are similar-isomorphic to $\mathfrak{K}_\mu$, and therefore also to one another. Under the similar relation between Ω_μ and Ω'_μ, the elements c_λ, c_ρ, c_α, c_β of Ω_μ are assigned the elements c'_λ, c'_ρ, c'_α, c'_β of Ω'_μ. Since this relation is also an isomorphism, it follows that

$$c'_\lambda + c'_\rho = c'_\alpha, \qquad c'_\lambda c'_\rho = c'_\beta.$$

Thus the similar relation π between Ω and Ω' is itself an isomorphism, and this proves (4.) completely.

Definition 4. Well-ordering of the integral functions in a well-ordered field. – Let $\mathfrak{K}_0 = \{c_\nu\}$ $(c_0 = 0, c_1 = \varepsilon)$ be a well-ordered field, and let $g(x)$ and $h(x)$ be two distinct integral functions from $\mathfrak{K}_0$. We write them in the form of infinite series

$$g(x) = c_{\mu_0} + c_{\mu_1} x + c_{\mu_2} x^2 + \cdots\,,$$

$$h(x) = c_{\nu_0} + c_{\nu_1}x + c_{\nu_2}x^2 + \cdots .$$

Then, once the natural number n exceeds a certain value,

$$c_{\mu_n} = c_{\nu_n} = 0 = c_0, \qquad \mu_n = \nu_n = 0$$

will hold. If k (≥ 0) is the largest number for which $\mu_k \neq \nu_k$, then in the case $\mu_k < \nu_k$ we say that the function $g(x)$ precedes the function $h(x)$ within $\mathfrak{K}_0$. This convention turns the system of integral functions from $\mathfrak{K}_0$ into a well-ordered system. Indeed, let S be any system of integral functions from $\mathfrak{K}_0$. Since $c_0 = 0$, the functions of least degree in S precede the others. Let n be this degree; let S_n be the subsystem of S consisting of the functions of degree n; let S_{n-1} be the system of functions in S_n in which the coefficient c_ν of x^n has the smallest possible index ν; let S_{n-2} be the system of those functions in S_{n-1} in which the coefficient of x^{n-1} has the smallest possible index; and so on. Finally, let $f(x)$ be the function in S_0 for which the constant term has the smallest possible index. Then $f(x)$ is the first function of S. The existence of a first function inside every system S of functions proves the asserted well-ordering.

Definition 5. Primitive well-ordering. – Let $\mathfrak{K}_0 = \{c_\nu\}$ be a well-ordered field, let $\mathfrak{K}_1$ be a simple algebraic extension of degree n ($n > 1$) of $\mathfrak{K}_0$, and let j be a primitive element of $\mathfrak{K}_1$. Every element of $\mathfrak{K}_1$ is then represented uniquely in the form $g(j)$, where $g(x)$ denotes any integral function from $\mathfrak{K}_0$ of degree $< n$. For any two elements $g(j)$ and $h(j)$, stipulate that $g(j)$ is the earlier one if the function $g(x)$ precedes the function $h(x)$ within $\mathfrak{K}_0$. By the preceding discussion it is clear that this makes $\mathfrak{K}_1$ into a well-ordered field. From $c_0 = 0$ it follows that the well-ordered field $\mathfrak{K}_0$ represents a segment of the well-ordered field $\mathfrak{K}_1$; from $c_1 = \varepsilon$ it follows that j is the first element following the segment $\mathfrak{K}_0$. We say: $\mathfrak{K}_1$ is *primitively ordered with respect to* $\mathfrak{K}_0$, and we call j the principal element of the primitive well-ordering. – Then the following theorem holds:

5. *If $\mathfrak{K}_1$ is a simple algebraic extension of a well-ordered field $\mathfrak{K}_0$, and j is a primitive element, then $\mathfrak{K}_1$ can be ordered primitively with respect to $\mathfrak{K}_0$ in one and only one way such that j becomes the principal element. Then $\mathfrak{K}_0$ is the segment of $\mathfrak{K}_1$ determined by j.*

Definition 6. Regular well-ordering. – Let $\mathfrak{K}_0$ be a well-ordered field and let Ω be an arbitrary algebraic extension of $\mathfrak{K}_0$. We shall assume that Ω is also well-ordered, and that this well-ordering has the following properties. The field $\mathfrak{K}_0$, and also every field obtained from $\mathfrak{K}_0$ by adjoining a segment of Ω, shall be a segment of Ω whenever it does not contain all elements of Ω. The segments of Ω so obtained – plainly, all those segments containing $\mathfrak{K}_0$ which are at the same time fields – shall be called *principal segments*; the element immediately following a principal segment will be called the corresponding *principal element*. The principal elements form a well-ordered subset $\mathfrak{J} = \{j_\nu\}$ ($0 \leq \nu < \sigma$) of Ω. The field $\mathfrak{K}_0$ is the principal segment determined by j_0; in general we shall denote by $\mathfrak{K}_\nu$ ($0 \leq \nu < \sigma$) the principal segment determined by j_ν, and by $\mathfrak{K}_\sigma$ the field Ω itself. – From the preceding assumptions it follows that every field $\mathfrak{K}_\nu(j_\nu)$ ($0 \leq \nu < \sigma$) also represents a principal segment (or is equal to $\mathfrak{K}_\sigma = \Omega$). We now make the additional assumption that $\mathfrak{K}_\nu(j_\nu)$ is primitively ordered with respect to $\mathfrak{K}_\nu$. Thus j_ν appears as the principal element of the primitive ordering. – If the well-ordered field Ω has all these properties, we call it *regularly ordered with respect to* $\mathfrak{K}_0$. – The primitive well-ordering is therefore the simplest case of the regular one, characterized by $\sigma = 1$.

The preceding explanations immediately imply several facts, which we therefore record without proof:

6. *If the field $\Omega = \mathfrak{K}_\sigma$ is regularly ordered with respect to $\mathfrak{K}_0$, and $\mathfrak{K}_\nu$ ($0 \leq \nu < \sigma$) is a principal segment, then both $\mathfrak{K}_\nu$ with respect to $\mathfrak{K}_0$ and Ω with respect to $\mathfrak{K}_\nu$ are regularly ordered. If here $\mathfrak{J} = \{j_\mu\}$ ($0 \leq \mu < \sigma$) is the system of principal elements of Ω with respect to $\mathfrak{K}_0$, $\mathfrak{J}'$ is the segment of $\mathfrak{J}$ determined by j_ν, and $\mathfrak{J}''$ is the corresponding remainder, then $\mathfrak{J}'$ represents the system of principal elements of $\mathfrak{K}_\nu$ with respect to $\mathfrak{K}_0$, while $\mathfrak{J}''$ represents the system of principal elements of Ω with respect to $\mathfrak{K}_\nu$.*

7. *If Ω is regularly ordered with respect to $\mathfrak{K}_0$, and $\mathfrak{M}$ is regularly ordered with respect to Ω, then $\mathfrak{M}$ is regularly ordered with respect to $\mathfrak{K}_0$, and Ω is a principal segment of $\mathfrak{M}$.*

Keep the notation of Definition 6, and let Ω' denote any field between $\mathfrak{K}_0$ and Ω. Suppose that Ω' is different from Ω, let a be the first element of Ω not belonging to Ω', and let $\mathfrak{A}$ be the segment of Ω determined by a. Then Ω' contains the field $\mathfrak{K}_0(\mathfrak{A})$, which by Definition 6 is a principal segment of Ω. Since $\mathfrak{K}_0(\mathfrak{A})$ contains the segment $\mathfrak{A}$ but not the element a following it, a must be a principal element. Hence, if Ω' is to contain all principal elements, then one must have $\Omega' = \Omega$, and it follows that $\mathfrak{K}_0(\mathfrak{J}) = \Omega$. Moreover, let $\mathfrak{J}_\nu$ $(0 < \nu < \sigma)$ denote the segment of $\mathfrak{J}$ determined by the principal element j_ν, and put $\Omega' = \mathfrak{K}_0(\mathfrak{J}_\nu)$. Then Ω' is part of $\mathfrak{K}_\nu$, since $\mathfrak{K}_\nu$ contains the system $\mathfrak{J}_\nu$. Since, moreover, j_ν does not occur in $\mathfrak{K}_\nu$ and therefore also not in Ω', and since the first element a of Ω not contained in Ω' must be a principal element, it follows that $a = j_\nu$ and $\Omega' = \mathfrak{K}_\nu = \mathfrak{K}_0(\mathfrak{J}_\nu)$. Taking Definition 2 into account, we thus obtain:

8. *If the field Ω is regularly ordered with respect to $\mathfrak{K}_0$, then the principal elements form a progressive system $\mathfrak{J}$, one has $\Omega = \mathfrak{K}_0(\mathfrak{J})$, and the principal segment of Ω determined by the principal element j_ν is obtained from $\mathfrak{K}_0$ by adjoining the segment of $\mathfrak{J}$ determined by j_ν.*

Keeping the same notation, let a be any element of Ω, and let $\mathfrak{K}_\nu$ $(0 \leq \nu \leq \sigma)$ be the first principal segment of Ω containing a, or, if no such segment exists, the field $\Omega = \mathfrak{K}_\sigma$ itself. If $\nu > 0$, then from $\mathfrak{K}_\nu = \mathfrak{K}_0(\mathfrak{J}_\nu)$ it follows that there must be a finite subsystem $\mathfrak{S}$ of $\mathfrak{J}_\nu$ such that the element a already occurs in $\mathfrak{K}_0(\mathfrak{S})$. If j_μ is the last element of $\mathfrak{S}$, then $\mu < \nu$. Furthermore, there can be no ordinal number ρ for which $\mu < \rho < \nu$. Otherwise $\mathfrak{S}$ would be part of $\mathfrak{J}_\rho$, and a would be an element of $\mathfrak{K}_0(\mathfrak{J}_\rho) = \mathfrak{K}_\rho$, contrary to the assumption made about $\mathfrak{K}_\nu$. Hence $\nu = \mu + 1$, and we have the theorem:

9. *Let $\{\mathfrak{K}_\nu\}$ $(0 \leq \nu \leq \sigma)$ be the system of principal segments of the field $\Omega = \mathfrak{K}_\sigma$ regularly ordered with respect to $\mathfrak{K}_0$, and let $\mathfrak{K}_\nu$ be the first field in the sequence $\{\mathfrak{K}_\nu\}$ $(0 \leq \nu \leq \sigma)$ which contains an arbitrarily given element a of Ω. Then ν cannot be a limit number.*

From this it follows easily that:

10. *The regular well-ordering of a field Ω is completely determined by the well-ordering of the base field $\mathfrak{K}_0$ and by the well-ordered system $\mathfrak{J} = \mathfrak{J}_\sigma = \{j_\nu\}$ $(0 \leq \nu < \sigma)$ of principal elements.*

Indeed, suppose that we had two different regular well-orderings of Ω. Denote them by $\Omega' = \{c'_\lambda\}$ and $\Omega'' = \{c''_\lambda\}$, and denote by $\mathfrak{K}'_\nu$, $\mathfrak{K}''_\nu$ $(0 \leq \nu < \sigma)$ their principal segments corresponding to the segments $\mathfrak{J}_\nu$ of the system $\mathfrak{J}$; put $\Omega' = \mathfrak{K}'_\sigma$, $\Omega'' = \mathfrak{K}''_\sigma$. Then the fields $\mathfrak{K}'_\nu$, $\mathfrak{K}''_\nu$, if one disregards the order of their elements, are of course identical, namely equal to $\mathfrak{K}_0(\mathfrak{J}_\nu)$. But if the equality sign is also to express agreement in the ordering, then certainly $\mathfrak{K}'_\sigma \neq \mathfrak{K}''_\sigma$, whereas $\mathfrak{K}'_0 = \mathfrak{K}_0 = \mathfrak{K}''_0$. Let c'_λ be the first element of Ω' which no longer has the same index inside Ω'', and let $\mathfrak{K}'_\nu$ be the first field of the sequence $\{\mathfrak{K}'_\nu\}$ $(0 < \nu \leq \sigma)$ containing the element c'_λ. Then $\nu > 0$, and by 9. ν is not a limit number. We have $\mathfrak{K}'_\nu \neq \mathfrak{K}''_\nu$, but $\mathfrak{K}'_{\nu-1} = \mathfrak{K}''_{\nu-1}$. However, by the rules of regular well-ordering, $\mathfrak{K}'_\nu = \mathfrak{K}'_{\nu-1}(j_{\nu-1})$ must be primitively ordered with respect to $\mathfrak{K}'_{\nu-1}$, and $\mathfrak{K}''_\nu = \mathfrak{K}''_{\nu-1}(j_{\nu-1})$ must be primitively ordered with respect to $\mathfrak{K}''_{\nu-1} = \mathfrak{K}'_{\nu-1}$; in both cases $j_{\nu-1}$ is the principal element. It follows from 5. that $\mathfrak{K}'_\nu = \mathfrak{K}''_\nu$, and we obtain a contradiction. Thus 10. is proved.

11. *Let Ω be an algebraic extension of the well-ordered field $\mathfrak{K}_0$, let $\mathfrak{J} = \mathfrak{J}_\sigma = \{j_\nu\}$ $(0 \leq \nu < \sigma)$ be a progressive system of elements of Ω, and suppose that $\mathfrak{K}_0(\mathfrak{J}) = \Omega$. Then there is one and only one regular well-ordering of Ω with respect to $\mathfrak{K}_0$ having $\mathfrak{J}$ as its system of principal elements.*

The assertion that there cannot be more than one well-ordering of the indicated kind is precisely Theorem 10. That there is one at all is expressed, in the case $\sigma = 1$, by Theorem 5. For $\sigma > 1$ we prove this by induction. – If σ is not a limit number, first order the field $\mathfrak{K}_{\sigma-1} = \mathfrak{K}_0(\mathfrak{J}_{\sigma-1})$ regularly with respect to $\mathfrak{K}_0$, in such a way that $\mathfrak{J}_{\sigma-1}$ becomes the system of principal elements; then order $\Omega = \mathfrak{K}_0(\mathfrak{J}) = \mathfrak{K}_{\sigma-1}(j_{\sigma-1})$ primitively with respect to $\mathfrak{K}_{\sigma-1}$, with $j_{\sigma-1}$ as principal element. By 6. and 7., the field Ω so ordered satisfies the required conditions. –

If σ is a limit number, then first, for each $\nu < \sigma$, we have a field $\mathfrak{K}_\nu = \mathfrak{K}_0(\mathfrak{J}_\nu)$ which is regularly ordered with respect to $\mathfrak{K}_0$ and has $\mathfrak{J}_\nu$ as its system of principal elements. If $\mu < \nu < \sigma$, then the field $\mathfrak{K}_0(\mathfrak{J}_\mu)$ occurs as a principal segment of $\mathfrak{K}_\nu$; by 6. and 7. it is regularly ordered with respect to $\mathfrak{K}_0$ and has $\mathfrak{J}_\mu$ as its system of principal elements. By 10. the indicated segment also agrees with $\mathfrak{K}_\mu$ in the order of its elements. Thus, among the fields in the sequence $\{\mathfrak{K}_\nu\}$ $(0 \leq \nu < \sigma)$, each is a segment of all later ones. From this, and because the sequence $\{\mathfrak{K}_\nu\}$ contains no last field, it is immediately seen that these fields in their totality constitute a well-ordered field Ω' of which they are all segments. Since $\mathfrak{J}$ has no last element, every element of $\mathfrak{J}$ occurs in some segment $\mathfrak{J}_\nu$, hence in some field $\mathfrak{K}_\nu$, and therefore in Ω'; hence Ω' contains all elements of $\mathfrak{K}_0(\mathfrak{J}) = \Omega$, and so gives a well-ordering of the field Ω. If $\mathfrak{A}$ is any segment of Ω', a the element of Ω' following this segment, and $\mathfrak{K}_\nu$ a field in the sequence $\{\mathfrak{K}_\nu\}$ containing a, then $\mathfrak{K}_0(\mathfrak{A})$ is a principal segment of $\mathfrak{K}_\nu$, hence a field $\mathfrak{K}_\mu$ $(\mu < \nu)$. Thus adjoining a segment of Ω' to $\mathfrak{K}_0$ always again yields a segment of Ω', and the segments so obtained are the fields in the sequence $\{\mathfrak{K}_\nu\}$ $(0 \leq \nu < \sigma)$. These therefore form the only principal segments of Ω', and the elements from $\mathfrak{J}$, since they follow these segments, are the principal elements. Since none of them is last, $\mathfrak{K}_\nu(j_\nu)$ is each time again a field in the sequence $\{\mathfrak{K}_\nu\}$ $(0 \leq \nu < \sigma)$, and as such is regularly ordered with respect to $\mathfrak{K}_0$, hence primitively ordered with respect to $\mathfrak{K}_\nu$. Thus Ω' is a regular well-ordering of Ω with respect to $\mathfrak{K}_0$, with $\mathfrak{J}$ as system of principal elements.

Together with 3., Theorem 11 gives:

12. *Every algebraic extension of a well-ordered field can be regularly ordered with respect to that field.*

§ 21. The Fundamental Theorem of Algebraic Extensions.

Definition 1. Principal functions and principal factors of the regular well-ordering. – Let $\mathfrak{N} = \mathfrak{K}_\sigma$ be an algebraic extension of the well-ordered field $\mathfrak{K}_0$, regularly ordered with respect to $\mathfrak{K}_0$, and let $\mathfrak{J} = \mathfrak{J}_\sigma = \{j_\nu\}$ $(0 \leq \nu < \sigma)$ be the system of principal elements; let $\mathfrak{K}_\nu$ and $\mathfrak{J}_\nu$ denote, respectively, the principal segment of $\mathfrak{N}$ and the segment of $\mathfrak{J}$ determined by j_ν. The element j_ν is a root of a certain irreducible function $f_\nu(x)$ over $\mathfrak{K}_0$ (with ε as coefficient of the highest power), and likewise a root of a certain irreducible function $\varphi_\nu(x)$ of the field $\mathfrak{K}_\nu$. This $\varphi_\nu(x)$ may be equal to $f_\nu(x)$, and for $\nu = 0$ is always equal to $f_\nu(x)$; in every case it is a factor of $f_\nu(x)$. We thus arrive at two systems of functions

$$F_\sigma = F = \{f_\nu(x)\}, \qquad \Phi_\sigma = \Phi = \{\varphi_\nu(x)\} \qquad (0 \leq \nu < \sigma).$$

We call the functions $f_\nu(x)$ the *principal functions*, and the functions $\varphi_\nu(x)$ the *principal factors* of the regular well-ordering, and denote by F_ν and Φ_ν the segments of F and Φ determined by $f_\nu(x)$ and $\varphi_\nu(x)$, respectively $(\nu \geq 1)$.[1] – With this terminology the following is immediate.

1. *If all principal functions are normal (with respect to $\mathfrak{K}_0$), then the regularly ordered field $\mathfrak{N} = \mathfrak{K}_\sigma$ is itself normal. It is just large enough to split completely the system F of principal functions; likewise each principal segment $\mathfrak{K}_\nu$ is normal and is just sufficient for the splitting of F_ν. The system F is progressive.*

Indeed, each of the normal functions $f_\rho(x)$ $(0 \leq \rho < \nu)$ from F_ν $(0 < \nu \leq \sigma)$ has in $\mathfrak{K}_\nu = \mathfrak{K}_0(\mathfrak{J}_\nu)$ a root j_ρ; hence it must split completely in $\mathfrak{K}_\nu$. Since the field $\mathfrak{K}_\nu$ arises from $\mathfrak{K}_0$ solely by adjoining roots of functions from F_ν, it is just sufficient for the splitting of F_ν, and is therefore normal. Moreover, if $\nu < \sigma$, the field $\mathfrak{K}_\nu$ no longer contains the element j_ν; thus $f_\nu(x)$ does not split completely in $\mathfrak{K}_\nu$, and so F is a progressive system.

Definition 2. Normal well-ordering. – Let $\mathfrak{N}$ be a normal extension of the well-ordered field $\mathfrak{K}_0$; let $\mathfrak{N}$ be regularly ordered with respect to $\mathfrak{K}_0$, and suppose that all principal functions are normal. Keeping the letters in the same meaning as before, j_ν $(0 \leq \nu < \sigma)$ is a root of the irreducible function $\varphi_\nu(x)$ over $\mathfrak{K}_\nu$, and $\varphi_\nu(x)$ is a factor of the function $f_\nu(x)$ irreducible over $\mathfrak{K}_0$. Since $\mathfrak{K}_\nu$ is a well-ordered field, Def. 4 of § 20 assigns a definite order also to the functions inside $\mathfrak{K}_\nu$. This therefore also applies to the irreducible factors that $f_\nu(x)$ has inside $\mathfrak{K}_\nu$. We shall now assume that every function $\varphi_\nu(x)$, unless it is the sole irreducible factor of $f_\nu(x)$ inside $\mathfrak{K}_\nu$, precedes the remaining irreducible factors of $f_\nu(x)$ inside $\mathfrak{K}_\nu$. If this condition is fulfilled, we say that $\mathfrak{N}$ is *normally ordered with respect to* $\mathfrak{K}_0$. Thus normal well-ordering is a special case of regular well-ordering.

Definition 3. Distinguished roots of the principal functions belonging to a normal well-ordering. – Let $\mathfrak{N}$ be a normal and normally ordered extension of the well-ordered field $\mathfrak{K}_0$, and let the symbols used so far keep their meanings. Since the principal function $f_\nu(x)$ is normal with respect to $\mathfrak{K}_0$, the field $\mathfrak{K}_\nu(j_\nu)$ can be obtained by adjoining to $\mathfrak{K}_\nu$ any root of $f_\nu(x)$. All these roots are primitive elements of the field $\mathfrak{K}_\nu(j_\nu)$ over $\mathfrak{K}_\nu$; their degree is equal to the degree of the principal factor $\varphi_\nu(x)$, which in any case is greater than 1. In the case of imperfect fields the roots of $\varphi_\nu(x)$ need not all be distinct. Suppose therefore that $\varphi_\nu(x)$ has r distinct roots $j_\nu, j'_\nu, \ldots, j_\nu^{(r-1)}$ and has degree n; then $n \geq r \geq 1$, $n > 1$. If now i_ν is any root of $f_\nu(x)$ (in the field $\mathfrak{N}$), then we obtain all elements of the field $\mathfrak{K}_\nu(i_\nu)$ by replacing x by i_ν in all integral functions $g(x)$ over $\mathfrak{K}_\nu$ whose degree is smaller than n. Among these functions there are r which, for $x = i_\nu$, become roots of $\varphi_\nu(x)$. By Def. 4 of § 20 these r functions have a definite order within $\mathfrak{K}_\nu$. If $\psi(x)$ is the first among them and if $\psi(i_\nu) = j_\nu$, then we call i_ν a *distinguished root* of $f_\nu(x)$. (We also use this designation when $r = 1$, which can occur only for imperfect fields.) The following holds.

[1] To avoid misunderstandings, we add that within F the functions $f_\nu(x)$ have a definite order, indicated by the index. This order of course has nothing to do with the order assigned to functions in $\mathfrak{K}_0$ by § 20, Def. 4.

2. *The principal element j_ν is a distinguished root of $f_\nu(x)$, and among the roots of the principal factor $\varphi_\nu(x)$ there is no other distinguished one.*

Indeed, since the roots $j_\nu, j'_\nu, \ldots, j_\nu^{(r-1)}$ of $\varphi_\nu(x)$ do not occur in $\mathfrak{K}_\nu$, the r functions over $\mathfrak{K}_\nu$ that for $x = i_\nu$ become these roots all have degree at least 1. From the fact that the well-ordered field $\mathfrak{K}_\nu$ begins with the elements $0, \varepsilon$, it follows that among all functions over $\mathfrak{K}_\nu$ of degree ≥ 1, the first is x. Thus, if for i_ν we choose one of the roots of $\varphi_\nu(x)$, then x is the first function over $\mathfrak{K}_\nu$ which for $x = i_\nu$ becomes a root of $\varphi_\nu(x)$. But this root is equal to j_ν if and only if we choose $i_\nu = j_\nu$, proving the assertion. – From 2. one further obtains:

3. *In the case $\varphi_\nu(x) = f_\nu(x)$, hence in particular for $\nu = 0$, j_ν is the unique distinguished root of $f_\nu(x)$.*

We now prove:

4. *Let $\mathfrak{N} = \mathfrak{K}_0(\overline{\mathfrak{J}})$ be a normal extension of the well-ordered field $\mathfrak{K}_0$. Suppose every element i_ν of the system $\mathfrak{J}_\sigma = \overline{\mathfrak{J}} = \{i_\nu\}$ $(0 \leq \nu < \sigma)$ is a root of a normal function $f_\nu(x)$ over $\mathfrak{K}_0$, and suppose the function system $F_\sigma = F = \{f_\nu(x)\}$ is progressive (or, what here amounts to the same thing, suppose the system $\overline{\mathfrak{J}}$ is progressive). Then the field $\mathfrak{N}$ can be normally ordered with respect to $\mathfrak{K}_0$ in one and only one way so that F is the system of principal functions and the elements i_ν become distinguished roots of the corresponding principal functions $f_\nu(x)$.*

We first show that two distinct normal orderings of the indicated kind cannot exist. Assume that there are two such normal orderings $\mathfrak{N}'$ and $\mathfrak{N}''$ of the field $\mathfrak{N}$. Denote by $\mathfrak{J}'_\sigma = \mathfrak{J}' = \{j'_\nu\}$, $\{\mathfrak{K}'_\nu\}$, $\Phi'_\sigma = \Phi' = \{\varphi'_\nu(x)\}$ the systems of principal elements, principal segments, and principal factors of $\mathfrak{N}'$, and by $\mathfrak{J}''_\sigma = \mathfrak{J}'' = \{j''_\nu\}$, $\{\mathfrak{K}''_\nu\}$, $\Phi''_\sigma = \Phi'' = \{\varphi''_\nu(x)\}$ the corresponding systems for $\mathfrak{N}''$. Let $\overline{\mathfrak{J}}_\nu$, F_ν, $\mathfrak{J}'_\nu$, $\mathfrak{J}''_\nu$, Φ'_ν, Φ''_ν denote the segments of $\overline{\mathfrak{J}}$, F, and so on determined by i_ν, $f_\nu(x)$, and so forth. Then $\mathfrak{K}'_\nu$ and $\mathfrak{K}''_\nu$ are two normal orderings of the same field $\mathfrak{K}_0(\overline{\mathfrak{J}}_\nu) = \mathfrak{K}_0(\mathfrak{J}'_\nu) = \mathfrak{K}_0(\mathfrak{J}''_\nu)$, which is just sufficient for splitting the system F_ν. Since i_0 is to be a distinguished element of $f_0(x)$, it follows by 3. that $j'_0 = i_0 = j''_0$. If the equality $j'_\nu = j''_\nu$ does not hold for every index, take the least one for which $j'_\nu \neq j''_\nu$. Then $\mathfrak{J}'_\nu = \mathfrak{J}''_\nu$, and since $\mathfrak{J}'_\nu$ and $\mathfrak{J}''_\nu$ are the systems of principal elements of the fields $\mathfrak{K}'_\nu$, $\mathfrak{K}''_\nu$ regularly ordered with respect to $\mathfrak{K}_0$, we have (also taking order into account) $\mathfrak{K}'_\nu = \mathfrak{K}''_\nu$ (§ 20, 10.). It follows further that $\varphi'_\nu(x) = \varphi''_\nu(x)$, because $\varphi'_\nu(x)$ and $\varphi''_\nu(x)$ are the first irreducible factor of $f_\nu(x)$ inside $\mathfrak{K}'_\nu = \mathfrak{K}''_\nu$. If, furthermore, inside $\mathfrak{K}'_\nu = \mathfrak{K}''_\nu$, $\psi(x)$ is the first function that for $x = i_\nu$ becomes a root of $\varphi'_\nu(x) = \varphi''_\nu(x)$, then, since i_ν is to be a distinguished root of $f_\nu(x)$, we get $j'_\nu = \psi(i_\nu) = j''_\nu$, contrary to the assumption $j'_\nu \neq j''_\nu$. Hence $\mathfrak{J}' = \mathfrak{J}''$, and from § 20, 10. it follows that $\mathfrak{N}' = \mathfrak{N}''$.

It remains to show that a normal ordering of the required kind exists. – For $\sigma = 1$ we obtain it by ordering $\mathfrak{N}$ primitively with respect to $\mathfrak{K}_0$, with i_0 as principal element. For $\sigma > 1$ the proof is by induction. If σ is not a limit number, let $\mathfrak{K}_{\sigma-1}$ denote the normal ordering of the field $\mathfrak{K}_0(\overline{\mathfrak{J}}_{\sigma-1})$ for which $F_{\sigma-1}$ is the system of principal functions and every element i_ν $(\nu < \sigma - 1)$ is a distinguished root of $f_\nu(x)$. If then, within the well-ordered field $\mathfrak{K}_{\sigma-1}$, $\varphi_{\sigma-1}(x)$ is the first irreducible factor of $f_{\sigma-1}(x)$, and $\psi(x)$ is the first function that for $x = i_{\sigma-1}$ becomes a root of $\varphi_{\sigma-1}(x)$, put $\psi(i_{\sigma-1}) = j_{\sigma-1}$ and order $\mathfrak{N}$ primitively with respect to $\mathfrak{K}_{\sigma-1}$, with $j_{\sigma-1}$ as principal element. This plainly gives the required normal ordering. – If σ is a limit number, then for every $\nu < \sigma$ let $\mathfrak{K}_\nu$ be the normal ordering of the field $\mathfrak{K}_0(\overline{\mathfrak{J}}_\nu)$ for which F_ν is the system of principal functions, while every element i_μ $(\mu < \nu)$ is a distinguished root of $f_\mu(x)$. A simple consideration then shows that each field of the sequence $\{\mathfrak{K}_\nu\}$ $(0 \leq \nu < \sigma)$ is a segment of every later one, and that the totality of the elements occurring in these fields yields a well-ordering of $\mathfrak{N}$ satisfying all conditions of the problem.

5. *Let $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$ be similarly isomorphic fields. Let $\mathfrak{N} = \mathfrak{K}_\sigma$ be a normal and normally ordered extension of $\mathfrak{K}_0$, and $\overline{\mathfrak{N}} = \overline{\mathfrak{K}}_\sigma$ such an extension of $\overline{\mathfrak{K}}_0$. Let $F = \{f_\nu(x)\}$ and $\overline{F} = \{\overline{f}_\nu(x)\}$ $(0 \leq \nu < \sigma)$ be the corresponding systems of principal functions, and suppose that the similarly isomorphic correspondence π_0 between $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$ assigns the functions $f_\nu(x)$ and $\overline{f}_\nu(x)$ to one* *another. Then $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are similarly isomorphic.*

Proof by induction. If $\sigma = 1$, if n is the degree of $f_0(x)$ and $\overline{f}_0(x)$, and if j_0, $\overline{j}_0$ are the principal elements, then every element of $\mathfrak{N}$ or $\overline{\mathfrak{N}}$ is obtained by replacing x by j_0 or $\overline{j}_0$ in all functions $g(x)$ or $\overline{g}(x)$ over $\mathfrak{K}_0$ or $\overline{\mathfrak{K}}_0$ of degree smaller than n. The functions $g(x)$ in $\mathfrak{K}_0$ and the functions $\overline{g}(x)$ in $\overline{\mathfrak{K}}_0$ have definite orders, and since π_0 is a similar correspondence between $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$, it follows at once that the correspondence between the functions over $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$ induced by π_0 is also similar. Further, from § 20, Def. 5, we obtain a similar correspondence π_1 between $\mathfrak{N} = \mathfrak{K}_0(j_0)$ and $\overline{\mathfrak{N}} = \overline{\mathfrak{K}}_0(\overline{j}_0)$ extending π_0, by making the element $\overline{g}(\overline{j}_0)$ of $\overline{\mathfrak{N}}$ correspond to the element $g(j_0)$ of $\mathfrak{N}$, where $\overline{g}(x)$ denotes the function over $\overline{\mathfrak{K}}_0$ corresponding under π_0 to $g(x)$ over $\mathfrak{K}_0$. Since π_0 is moreover an isomorphism, it follows, as in § 8, 1., that π_1 is also an isomorphism. – If $\sigma > 1$ and if $\mathfrak{K}_\nu$, $\overline{\mathfrak{K}}_\nu$ $(\nu < \sigma)$ denote the principal segments of $\mathfrak{N}$ and $\overline{\mathfrak{N}}$, then $\mathfrak{K}_\nu$ and $\overline{\mathfrak{K}}_\nu$ are normal and normally ordered extensions of $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$ whose principal functions are carried into each other by π_0. We may therefore regard it as proved that $\mathfrak{K}_\nu$ and $\overline{\mathfrak{K}}_\nu$ are similarly isomorphic, i.e. that the unique similar correspondence π_ν between $\mathfrak{K}_\nu$ and $\overline{\mathfrak{K}}_\nu$ is also an isomorphism. If $\mu < \nu < \sigma$, then $\mathfrak{K}_\mu$ and $\overline{\mathfrak{K}}_\mu$ are corresponding segments of $\mathfrak{K}_\nu$ and $\overline{\mathfrak{K}}_\nu$, and hence are mapped similarly onto one another by π_ν, just as by π_μ. Thus π_μ is a part of π_ν. If σ is a limit number, so that every element of $\mathfrak{N}$ or $\overline{\mathfrak{N}}$ occurs in some field $\mathfrak{K}_\nu$ or $\overline{\mathfrak{K}}_\nu$, it follows that the correspondences π_ν determine a one-to-one mapping π_σ from $\mathfrak{N}$ onto $\overline{\mathfrak{N}}$. A simple consideration shows that π_σ is similar, because each π_ν is similar; likewise π_σ is an isomorphism, because the π_ν are isomorphisms. – If σ is not a limit number, then first there is the similarly isomorphic correspondence $\pi_{\sigma-1}$ between $\mathfrak{K}_{\sigma-1}$ and $\overline{\mathfrak{K}}_{\sigma-1}$ extending π_0, and it assigns the principal functions $f_{\sigma-1}(x)$ and $\overline{f}_{\sigma-1}(x)$ to one another. Moreover, $\pi_{\sigma-1}$ transforms the principal factor $\varphi_{\sigma-1}(x)$, the first factor of $f_{\sigma-1}(x)$ inside $\mathfrak{K}_{\sigma-1}$, into the first factor of $\overline{f}_{\sigma-1}(x)$ inside $\overline{\mathfrak{K}}_{\sigma-1}$, namely $\overline{\varphi}_{\sigma-1}(x)$. The fields $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are normal and normally ordered extensions of $\mathfrak{K}_{\sigma-1}$ and $\overline{\mathfrak{K}}_{\sigma-1}$ with only one principal function, $\varphi_{\sigma-1}(x)$ and $\overline{\varphi}_{\sigma-1}(x)$ respectively. Since $\mathfrak{K}_{\sigma-1}$ and $\overline{\mathfrak{K}}_{\sigma-1}$ are mapped similarly and isomorphically by $\pi_{\sigma-1}$ and $\pi_{\sigma-1}$ also carries the principal functions into one another, the existence of the similarly isomorphic correspondence between $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ follows as in the case $\sigma = 1$.

Letting in Theorem 5 $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$ coincide as one and the same well-ordered field, we obtain:

6. *If $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are normal and normally ordered extensions of the well-ordered field $\mathfrak{K}_0$ and the same system $\{f_\nu(x)\}$ of principal functions belongs to both, then $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are similarly isomorphic.*

7. *If $\mathfrak{K}_0$ is a well-ordered field and $F = F_\sigma = \{f_\nu(x)\}$ $(0 \leq \nu < \sigma)$ is a progressive system of normal functions over $\mathfrak{K}_0$, then there exist normal and normally ordered extensions of $\mathfrak{K}_0$ with F as their system of principal functions.*

Proof by induction. In the case $\sigma = 1$, by § 8, 1. we can determine an extension $\mathfrak{N}$ that is just sufficient for the splitting of $f_0(x)$. It is normal; and since $f_0(x)$ is normal, we have $\mathfrak{N} = \mathfrak{K}_0(j_0)$ if j_0 is any root of $f_0(x)$ in $\mathfrak{N}$. If we order $\mathfrak{N}$ primitively with respect to $\mathfrak{K}_0$ so that j_0 is principal element, then it is immediately clear that $\mathfrak{N}$ satisfies the conditions of the theorem. – If $\sigma > 1$ and is not a limit number, let $\mathfrak{K}_{\sigma-1}$ be a normal and normally ordered extension of $\mathfrak{K}_0$ with principal-function system $F_{\sigma-1} = \{f_\nu(x)\}$ $(0 \leq \nu < \sigma - 1)$. Among the irreducible factors that the normal function $f_{\sigma-1}(x)$ has in the well-ordered field $\mathfrak{K}_{\sigma-1}$, let $\varphi_{\sigma-1}(x)$ be the first. Its degree must be greater than 1, because $\mathfrak{K}_{\sigma-1}$ is just sufficient for the splitting of $F_{\sigma-1}$ (Theorem 1) and the normal-function system F is progressive. If one extends $\mathfrak{K}_{\sigma-1}$ by adjoining a root $j_{\sigma-1}$ of $\varphi_{\sigma-1}(x)$ to a field $\mathfrak{N}$ and orders this field primitively with respect to $\mathfrak{K}_{\sigma-1}$ with $j_{\sigma-1}$ as principal element, then $\mathfrak{N}$ plainly satisfies the conditions of Theorem 7. – If σ is a limit number, let $\mathfrak{K}'_\nu$ $(0 < \nu < \sigma)$ be a normal and normally ordered extension of $\mathfrak{K}_0$ with principal-function system $F_\nu = \{f_\lambda(x)\}$ $(0 \leq \lambda < \nu)$. If $\mu < \nu$, then $\mathfrak{K}'_\nu$ has a principal segment that is normal and normally ordered with respect to $\mathfrak{K}_0$ and has principal-function system F_μ. By 6 this segment must be similarly isomorphic to $\mathfrak{K}'_\mu$. Thus, among the fields of the sequence $\mathfrak{K}_0, \{\mathfrak{K}'_\nu\}$ $(0 < \nu < \sigma)$, each is similarly isomorphic to a segment of every later one. Hence (§ 20, 4.) there is a well-ordered field $\overline{\mathfrak{N}}$ such that $\mathfrak{K}_0$ is similarly isomorphic to a segment $\overline{\mathfrak{K}}_0$, each field $\mathfrak{K}'_\nu$ $(0 < \nu < \sigma)$ is similarly isomorphic to a segment $\overline{\mathfrak{K}}_\nu$ of $\overline{\mathfrak{N}}$, and every element of $\overline{\mathfrak{N}}$ occurs in some such segment $\overline{\mathfrak{K}}_\nu$. Since a similarly isomorphic correspondence exists between $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$, it follows that one obtains a field $\mathfrak{N}$ similarly isomorphic to $\overline{\mathfrak{N}}$, and at the same time an extension field of $\mathfrak{K}_0$, by replacing the segment $\overline{\mathfrak{K}}_0$ of $\overline{\mathfrak{N}}$ by the well-ordered system $\mathfrak{K}_0$ and requiring that every composition equation $(a + b = c,\ a \cdot b = d)$ in $\overline{\mathfrak{N}}$ become a composition equation in $\mathfrak{N}$, whenever the elements of $\overline{\mathfrak{K}}_0$ that may occur in it are substituted by the corresponding elements of $\mathfrak{K}_0$. By this substitution each segment $\overline{\mathfrak{K}}_\nu$ of $\overline{\mathfrak{N}}$ passes into a segment $\mathfrak{K}_\nu$ of $\mathfrak{N}$ similarly isomorphic to it, and hence also to $\mathfrak{K}'_\nu$, and every element of $\mathfrak{N}$ occurs in such segments $\mathfrak{K}_\nu$. Through the similarly isomorphic correspondence π_ν between $\mathfrak{K}'_\nu$ and $\mathfrak{K}_\nu$, all order relations and equations of one field are carried into correct order relations and equations of the other. Since π_ν does not alter the elements of $\mathfrak{K}_0$, nor therefore the functions of the system F, $\mathfrak{K}_\nu$, like $\mathfrak{K}'_\nu$, is a normal and normally ordered extension of $\mathfrak{K}_0$ with F_ν as principal-function system. From the fact that the field $\mathfrak{N}$ is composed of the elements of the field sequence $\{\mathfrak{K}_\nu\}$, and that each field of this sequence is a segment of every later one and a segment of $\mathfrak{N}$, it is immediately clear that $\mathfrak{N}$ itself is normally ordered with respect to $\mathfrak{K}_0$, and that F is the corresponding system of principal functions.

From what has gone before one easily obtains the fundamental theorem:

8. *If $\mathfrak{K}_0$ is an arbitrary field and S is an arbitrary system of integral functions over $\mathfrak{K}_0$, then there is an extension of $\mathfrak{K}_0$, unique up to equivalence, which is just large enough to split all functions of the system S into linear factors.*

Proof. For the question considered here, the system S may be replaced by any equivalent system (§ 18, 3.); hence it may be assumed that S contains only normal functions (§ 18, 6.). Moreover we may regard S as well-ordered and, if it is not progressive, replace it by an equivalent progressive subsystem (§ 20, 1.). Thus from the outset we may assume that a progressive system of normal functions $F = \{f_\nu(x)\}$ is given. It is further permissible to suppose that the field $\mathfrak{K}_0$ is well-ordered. But then by 7. there is a normal and normally ordered extension $\mathfrak{N}$ of $\mathfrak{K}_0$ with F as principal-function system, and by 1. this extension is just sufficient for the complete splitting of F. – Now suppose that there is a second extension $\overline{\mathfrak{N}}$ of $\mathfrak{K}_0$ which is just sufficient for the complete splitting of F. We must prove the equivalence of the extensions $\mathfrak{N}$ and $\overline{\mathfrak{N}}$. For this purpose we use the principle of choice. Namely, from the roots that the functions $f_\nu(x)$ have in $\overline{\mathfrak{N}}$, select one for each ν, so that we obtain a system $\overline{\mathfrak{J}} = \{i_\nu\}$ where i_ν is a root of $f_\nu(x)$. Since these are normal functions, $\overline{\mathfrak{N}} = \mathfrak{K}_0(\overline{\mathfrak{J}})$. By 4. $\overline{\mathfrak{N}}$ can be normally ordered with respect to $\mathfrak{K}_0$ so that F becomes the system of principal functions (and so that the elements i_ν become distinguished roots of the principal functions. This latter point is now of no further importance, but without it Theorem 4 could not have been proved). Thus $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are normal and normally ordered extensions of $\mathfrak{K}_0$ with the same system of principal functions, and there exists between them a similarly isomorphic correspondence π (Theorem 6). Since under this correspondence every element of the common segment $\mathfrak{K}_0$ is assigned to itself, the extensions $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are equivalent.

By specialization, 8. further gives:

9. *Every field can be extended algebraically to an algebraically closed field, and essentially in only one way. – In this way to every arbitrary field type there corresponds a definite closed one.*

Indeed, if by S we understand the system of all integral functions over $\mathfrak{K}_0$, then an extension field of $\mathfrak{K}_0$ is algebraically closed and algebraic over $\mathfrak{K}_0$ if and only if it is just sufficient for the splitting of S. Hence the first part of 9. follows from 8. – Let any field type be given; let $\mathfrak{K}_0$ and $\mathfrak{K}'_0$ be two fields of this type, and let Ω and Ω' be algebraic and algebraically closed extensions of $\mathfrak{K}_0$ and $\mathfrak{K}'_0$. We must show that Ω and Ω' are of the same type. To this end introduce a system $\mathfrak{S}$ of new elements, which we put in one-to-one correspondence with the elements of Ω' that do not belong to $\mathfrak{K}_0'$. This assignment, together with an isomorphism π_0 between $\mathfrak{K}_0$ and $\mathfrak{K}_0'$, gives a mapping π_1 between Ω' and $\overline{\Omega} = \mathfrak{K}_0 + \mathfrak{S}$. If we require that under π_1 every composition equation $a' + b' = c'$, $a' \cdot b' = d'$ from Ω', even when a' and b' do not both belong to $\mathfrak{K}_0'$, shall pass into a composition equation in $\overline{\Omega}$, then $\overline{\Omega}$ becomes a field isomorphic to Ω', hence algebraically closed, and is an algebraic extension of $\mathfrak{K}_0$. Therefore, by the first part of 9. already proved, Ω and $\overline{\Omega}$ are equivalent extensions of $\mathfrak{K}_0$, and we have an isomorphic correspondence $\overline{\pi}_1$ between Ω and $\overline{\Omega}$ that leaves the elements of $\mathfrak{K}_0$ unchanged. The desired isomorphic correspondence between Ω and Ω' is obtained from $\overline{\pi}_1$ and π_1. It is an extension of the isomorphism π_0 existing between $\mathfrak{K}_0$ and $\mathfrak{K}_0'$.

§ 22. Transcendental Extensions. Algebraic Dependence Among Elements of an Extension Field.

Every non-algebraic extension of a field will be called transcendental. – In the following investigations, which are meant to lead us to a clear classification of all fields, we fix a field $\mathfrak{K}$ and refer to it expressions such as “extension field,” “transcendental,” and so on, unless something else is expressly stated. If $\mathfrak{S}$ is a system of elements of an extension field Ω, and a is an element of Ω, we call *a algebraically dependent* on $\mathfrak{S}$ (with respect to $\mathfrak{K}$) if a is algebraic over the field $\mathfrak{K}(\mathfrak{S})$. We call a system of elements $\mathfrak{S}'$ algebraically dependent on $\mathfrak{S}$ if every element of $\mathfrak{S}'$ is algebraically dependent on $\mathfrak{S}$.

1. *If the element a is algebraically dependent on the system $\mathfrak{S}$, then there is a finite subsystem $\mathfrak{S}'$ of $\mathfrak{S}$ on which a is algebraically dependent.*

For our assumption implies that a is a root of an integral function $\varphi(x)$ of the field $\mathfrak{K}(\mathfrak{S})$. Since $\varphi(x)$ has only finitely many coefficients, there is a finite subsystem $\mathfrak{S}'$ of $\mathfrak{S}$ such that these coefficients already occur in $\mathfrak{K}(\mathfrak{S}')$. Then a is algebraically dependent on $\mathfrak{S}'$.

2. *If $\mathfrak{S}_3$ is algebraically dependent on $\mathfrak{S}_2$, and $\mathfrak{S}_2$ on $\mathfrak{S}_1$, then $\mathfrak{S}_3$ is algebraically dependent on $\mathfrak{S}_1$.*

Indeed, put $\mathfrak{K}(\mathfrak{S}_1) = \mathfrak{K}_1$, $\mathfrak{K}(\mathfrak{S}_2) = \mathfrak{K}_2$. It follows from our assumption that $\mathfrak{K}_1(\mathfrak{S}_2)$ is algebraic over $\mathfrak{K}_1$, and $\mathfrak{K}_2(\mathfrak{S}_3)$ is algebraic over $\mathfrak{K}_2$ (§ 7, 7.). Since $\mathfrak{K}_2$ is a subfield of $\mathfrak{K}_1(\mathfrak{S}_2)$, the elements of $\mathfrak{K}_2(\mathfrak{S}_3)$ are algebraic over $\mathfrak{K}_1(\mathfrak{S}_2)$, hence also over $\mathfrak{K}_1 = \mathfrak{K}(\mathfrak{S}_1)$ (§ 7, 9.). Therefore the elements of $\mathfrak{S}_3$ are algebraic over $\mathfrak{K}(\mathfrak{S}_1)$, as required.

Accordingly, if we call two systems $\mathfrak{S}_1, \mathfrak{S}_2$ *algebraically equivalent* (with respect to $\mathfrak{K}$), or simply equivalent, when each is algebraically dependent on the other, then any two systems equivalent to a third are equivalent to each other, and so on.

We call a system $\mathfrak{S}$ of elements of an extension field *algebraically reducible* (with respect to $\mathfrak{K}$), or simply reducible, if it is equivalent to a proper part of itself or consists of a single element algebraic over $\mathfrak{K}$; otherwise we call $\mathfrak{S}$ (algebraically) irreducible. It is immediately clear from this definition that in a reducible system containing more than one element, at least one element occurs which is algebraically dependent on the system of the remaining elements. In conjunction with 1. this gives at once:

3. *Every subsystem of an irreducible system is irreducible.*

4. *Every reducible system contains a finite reducible subsystem.*

Purely transcendental extensions. – We call an extension of a field *purely transcendental* if it can be obtained by adjoining an irreducible system $\mathfrak{S}$. If $\mathfrak{S}$ consists of only finitely many elements $x_1, x_2, \ldots, x_n$, then we have an extension of the kind considered at the end of § 5: the field $\mathfrak{K}(\mathfrak{S})$ consists of rational functions of the transcendentals, or indeterminates, $x_1, x_2, \ldots, x_n$ with coefficients in $\mathfrak{K}$. If the system $\mathfrak{S}$ is infinite, then $\mathfrak{K}(\mathfrak{S})$ consists of the elements of all fields $\mathfrak{K}(\mathfrak{S}')$, where $\mathfrak{S}'$ ranges over finite subsystems of $\mathfrak{S}$. The most general purely transcendental extension is therefore obtained by introducing an arbitrary system $\mathfrak{S}$ of infinitely many "indeterminates." Each finite subsystem $\mathfrak{S}'$ of $\mathfrak{S}$ then gives an extension $\mathfrak{K}(\mathfrak{S}')$ of the kind previously considered, and the totality of all fields $\mathfrak{K}(\mathfrak{S}')$ constitutes a field (§ 2, 2.) which arises from $\mathfrak{K}$ by adjoining the irreducible system $\mathfrak{S}$. – Suppose now we have two extensions $\mathfrak{K}(\mathfrak{S}_1)$, $\mathfrak{K}(\mathfrak{S}_2)$, where $\mathfrak{S}_1$ and $\mathfrak{S}_2$ are irreducible systems and the finite or infinite number of their elements (cardinal number, cardinality) is the same; that is, suppose a one-to-one correspondence φ can be established between their elements. Then there is a definite isomorphism π between $\mathfrak{K}(\mathfrak{S}_1)$ and $\mathfrak{K}(\mathfrak{S}_2)$ under which every element of $\mathfrak{K}$ corresponds to itself and the elements of $\mathfrak{S}_1$ and $\mathfrak{S}_2$ correspond in the same way as under φ. This was shown in § 5 for finite systems, and in the infinite case follows at once from the same result by a simple consideration. Thus:

5. *Two purely transcendental extensions $\mathfrak{K}(\mathfrak{S}_1)$, $\mathfrak{K}(\mathfrak{S}_2)$ of a field $\mathfrak{K}$, in which the irreducible systems $\mathfrak{S}_1$, $\mathfrak{S}_2$ contain equally many elements, are equivalent.*

6. *If an irreducible system $\mathfrak{S}$ becomes reducible upon adding an element a, then a is algebraically dependent on $\mathfrak{S}$.*

For the reducible system $\mathfrak{S} + a$ contains a finite reducible system $\mathfrak{T}$. The element a must occur in $\mathfrak{T}$, since otherwise $\mathfrak{T}$ would be a part of $\mathfrak{S}$ and hence irreducible. Let $\mathfrak{T} = \mathfrak{T}' + a$, and suppose $\mathfrak{T}'$ consists of the elements $x_1, x_2, \ldots, x_n$. Since $\mathfrak{T}'$ is irreducible, every field in the sequence

$$\mathfrak{K}, \quad \mathfrak{K}_1 = \mathfrak{K}(x_1), \quad \mathfrak{K}_2 = \mathfrak{K}_1(x_2), \quad \ldots, \quad \mathfrak{K}_n = \mathfrak{K}_{n-1}(x_n) = \mathfrak{K}(\mathfrak{T}')$$

is transcendental over the preceding one. If $\mathfrak{K}_n(a)$ were also transcendental over $\mathfrak{K}_n$, then (§ 5, 9.) every element of $\mathfrak{T}$ would be algebraically independent of the system of the remaining elements, so that $\mathfrak{T}$ would be irreducible. Since this is not the case, a is algebraic over $\mathfrak{K}_n = \mathfrak{K}(\mathfrak{T}')$, hence algebraically dependent on $\mathfrak{T}'$ and therefore also on $\mathfrak{S}$.

7. *Let $\mathfrak{S}$ be an irreducible system (with respect to $\mathfrak{K}$), and let a be transcendental over $\mathfrak{K}$ but algebraically dependent on $\mathfrak{S}$. Then $\mathfrak{S}$ contains a definite finite subsystem $\mathfrak{T}$ with the following properties: a is algebraically dependent on $\mathfrak{T}$; every subsystem of $\mathfrak{S}$ on which a is algebraically dependent contains $\mathfrak{T}$; if any element of $\mathfrak{T}$ is replaced by a, then $\mathfrak{S}$ passes into an equivalent irreducible system; none of the other elements of $\mathfrak{S}$ has this property.*

Proof. Let $\mathfrak{S}'$ denote any subsystem of $\mathfrak{S}$ on which a is algebraically dependent (not excluding $\mathfrak{S}$ itself). Then $\mathfrak{S}'$ has finite subsystems on which a is algebraically dependent. Let $\mathfrak{T}$ be such a subsystem with as few elements as possible. Let its elements be $\alpha_1, \alpha_2, \ldots, \alpha_m$, and let $\mathfrak{T}_i$ $(i = 1, \ldots, m)$ denote the system that remains after omitting α_i from $\mathfrak{T}$ (in the case $m = 1$, the empty system). Since a is algebraically dependent on $\mathfrak{T}$ and algebraically independent of each of the systems $\mathfrak{T}_i$, the system $\mathfrak{T} + a$ is reducible, whereas the systems $\mathfrak{T}_i + a$ are irreducible (Theorem 6). This remains true in the case $m = 1$, because a is transcendental over $\mathfrak{K}$. Since the irreducible system $\mathfrak{T}_i + a$ passes into the reducible system $\mathfrak{T} + a$ by adjoining α_i, the element α_i is algebraically dependent on $\mathfrak{T}_i + a$, and the systems $\mathfrak{T}_i + a$ and $\mathfrak{T} + a$ are equivalent. Likewise $\mathfrak{T}$ and $\mathfrak{T} + a$ are equivalent. Thus we see: *The $m + 1$ systems $\mathfrak{T}_i + a$, $\mathfrak{T}$, and $\mathfrak{T} + a$, and therefore also any two of them, are equivalent.*

We now show that every subsystem $\mathfrak{S}''$ of $\mathfrak{S}$ on which a is algebraically dependent must contain the system $\mathfrak{T}$. To this end let $\mathfrak{U}$ be a finite subsystem of $\mathfrak{S}''$ with as few elements as possible among those on which a is algebraically dependent, and prove that $\mathfrak{U}$ and $\mathfrak{T}$ are identical. Suppose $\mathfrak{U}$ consists of the elements $\beta_1, \ldots, \beta_n$, and let $\mathfrak{U}_i$ $(i = 1, \ldots, n)$ be the system left after omitting β_i from $\mathfrak{U}$. Then the systems $\mathfrak{U}_i + a$ and $\mathfrak{U}$ are irreducible and are equivalent to one another and to the reducible system $\mathfrak{U} + a$. – Because $\mathfrak{T}$ and $\mathfrak{U}$ are smallest systems on which a is algebraically dependent, neither of them can be a proper part of the other. If they were different, we could therefore assume that α_1 is not in $\mathfrak{U}$ and β_1 is not in $\mathfrak{T}$. Let $\mathfrak{V}$ be the subsystem of $\mathfrak{S}$ containing all elements of $\mathfrak{T}$ and all elements of $\mathfrak{U}$, and no other elements. Since $\mathfrak{T}_1 + a$ is equivalent to the system $\mathfrak{T}$ contained in $\mathfrak{V}$, one obtains from $\mathfrak{V}$ a system $\mathfrak{V}'$ equivalent to $\mathfrak{V}$ by replacing α_1 by a. The system $\mathfrak{B}'$ contains $\mathfrak{U} + a$, and since this is equivalent to $\mathfrak{U}$, one obtains a system $\mathfrak{B}''$ equivalent to $\mathfrak{B}'$ by omitting the element a from $\mathfrak{B}'$. Thus $\mathfrak{B}$ would be equivalent to a proper part $\mathfrak{B}''$ of itself, hence reducible. But this is impossible, because $\mathfrak{B}$ is a part of the irreducible system $\mathfrak{S}$.

Thus $\mathfrak{T}$ *must indeed be contained in every subsystem of* $\mathfrak{S}$ *on which* a *is algebraically dependent.* Therefore, if $\mathfrak{S}_i$ $(i = 1, \dots, m)$ denotes the system left after omitting α_i from $\mathfrak{S}$, then a is algebraically independent of the irreducible system $\mathfrak{S}_i$, so that $\mathfrak{S}_i + a$ is likewise irreducible. But $\mathfrak{S}_i + a$ arises from $\mathfrak{S}$ by replacing α_i by a, or, what is the same thing, by replacing $\mathfrak{T} = \mathfrak{T}_i + \alpha_i$ by the equivalent system $\mathfrak{T}_i + a$. Hence $\mathfrak{S}$ and $\mathfrak{S}_i + a$ are equivalent systems. Thus: *One obtains an irreducible system equivalent to* $\mathfrak{S}$ *by replacing one of the elements of* $\mathfrak{T}$ *by* a. – On the other hand, if γ is an element of $\mathfrak{S}$ not occurring in $\mathfrak{T}$, and if $\mathfrak{S}'$ remains after omitting it, then $\mathfrak{S}'$ contains $\mathfrak{T}$; hence a is algebraically dependent on $\mathfrak{S}'$, and the system $\mathfrak{S}' + a$ arising from $\mathfrak{S}$ by replacing γ by a is reducible. It is equivalent to $\mathfrak{S}'$, and since $\mathfrak{S}'$, as a proper part of the irreducible system $\mathfrak{S}$, cannot be equivalent to $\mathfrak{S}$, neither can $\mathfrak{S}' + a$ be equivalent to $\mathfrak{S}$. This completes the proof of 7 in all parts.

8. *Let* $\mathfrak{U}$ *and* $\mathfrak{B}$ *be finite irreducible systems of* m *and* n *elements, respectively; suppose* $n \leq m$ *and that* $\mathfrak{B}$ *is algebraically dependent on* $\mathfrak{U}$. *Then, in the case* $m = n$, *the systems* $\mathfrak{U}$ *and* $\mathfrak{B}$ *are equivalent; in the case* $n < m$, $\mathfrak{U}$ *is equivalent to an irreducible system consisting of* $\mathfrak{B}$ *and* $m - n$ *elements from* $\mathfrak{U}$.

Let $u_1, \dots, u_m$ be the elements of $\mathfrak{U}$, and $v_1, \dots, v_n$ those of $\mathfrak{B}$. Since v_1 is transcendental over $\mathfrak{K}$ and algebraically dependent on $\mathfrak{U}$, Theorem 7 gives at least one element of $\mathfrak{U}$ such that replacing it by v_1 changes $\mathfrak{U}$ into an equivalent irreducible system. This proves the assertion for $n = 1$. For $n > 1$ the proof is by complete induction. Let $\mathfrak{B}'$ be the system consisting of $v_1, \dots, v_{n-1}$. Since it is irreducible and algebraically dependent on $\mathfrak{U}$, we may regard it as proved that $\mathfrak{U}$ is equivalent to an irreducible system $\mathfrak{U}'$ consisting of $\mathfrak{B}'$ and $m - n + 1$ elements from $\mathfrak{U}$. The element v_n is algebraically dependent on $\mathfrak{U}'$, but not on $\mathfrak{B}'$. Hence, if $\mathfrak{T}$ is the smallest subsystem of $\mathfrak{U}'$ on which v_n is algebraically dependent, then $\mathfrak{T}$ contains at least one of the $m - n + 1$ elements from $\mathfrak{U}$. Replacing such an element by v_n turns $\mathfrak{U}'$ into an irreducible system equivalent to $\mathfrak{U}$, consisting of $\mathfrak{B}$ and $m - n$ elements of $\mathfrak{U}$. In the case $m = n$, therefore, $\mathfrak{U}$ and $\mathfrak{B}$ are equivalent.

We also see from this that an irreducible system $\mathfrak{B}$ algebraically dependent on $\mathfrak{U}$ cannot contain more than m elements. Otherwise a proper part $\mathfrak{B}'$ of $\mathfrak{B}$ consisting of m elements would, by what has just been proved, be equivalent to $\mathfrak{U}$. Then $\mathfrak{B}$ would be algebraically dependent on $\mathfrak{B}'$, hence reducible. Thus:

9. *An irreducible system* $\mathfrak{B}$ *algebraically dependent on a finite system* $\mathfrak{U}$ *cannot contain more elements than that system; in the case of equal numbers of elements the two systems are equivalent.*

§ 23. Transcendence Degree and the Division of Fields into Classes.

In the following considerations, in which as before we refer to a ground field $\mathfrak{K}$, the well-ordering theorem again enters into use.

1. *Every system $\mathfrak{S}$ whose elements are not all algebraic with respect to $\mathfrak{K}$ is equivalent to an irreducible subsystem.*

Proof. If $\mathfrak{S}$ contains algebraic elements, then after omitting them a system equivalent to $\mathfrak{S}$ remains, and we may continue with it. We may therefore assume from the outset that every element of $\mathfrak{S}$ is transcendental with respect to $\mathfrak{K}$. If first $\mathfrak{S}$ is finite, and $\mathfrak{T}$ is an equivalent subsystem of $\mathfrak{S}$ with as few elements as possible, then $\mathfrak{T}$ is plainly irreducible.

If the system $\mathfrak{S}$ is infinite, we regard it as well-ordered, $\mathfrak{S} = \{a_\nu\}$, and denote by $\mathfrak{S}_\nu$ the segment of $\mathfrak{S}$ determined by a_ν. We omit from $\mathfrak{S}$ every element a_ν $(\nu > 0)$ which is algebraically dependent on its associated segment $\mathfrak{S}_\nu$. There remains a system $\mathfrak{S}'$ which is irreducible and equivalent to $\mathfrak{S}$. First, a_0, since it belongs to $\mathfrak{S}'$, is algebraically dependent on $\mathfrak{S}'$. Suppose further that it has already been proved that the elements of the segment $\mathfrak{S}_\nu$ are algebraically dependent on $\mathfrak{S}'$. Then the same is true of a_ν, because a_ν either is algebraically dependent on $\mathfrak{S}_\nu$ or belongs to $\mathfrak{S}'$. This gives the equivalence of $\mathfrak{S}$ and $\mathfrak{S}'$. If $\mathfrak{S}'$ were reducible, and if $\mathfrak{L}$ were a finite reducible subsystem of $\mathfrak{S}'$ with as few elements as possible, and a_ν the last element of $\mathfrak{L}$, then, since a_ν is transcendental with respect to $\mathfrak{K}$, it could not be the sole element of $\mathfrak{L}$; and the system $\mathfrak{L}'$ which remains after omitting a_ν from $\mathfrak{L}$, being part of $\mathfrak{S}_\nu$, would be irreducible. Hence, by § 22, 6, a_ν would be algebraically dependent on $\mathfrak{L}'$, and therefore on $\mathfrak{S}_\nu$. But then a_ν would not belong to $\mathfrak{S}'$, a contradiction. Thus $\mathfrak{S}'$ is irreducible, and Theorem 1 is proved.

Now let $\Omega = \mathfrak{K}(\mathfrak{S})$ denote an arbitrary transcendental extension of $\mathfrak{K}$. Then $\mathfrak{S}$ is equivalent to an irreducible system $\mathfrak{S}'$; hence the elements of $\mathfrak{S}$ are algebraic with respect to $\mathfrak{K}(\mathfrak{S}')$. Thus $\mathfrak{K}(\mathfrak{S})$ is an algebraic extension of $\mathfrak{K}(\mathfrak{S}')$, while $\mathfrak{K}(\mathfrak{S}')$ is a purely transcendental extension of $\mathfrak{K}$. We obtain:

2. *Every transcendental extension of a field can be decomposed into a purely transcendental extension followed by an algebraic extension.*

Of course such a decomposition can be made in infinitely many ways. Suppose that $\mathfrak{S}_1$ and $\mathfrak{S}_2$ are two irreducible systems in Ω, on which all elements of Ω are algebraically dependent. Then $\mathfrak{S}_1$ and $\mathfrak{S}_2$ are equivalent. If one of these systems is finite, it was shown above that the other is also finite and contains the same number of elements (§ 22, 9). We shall show that in the case of infinite systems as well the cardinal numbers agree, and hence, by § 22, 5, that $\mathfrak{K}(\mathfrak{S}_1)$ and $\mathfrak{K}(\mathfrak{S}_2)$ are equivalent extensions of $\mathfrak{K}$. For this purpose we first introduce some notation.

Let $\{c_\mu\}$ $(0 \leq \mu < \tau)$ be a well-ordered sequence of elements of an extension field, let τ be a limit number, and suppose that from a certain μ onward all elements c_μ are equal, say equal to a. We then say that the sequence has a limit, and that this limit is a; in symbols

$$\lim_{\mu=\tau} c_\mu = a.$$

Suppose there is a well-ordered double sequence $\{c_{\mu\nu}\}$ $(0 \leq \mu < \tau,\ 0 \leq \nu < \sigma)$, which we think of as ordered in rows and columns, and let the rows be denoted by $\mathfrak{W}_\mu$, so that $\mathfrak{W}_\mu = \{c_{\mu\nu}\}$ $(0 \leq \nu < \sigma)$. If τ is a limit number and every column has a limit

$$\lim_{\mu=\tau} c_{\mu\nu} = c'_\nu \qquad (0 \leq \nu < \sigma),$$

then we say that the sequence $\{\mathfrak{W}_\mu\}$ has the limit $\mathfrak{W}' = \{c'_\nu\}$.

Let $\mathfrak{U}$ and $\mathfrak{V} = \{v_\mu\}$ $(0 \leq \mu < \tau)$ be two irreducible systems, well-ordered in arbitrary ways; suppose that $\mathfrak{V}$ is algebraically dependent on $\mathfrak{U}$. Let v_μ be an arbitrary element of $\mathfrak{V}$, and let

$\mathfrak{B}_\mu$ denote the segment of $\mathfrak{B}$ determined by v_μ. By § 22, 7, $\mathfrak{U}$ contains a definite least subsystem $\mathfrak{L}$ on which v_μ is algebraically dependent. The system $\mathfrak{L}$ may well contain elements that occur in $\mathfrak{B}_\mu$, but it certainly also contains elements that do not occur in $\mathfrak{B}_\mu$. If we replace the last of these by v_μ, then $\mathfrak{U}$ is transformed into an irreducible system, equivalent to $\mathfrak{U}$, with the same ordinal number. For this system, completely determined by $\mathfrak{U}$, $\mathfrak{B}$, and μ, we introduce the notation $\vartheta_\mu(\mathfrak{U}, \mathfrak{B})$.

Now let $\mathfrak{U} = \{u_\nu\}$ $(0 \leq \nu < \sigma)$ and $\mathfrak{B} = \{v_\mu\}$ $(0 \leq \mu < \tau)$ be irreducible well-ordered systems, and suppose that $\mathfrak{B}$ is algebraically dependent on $\mathfrak{U}$. Suppose further that there is a double sequence $\{c_{\mu\nu}\}$ $(0 \leq \mu \leq \tau,\ 0 \leq \nu < \sigma)$, whose rows we denote by $\mathfrak{W}_\mu$. We shall assume the following relations:

1) Each row $\mathfrak{W}_\mu = \{c_{\mu\nu}\}$ $(0 \leq \nu < \sigma)$ is an irreducible system and is equivalent to $\mathfrak{U}$, so that $\mathfrak{B}$ is algebraically dependent on every system $\mathfrak{W}_\mu$.

2) $\mathfrak{W}_0 = \mathfrak{U}$, hence $c_{0\nu} = u_\nu$ $(0 \leq \nu < \sigma)$.

3) If $0 < \mu \leq \tau$ and μ is not a limit number, then

$$\mathfrak{W}_\mu = \vartheta_{\mu-1}(\mathfrak{W}_{\mu-1}, \mathfrak{B}).$$

4) If $0 < \mu \leq \tau$ and μ is a limit number, then the sequence $\{\mathfrak{W}_\lambda\}$ $(0 \leq \lambda < \mu)$ has a limit, and this limit is $\mathfrak{W}_\mu$.

When these conditions are fulfilled, we say that the system sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \tau)$, or also the double sequence $\{c_{\mu\nu}\}$, belongs to the system pair $\mathfrak{U}, \mathfrak{B}$. The premise for this language is therefore always that $\mathfrak{U}$ and $\mathfrak{B}$ are well-ordered irreducible systems and that $\mathfrak{B}$ is algebraically dependent on $\mathfrak{U}$.

The conditions 1) through 4) determine every row by the preceding rows; and since $\mathfrak{W}_0$ is to be $\mathfrak{U}$, we first obtain:

a) *To a system pair* $\mathfrak{U}, \mathfrak{B}$ *there can belong at most one system sequence.* Moreover our conditions immediately give:

b) If the system sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \tau)$ belongs to the system pair $\mathfrak{U}, \mathfrak{B}$, and if $0 < \rho < \tau$, while $\mathfrak{B}_\rho$ is the segment of $\mathfrak{B}$ determined by v_ρ, then the system sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \rho)$ belongs to the system pair $\mathfrak{U}, \mathfrak{B}_\rho$.

Assume now that the system sequence $\{\mathfrak{W}_\mu\}$ belongs to $\mathfrak{U}, \mathfrak{B}$, and select a column $\{c_{\mu\nu}\}$ $(0 \leq \mu \leq \tau)$ from the double sequence. If not all elements of the column are equal to the initial element $c_{0\nu} = u_\nu$, let $c_{\rho\nu}$ be the first one different from u_ν. Then ρ cannot be a limit number; for otherwise, by 4), $c_{\rho\nu}$ would be the limit of $\{c_{\mu\nu}\}$ $(0 \leq \mu < \rho)$, and so would be equal to preceding elements of the column. Since ρ is not a limit number, by 3) $\mathfrak{W}_\rho = \vartheta_{\rho-1}(\mathfrak{W}_{\rho-1}, \mathfrak{B})$; and $\mathfrak{W}_\rho$ is obtained from $\mathfrak{W}_{\rho-1}$ by replacing some element of $\mathfrak{W}_{\rho-1}$ by $v_{\rho-1}$. Since $c_{\rho\nu}$ is different from $c_{\rho-1,\nu}$, it follows that $c_{\rho\nu} = v_{\rho-1}$. But then $c_{\mu\nu} = v_{\rho-1}$ for every $\mu \geq \rho$. For otherwise let μ be the least ordinal number $> \rho$ for which $c_{\mu\nu} \neq v_{\rho-1}$. As before, it follows that μ is not a limit number and that $\mathfrak{W}_\mu$ is obtained from $\mathfrak{W}_{\mu-1}$ by replacing $c_{\mu-1,\nu} = v_{\rho-1}$ by $c_{\mu\nu} = v_{\mu-1}$. But by the definition of ϑ, $\vartheta_{\mu-1}(\mathfrak{W}_{\mu-1}, \mathfrak{B})$ is obtained from $\mathfrak{W}_{\mu-1}$ by replacing a certain element of $\mathfrak{W}_{\mu-1}$ which does not belong to the segment of $\mathfrak{B}$ determined by $v_{\mu-1}$, by $v_{\mu-1}$. The element $v_{\rho-1}$, however, does belong to this segment. Hence:

c) If the system sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \tau)$, with $\mathfrak{W}_\mu = \{c_{\mu\nu}\}$, belongs to the system pair $\mathfrak{U}, \mathfrak{B}$, then in a column $\{c_{\mu\nu}\}$ $(0 \leq \mu \leq \tau)$ either all elements are equal to u_ν, or there is an ordinal number ρ, $0 < \rho \leq \tau$, not a limit number, such that $c_{\mu\nu} = u_\nu$ for $\mu < \rho$, and $c_{\mu\nu} = v_{\rho-1}$ for $\mu \geq \rho$.

Under the same assumptions, select any element $v_{\rho-1}$ from $\mathfrak{B}$. By 3) it occurs in $\mathfrak{W}_\rho$, say as $c_{\rho\nu}$; and, as we have seen, when μ runs through the ordinal numbers $> \rho$, it cannot be replaced by any $v_{\mu-1}$. Thus $v_{\rho-1}$ occurs in every system $\mathfrak{W}_\mu$ whose index satisfies $\mu \geq \rho$. Therefore:

d) If the system sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \tau)$ belongs to the system pair $\mathfrak{U}, \mathfrak{B} = \{v_\mu\}$ $(0 \leq \mu < \tau)$, and if $\mathfrak{B}_\mu$ $(0 < \mu < \tau)$ denotes the segment determined by v_μ, while $\mathfrak{B} = \mathfrak{B}_\tau$, then $\mathfrak{W}_\mu$ $(0 < \mu \leq \tau)$ contains all elements of $\mathfrak{B}_\mu$ (of course not necessarily in the same order as $\mathfrak{B}_\mu$).

We now assert:

e) *To every pair* $\mathfrak{U}, \mathfrak{B}$ *of irreducible well-ordered systems, the second of which is algebraically dependent on the first, there belongs a system sequence.*

As before, put $\mathfrak{U} = \{u_\nu\}$ $(0 \leq \nu < \sigma)$, $\mathfrak{B} = \mathfrak{B}_\tau = \{v_\mu\}$ $(0 \leq \mu < \tau)$, and denote by $\mathfrak{B}_\mu$ the segment of $\mathfrak{B}$ determined by v_μ. In the case $\tau = 1$ we plainly obtain the required system sequence by putting $\mathfrak{W}_0 = \mathfrak{U}$, $\mathfrak{W}_1 = \vartheta_0(\mathfrak{U}, \mathfrak{B})$. For $\tau > 1$, e) is to be proved by induction. If τ is not a limit number, then to begin with there belongs to the system pair $\mathfrak{U}, \mathfrak{B}_{\tau-1}$ a system sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \tau - 1)$; and if we put $\vartheta_{\tau-1}(\mathfrak{W}_{\tau-1}, \mathfrak{B}) = \mathfrak{W}_\tau$, it follows at once that the system sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \tau)$ satisfies all conditions 1) through 4), and hence belongs to $\mathfrak{U}, \mathfrak{B}$.

Let τ be a limit number. If $0 < \lambda \leq \rho < \tau$, then to the pair $\mathfrak{U}, \mathfrak{B}_\rho$ there belongs one, and by a) only one, system sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \rho)$; and by b) the subsequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \lambda)$ belongs to $\mathfrak{U}, \mathfrak{B}_\lambda$. Hence for each $\mu < \tau$ there exists a uniquely determined system $\mathfrak{W}_\mu$ such that the segment of the sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu < \tau)$ extending through some $\mathfrak{W}_\rho$ inclusive is the system sequence belonging to $\mathfrak{U}, \mathfrak{B}_\rho$. We put $\mathfrak{W}_\mu = \{c_{\mu\nu}\}$ and consider a column $\{c_{\mu\nu}\}$ $(0 \leq \mu < \tau)$. If not all elements are equal to u_ν, and if $c_{\rho\nu}$ is the first one different from u_ν, then all elements of the column following $c_{\rho\nu}$ are equal to $c_{\rho\nu}$; for the assumption $c_{\rho\nu} \neq c_{\chi\nu}$, $\rho < \chi$, is incompatible, by c), with the fact that the sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \chi)$ belongs to $\mathfrak{U}, \mathfrak{B}_\chi$. Thus every column, and hence also the sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu < \tau)$, has a limit. We put

$$\lim_{\mu=\tau} \mathfrak{W}_\mu = \mathfrak{W}_\tau, \qquad \mathfrak{W}_\tau = \{c_{\tau\nu}\} \quad (0 \leq \nu < \sigma)$$

and assert that the system sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \tau)$ belongs to $\mathfrak{U}, \mathfrak{B}$. To prove this, we have to show two things:

f) $\mathfrak{W}_\tau$ is irreducible,

g) $\mathfrak{W}_\tau$ is equivalent to $\mathfrak{W}_0 = \mathfrak{U}$.

If the system $\mathfrak{W}_\tau$ were reducible, it would contain a finite reducible subsystem $\mathfrak{L}$. Each element of $\mathfrak{L}$, being the limit of its column, is equal to preceding elements of that column; and since $\mathfrak{L}$ contains only finitely many elements, one can choose $\rho < \tau$ so that all elements of $\mathfrak{L}$ already occur in $\mathfrak{W}_\rho$. But then $\mathfrak{W}_\rho$ would be reducible, contrary to the fact that the system sequence $\{\mathfrak{W}_\mu\}$ $(0 \leq \mu \leq \rho)$ belongs to $\mathfrak{U}, \mathfrak{B}_\rho$. This proves f).

An arbitrary element of $\mathfrak{B}$ may be denoted by $v_{\rho-1}$, where $\rho < \tau$. The element $v_{\rho-1}$ occurs in $\mathfrak{W}_\rho$, say as $c_{\rho\nu}$; and then in $\{c_{\mu\nu}\}$ $(0 \leq \mu < \tau)$, for every $\mu \geq \rho$, one has $c_{\mu\nu} = v_{\rho-1}$, hence also $c_{\tau\nu} = \lim_{\mu=\tau} c_{\mu\nu} = v_{\rho-1}$. Thus $\mathfrak{W}_\tau$ contains all elements of $\mathfrak{B}$. Further, every element of $\mathfrak{W}_\tau$ already occurs in some system $\mathfrak{W}_\rho$ $(\rho < \tau)$, and $\mathfrak{W}_\rho$ is equivalent to $\mathfrak{W}_0$; hence $\mathfrak{W}_\tau$ is algebraically dependent on $\mathfrak{W}_0$. To prove g) it remains only to show that $\mathfrak{W}_0$ is also algebraically dependent on $\mathfrak{W}_\tau$. For this purpose we give a more general proof.

We denote by $\mathfrak{W}_{\mu\nu}$ $(0 \leq \mu \leq \tau,\ 0 < \nu < \sigma)$ the segment of $\mathfrak{W}_\mu$ determined by $c_{\mu\nu}$, and put $\mathfrak{W}_\mu = \mathfrak{W}_{\mu\sigma}$. We further denote by $\mathfrak{W}'_{\mu\nu}$ $(0 \leq \mu \leq \tau,\ 0 < \nu \leq \sigma)$ that subsystem of $\mathfrak{W}_\mu$ which contains all elements of $\mathfrak{W}_{\mu\nu}$, all elements of $\mathfrak{B}_\mu$, and no other element; and put $\mathfrak{W}'_{0\nu} = \mathfrak{W}_{0\nu}$ $(0 < \nu \leq \sigma)$. We now assert:

h) Every system $\mathfrak{W}'_{\mu\nu}$ $(0 \leq \mu \leq \tau,\ 0 < \nu \leq \sigma)$ is algebraically dependent on every system $\mathfrak{W}'_{\rho\nu}$ $(\mu \leq \rho < \tau)$.

This assertion includes, when we choose $\mu = 0$, $\nu = \sigma$, $\rho = \tau$, the assertion that $\mathfrak{W}_0$ is algebraically dependent on $\mathfrak{W}_\tau$. Thus h) proves g), and hence e). If h) did not hold for every system $\mathfrak{W}'_{\mu\nu}$, we would choose among the systems for which it fails one for which the second index ν is as small as possible. We then choose for this $\mathfrak{W}'_{\mu\nu}$ the system $\mathfrak{W}'_{\rho\nu}$ with least index $\rho > \mu$ on which $\mathfrak{W}'_{\mu\nu}$ is not algebraically dependent. Let a be an element of $\mathfrak{W}'_{\mu\nu}$ which is not algebraically dependent on $\mathfrak{W}'_{\rho\nu}$. Then a does not belong to $\mathfrak{W}'_{\rho\nu}$, hence not to $\mathfrak{W}_\rho$, and also not to $\mathfrak{B}_\rho$; consequently it belongs to $\mathfrak{W}_{\mu\nu}$. It must moreover be the last element of $\mathfrak{W}_{\mu\nu}$. Otherwise a would belong to a segment $\mathfrak{W}_{\mu\lambda}$ of $\mathfrak{W}_{\mu\nu}$, and hence also to the system $\mathfrak{W}'_{\mu\lambda}$; since this system, by the minimal property of ν, is algebraically dependent on $\mathfrak{W}'_{\rho\lambda}$, and $\mathfrak{W}'_{\rho\lambda}$ is a part of $\mathfrak{W}'_{\rho\nu}$, the element a would be algebraically dependent on $\mathfrak{W}'_{\rho\nu}$. Thus a is the last element of $\mathfrak{W}_{\mu\nu}$; hence ν is not a limit number, and

$$a = c_{\mu,\nu-1}, \qquad \mathfrak{W}_{\mu\nu} = \mathfrak{W}_{\mu,\nu-1} + c_{\mu,\nu-1},$$

and, since a does not belong to $\mathfrak{B}_\mu$,

$$\mathfrak{W}'_{\mu\nu} = \mathfrak{W}'_{\mu,\nu-1} + c_{\mu,\nu-1},$$

where in the case $\nu = 1$ both $\mathfrak{W}_{\mu,\nu-1}$ and $\mathfrak{W}'_{\mu,\nu-1}$ denote the empty system. Let λ run through all numbers $\geq \mu$ and $< \rho$. By the minimal property of ρ, $\mathfrak{W}'_{\mu\nu}$ is algebraically dependent on every system $\mathfrak{W}'_{\lambda\nu}$; therefore none of these systems may be algebraically dependent on $\mathfrak{W}'_{\rho\nu}$. But, apart from $c_{\lambda,\nu-1}$, all elements of $\mathfrak{W}'_{\lambda\nu}$ belong to $\mathfrak{W}'_{\lambda,\nu-1}$, and hence are algebraically dependent on $\mathfrak{W}'_{\rho,\nu-1}$, and therefore on $\mathfrak{W}'_{\rho\nu}$. Consequently $c_{\lambda,\nu-1}$ cannot be algebraically dependent on $\mathfrak{W}'_{\rho\nu}$. Thus the elements $c_{\lambda,\nu-1}$ must be distinct from the element $c_{\rho,\nu-1}$ contained in $\mathfrak{W}'_{\rho\nu}$. It follows that ρ is not a limit number and

$$(1) \qquad c_{\rho-1,\nu-1} = u_{\nu-1}, \qquad c_{\rho,\nu-1} = v_{\rho-1}, \qquad \mathfrak{W}_\rho = \vartheta_{\rho-1}(\mathfrak{W}_{\rho-1}, \mathfrak{B})$$

holds. Let $\mathfrak{L}$ denote the least subsystem of $\mathfrak{W}_{\rho-1}$ on which $v_{\rho-1}$ is algebraically dependent. By the definition of $\vartheta_{\rho-1}$, $\mathfrak{W}_\rho$ is obtained from $\mathfrak{W}_{\rho-1}$ by replacing the last element of $\mathfrak{L}$ not belonging to $\mathfrak{B}_{\rho-1}$ by $v_{\rho-1}$. In view of (1), this last element must be $c_{\rho-1,\nu-1} = u_{\nu-1}$; that is, every element of $\mathfrak{L}$ belongs either to $\mathfrak{W}_{\rho-1,\nu}$ or to $\mathfrak{B}_{\rho-1}$. If therefore $\mathfrak{L}'$ denotes the system into which $\mathfrak{L}$ is transformed by substituting $v_{\rho-1}$ for $u_{\nu-1}$, then every element of $\mathfrak{L}'$ belongs either to $\mathfrak{W}'_{\rho\nu}$ or to $\mathfrak{B}_{\rho-1}$; thus $\mathfrak{L}'$ is part of $\mathfrak{W}'_{\rho\nu}$. On the other hand $\mathfrak{L}'$ is equivalent to $\mathfrak{L}$, and the element $a = u_{\nu-1} = c_{\rho-1,\nu-1}$ contained in $\mathfrak{L}$ is therefore algebraically dependent on $\mathfrak{W}'_{\rho\nu}$. This contradicts the preceding conclusion. Hence the assertion h) is true for all systems $\mathfrak{W}'_{\mu\nu}$, and e) is proved.

For our purpose the only point of interest is that the system $\mathfrak{B}$, as part of $\mathfrak{W}_\tau$, is mapped one-to-one onto a part of $\mathfrak{U}$ in the similar relation existing between $\mathfrak{U} = \mathfrak{W}_0$ and $\mathfrak{W}_\tau$. If the systems $\mathfrak{U}$ and $\mathfrak{B}$ are equivalent, then in the same way there is a one-to-one mapping of $\mathfrak{U}$ onto a part of $\mathfrak{B}$. By a known theorem of Bernstein it follows that there is also a one-to-one correspondence between $\mathfrak{U}$ and $\mathfrak{B}$, and hence:

3. *Equivalent irreducible systems have the same cardinal number.*

When a transcendental extension Ω of a field $\mathfrak{K}$ is decomposed, by inserting a field $\mathfrak{K}(\mathfrak{S})$, into a purely transcendental extension followed by an algebraic one, all possible such decompositions are obtained from this one by replacing the irreducible system $\mathfrak{S}$ by all possible equivalent irreducible systems. The cardinal number of $\mathfrak{S}$ is therefore completely determined by $\mathfrak{K}$ and Ω. We call it the *transcendence degree of the extension*, or the *transcendence degree of* Ω *with respect to* $\mathfrak{K}$. We assign transcendence degree 0 to algebraic extensions.

Let Ω be a field between $\mathfrak{K}$ and $\mathfrak{M}$, let $\mathfrak{S}$ be a system of elements of Ω irreducible with respect to $\mathfrak{K}$, and let $\mathfrak{T}$ be a system of elements of $\mathfrak{M}$ irreducible with respect to Ω. Suppose Ω is algebraic with respect to $\mathfrak{K}(\mathfrak{S})$, and $\mathfrak{M}$ algebraic with respect to $\Omega(\mathfrak{T})$, so that the cardinal numbers s and t of $\mathfrak{S}$ and $\mathfrak{T}$ are the transcendence degrees of Ω with respect to $\mathfrak{K}$, and of $\mathfrak{M}$ with respect to Ω. Then, with respect to $\mathfrak{K}(\mathfrak{S} + \mathfrak{T})$, all elements of Ω and all elements of $\mathfrak{T}$ are algebraic. But by adjoining all these elements to $\mathfrak{K}(\mathfrak{S} + \mathfrak{T})$ one obtains the field $\Omega(\mathfrak{T})$, which is therefore algebraic with respect to $\mathfrak{K}(\mathfrak{S}+\mathfrak{T})$. Since likewise $\mathfrak{M}$ is algebraic with respect to $\Omega(\mathfrak{T})$, $\mathfrak{M}$ is an algebraic extension of $\mathfrak{K}(\mathfrak{S}+\mathfrak{T})$.

Consider an arbitrary finite subsystem of $\mathfrak{S}+\mathfrak{T}$, and write it in the form $\mathfrak{S}'+\mathfrak{T}'$, where $\mathfrak{S}'$ is part of $\mathfrak{S}$ and $\mathfrak{T}'$ part of $\mathfrak{T}$. The system $\mathfrak{S}'$ is irreducible with respect to $\mathfrak{K}$, and $\mathfrak{T}'$ is irreducible with respect to Ω, hence also with respect to $\mathfrak{K}(\mathfrak{S}')$. If therefore $\mathfrak{S}'$ consists of the elements $x_1,\dots,x_m$, and $\mathfrak{T}'$ of the elements $y_1,\dots,y_n$, then of the elements $x_1,\dots,x_m,y_1,\dots,y_n$, each is transcendental over the field obtained by adjoining the preceding ones to $\mathfrak{K}$. Thus, with respect to $\mathfrak{K}$, none of these elements is algebraically dependent on the system of the remaining ones. Hence the system $\mathfrak{S}'+\mathfrak{T}'$ is irreducible with respect to $\mathfrak{K}$; since it was an arbitrary finite subsystem of $\mathfrak{S}+\mathfrak{T}$, the system $\mathfrak{S}+\mathfrak{T}$ itself is irreducible with respect to $\mathfrak{K}$ (§ 22, 4). The cardinal number of $\mathfrak{S}+\mathfrak{T}$ is denoted by $s+t$, so that the following theorem holds:

4. *An extension composed of two successive extensions of transcendence degrees s and t has transcendence degree $s+t$.*

It is immediate that this remains valid when s or t is 0.

By the *absolute transcendence degree* of a field $\mathfrak{K}$ we mean its transcendence degree over the prime field; and we collect in a class $K_{c,t}$ all fields (or field types) which have the same characteristic c and the same absolute transcendence degree t. Here c may be 0 or any prime number, and t may be 0, a natural number, or any transfinite cardinal number. In each class $K_{c,t}$ there are two notable types. Namely, if one takes a prime field $\mathfrak{P}_c$ and forms a purely transcendental extension of transcendence degree t, one reaches a well-determined type (§ 22, 5), which belongs to the class $K_{c,t}$ and will be denoted by $\mathfrak{R}_{c,t}$. To this type there further belongs, by § 21, 9, a definite algebraically closed type; call it $\mathfrak{A}_{c,t}$. It arises from $\mathfrak{R}_{c,t}$ by algebraic extension, and therefore also belongs to the class $K_{c,t}$.

5. *If a field $\mathfrak{R}'_{c,t}$ of type $\mathfrak{R}_{c,t}$ is algebraically extended to an algebraically closed field $\mathfrak{A}'_{c,t}$, then all fields between $\mathfrak{R}'_{c,t}$ and $\mathfrak{A}'_{c,t}$ belong to the class $K_{c,t}$. Moreover every type of the class $K_{c,t}$ is represented by at least one field between $\mathfrak{R}'_{c,t}$ and $\mathfrak{A}'_{c,t}$.*

That all fields between $\mathfrak{R}'_{c,t}$ and $\mathfrak{A}'_{c,t}$ belong to the class $K_{c,t}$ follows from the fact that they are algebraic with respect to $\mathfrak{R}'_{c,t}$, and that $\mathfrak{R}'_{c,t}$ belongs to this class. Next it follows from our hypotheses that $\mathfrak{A}'_{c,t}$ is of type $\mathfrak{A}_{c,t}$. Let now $\mathfrak{K}$ be any field of the class $K_{c,t}$, and let $\mathfrak{P}_c$ be the prime field contained in $\mathfrak{K}$. We may choose from $\mathfrak{K}$ a system $\mathfrak{S}$, irreducible with respect to $\mathfrak{P}_c$, such that $\mathfrak{K}$ is algebraic with respect to $\overline{\mathfrak{R}}_{c,t}=\mathfrak{P}_c(\mathfrak{S})$. If we algebraically extend $\mathfrak{K}$ to an algebraically closed field $\overline{\mathfrak{A}}_{c,t}$, then this field is also algebraic with respect to $\overline{\mathfrak{R}}_{c,t}$. The fact that $\mathfrak{K}$ belongs to $K_{c,t}$ implies that $\mathfrak{S}$ has cardinal number t, that $\overline{\mathfrak{R}}_{c,t}$ is of type $\mathfrak{R}_{c,t}$, and that $\overline{\mathfrak{A}}_{c,t}$ is of type $\mathfrak{A}_{c,t}$. It follows further (compare the proof of § 21, 9) that $\overline{\mathfrak{A}}_{c,t}$ can be mapped isomorphically onto $\mathfrak{A}'_{c,t}$ in such a way that $\overline{\mathfrak{R}}_{c,t}$ passes into $\mathfrak{R}'_{c,t}$. Under this mapping $\mathfrak{K}$ is placed in isomorphic correspondence with a field between $\mathfrak{R}'_{c,t}$ and $\mathfrak{A}'_{c,t}$; hence the type of $\mathfrak{K}$ is represented among these fields. If $\mathfrak{K}$ itself is algebraically closed, then $\mathfrak{K}=\overline{\mathfrak{A}}_{c,t}$ is of type $\mathfrak{A}_{c,t}$. Therefore:

6. *The type $\mathfrak{A}_{c,t}$ is the only algebraically closed type in the class $K_{c,t}$.*

§ 24. Additional Remarks of Various Kinds.

The Cardinal Number (Power, Number of Elements) of a Field. – From the well-ordering theorem it follows easily that cardinal numbers also form a well-ordered sequence; that the sum and the product of two cardinal numbers, at least one of which is transfinite, is equal to the larger one; and that the system of all finite subsets of an infinite set M has the same power as M. From this one further concludes that the number of integral functions in one indeterminate x over an infinite field is no greater than the number of elements of the field, and that the same is true for the roots of these functions. Algebraic extension of an infinite field therefore does not increase its power. Another simple consequence of the cited theorems is that, in the case $t \geq \aleph_0$ (where $\aleph_0$ denotes the least transfinite cardinal number, the power of the countable sets), the field $\mathfrak{R}_{c,t}$, and hence every field of the class $K_{c,t}$, has cardinal number t; while in the case of finite t, a field of the class $K_{c,t}$, if it is not finite (which can occur only for $c = p$, $t = 0$), has $\aleph_0$ elements. Thus for a field whose elements are not countable, the cardinal number agrees with the absolute transcendence degree. Since the continuum is not countable, it follows (§ 23, 6) that all algebraically closed fields of the power of the continuum and of characteristic 0 are isomorphic. Among these fields are: the field of complex numbers, the field of all algebraic functions of one or more variables, and others.

Lüroth's Theorem. – Beside Theorem II of § 14 stands the following:

If $\mathfrak{K}(x)$ is a simple transcendental extension of $\mathfrak{K}$, then every field between $\mathfrak{K}$ and $\mathfrak{K}(x)$ is also simple with respect to $\mathfrak{K}$.

We can also formulate this theorem as follows: if Ω is a purely transcendental extension of $\mathfrak{K}$ of transcendence degree 1, then every field between $\mathfrak{K}$ and Ω is likewise a purely transcendental extension of $\mathfrak{K}$. The natural question whether the same theorem holds for every purely transcendental extension has not yet been settled.

To arrive at a proof of Lüroth's theorem, introduce beside x a new indeterminate t, and consider the integral functions of t with coefficients in $\mathfrak{K}(x)$. To every such function $g(t)$ there belongs a function obtained from it by multiplication by a factor from $\mathfrak{K}(x)$, primitive with respect to x,

$$\overline{g}(t) = \varphi_0(x)t^n + \varphi_1(x)t^{n-1} + \cdots + \varphi_n(x),$$

which is distinguished by the fact that its coefficients are integral functions of x not all having a common divisor. This function is determined up to a factor from $\mathfrak{K}$; if one wants it to be completely determined, one may further require that in $\varphi_0(x)$ the coefficient of the highest power shall be ε. It is shown in the usual way that the product of two primitive functions is again primitive; from this it follows immediately that from an equation of the form $f(t) = g(t)h(t)$, where $f(t), g(t), h(t)$ are integral functions of t with coefficients in $\mathfrak{K}(x)$, one again obtains a correct equation by replacing these functions by the associated primitive ones.

Let $\varphi(t)/\psi(t)$ be a rational function of t, with coefficients in $\mathfrak{K}$, in reduced form; let m be its degree (> 0), and let y be an element transcendental with respect to $\mathfrak{K}(t)$. Then the function

$$\varphi(t) - y\psi(t)$$

has degree m in t; it is primitive in y, and, as we shall show at once, irreducible in the field $\mathfrak{K}(y)$. For if there were a decomposition

$$\varphi(t) - y\psi(t) = g(t)h(t),$$

then, on passing to the functions primitive in y, the left hand side would not change; hence we may assume at once that $g(t)$ and $h(t)$ are primitive. One of these functions, say $g(t)$, would then have to be free of y, and the other of first degree in y. Ordering by powers of y, one sees that $g(t)$ would have to divide both $\varphi(t)$ and $\psi(t)$, whereas $\varphi(t)$ and $\psi(t)$ were assumed relatively prime. Put

$$y = \frac{\varphi(x)}{\psi(x)}.$$

Then $\mathfrak{K}(y)$ is a field between $\mathfrak{K}$ and $\mathfrak{K}(x)$, and x is a root of $\varphi(t) - y\psi(t)$, which has degree m and is irreducible over $\mathfrak{K}(y)$. Hence $[\mathfrak{K}(x) : \mathfrak{K}(y)] = m$, i.e. the degree of $\mathfrak{K}(x)$ with respect to $\mathfrak{K}(y)$ is equal to the degree of y as a rational function of x.

We now show that the function

$$\psi(x)(\varphi(t) - y\psi(t)) = \varphi(t)\psi(x) - \psi(t)\varphi(x)$$

is primitive in x as an integral function of t. For this purpose put

$$\varphi(x) = a_0 + a_1 x + \cdots , \qquad \psi(x) = b_0 + b_1 x + \cdots ,$$

so that

$$\varphi(t)\psi(x) - \psi(t)\varphi(x) = (a_0\psi(x) - b_0\varphi(x)) + (a_1\psi(x) - b_1\varphi(x))t + \cdots .$$

If all coefficients $\gamma_i = a_i\psi(x) - b_i\varphi(x)$ had a common factor $\chi(x)$, then this factor would also divide all expressions

$$\gamma_i b_k - \gamma_k b_i = (a_i b_k - a_k b_i)\psi(x), \qquad \gamma_i a_k - \gamma_k a_i = (a_i b_k - a_k b_i)\varphi(x),$$

and hence, since (as $m > 0$) we can choose i and k so that $a_i b_k - a_k b_i \neq 0$, it would also divide $\varphi(x)$ and $\psi(x)$, contradicting the assumption that $\varphi(x)$ and $\psi(x)$ are relatively prime. Therefore, if $\varphi(t)\psi(x) - \psi(t)\varphi(x)$ decomposes in the integral domain $\mathfrak{K}[t, x]$ into two factors, neither of them can be a function of x alone. In precisely the same way one shows that neither of the two factors can be a function of t alone.

Let now $\mathfrak{M}$ be an arbitrary field, distinct from $\mathfrak{K}$, between $\mathfrak{K}$ and $\mathfrak{K}(x)$. Then $\mathfrak{K}(x)$ is finite over $\mathfrak{M}$; for if y is an element of $\mathfrak{M}$ not contained in $\mathfrak{K}$, then, as we saw, $\mathfrak{K}(x)$ is finite over $\mathfrak{K}(y)$, and $\mathfrak{M}$ lies between $\mathfrak{K}(y)$ and $\mathfrak{K}(x)$. Let

$$(1) \qquad [\mathfrak{K}(x) : \mathfrak{M}] = n.$$

Then x is a root of an irreducible function of degree n over $\mathfrak{M}$,

$$g(t) = t^n + u_1 t^{n-1} + \cdots + u_n,$$

whose coefficients, since x is transcendental with respect to $\mathfrak{K}$, cannot all occur in $\mathfrak{K}$. Let

$$y = \frac{\varphi(x)}{\psi(x)}$$

in reduced form be one of the coefficients of $g(t)$ which does not occur in $\mathfrak{K}$, and let m be its degree in x. Then $\mathfrak{M}$ lies between $\mathfrak{K}(y)$ and $\mathfrak{K}(x)$, and we obtain $[\mathfrak{K}(x) : \mathfrak{K}(y)] = m$, hence, by (1),

$$(2) \qquad [\mathfrak{M} : \mathfrak{K}(y)] = \frac{m}{n}, \qquad m \geq n.$$

To $g(t)$ there belongs a function primitive in x,

$$\overline{g}(t) = \varphi_0(x)t^n + \varphi_1(x)t^{n-1} + \cdots + \varphi_n(x),$$

where

$$\frac{\varphi_k(x)}{\varphi_0(x)} = u_k \qquad (k = 1, \ldots , n).$$

Since among the functions u_k there also occurs y, $\overline{g}(t)$ has degree at least m in x. Now x is a root of the function $\varphi(t) - y\psi(t)$ belonging to the field $\mathfrak{M}$; this function must therefore be divisible by the function $g(t)$ irreducible over $\mathfrak{M}$. Thus we get

$$\varphi(t) - \frac{\varphi(x)}{\psi(x)}\psi(t) = g(t)h(t),$$

and, on passing to the associated functions primitive in x,

$$(3) \qquad \varphi(t)\psi(x) - \psi(t)\varphi(x) = \overline{g}(t)\,\overline{h}(t).$$

The left hand side of (3) has degree m in x; $\overline{g}(t)$, as we saw, has degree at least m in x. It follows from (3) that $\overline{g}(t)$ has exactly degree m in x, and that $\overline{h}(t)$ is free of x. Since it was further shown that $\varphi(t)\psi(x) - \psi(t)\varphi(x)$ cannot have a factor that is a function of t alone, $\overline{h}(t)$ cannot contain t either. Comparing degrees in t on both sides of (3), we obtain

$$m = n,$$

and from (2)

$$\mathfrak{M} = \mathfrak{K}(y).$$

This proves Lüroth's theorem.

This "rational" proof of the theorem has the advantage that it holds without any restriction for all fields $\mathfrak{K}$, whereas the usual form of the proof contains the assumptions that the field $\mathfrak{K}$ is perfect and has infinitely many elements. In the case of an imperfect field that form of proof requires cumbersome modifications; for finite fields it fails altogether.

The Algebraically Closed Extensions. – Proving the existence of an algebraically closed extension belonging to a given field $\mathfrak{K}$ still gives no method for actually constructing such an extension. Even for arranging the elements of a field $\mathfrak{K}$ into a well-ordered system, to which that proof refers, there is naturally no general method. The construction of an algebraically closed extension is, however, by no means tied to an already carried-out well-ordering of $\mathfrak{K}$. Two examples follow. In both, the ground field has the power of the continuum, of which, as is known, it has not even been established whether it is equal to $\aleph_1$, the second transfinite cardinal number. Of this power are, among others, the field of real or complex numbers, the field of rational functions of one variable z with complex coefficients, and also the field $K(p)$ of the so-called p-adic numbers introduced by K. Hensel.[1]

In the case $\mathfrak{K} = K(p)$, one reaches the goal on the basis of various investigations which Hensel carried out in the cited work and in a paper recently appearing in this journal.[2] The essential point is the proof that every integral function $f(x)$ over $K(p)$ splits completely in a field obtained from $K(p)$ by adjoining a root of a function $g(x)$ which is irreducible over $K(p)$ and whose coefficients are ordinary integers. Thus it is proved that the system of all integral functions over $K(p)$ is equivalent to the system S of integral functions with integral coefficients. While the first-mentioned system has the power of the continuum, the system S is countable and may therefore be arranged in a simple sequence. Once one has decided on some order and denotes the functions so ordered by $f_n(x)$, where n runs through the natural numbers, Hensel's methods allow one, by a finite procedure, to compute the first function from S which is equivalent to the system

$$f_1(x), f_2(x), \ldots, f_n(x),$$

irreducible over $K(p)$, and moreover normal. If this function is denoted by $\varphi_n(x)$, one has a system of normal functions

$$\varphi_1(x), \varphi_2(x), \ldots,$$

in which each segment is subordinated to the following function. Introducing a system of symbols j_n, with the stipulation that j_n is to be a root of $\varphi_n(x)$, one obtains a system of extensions of the field $\mathfrak{K} = K(p)$:

$$(4) \qquad \mathfrak{K}_1, \mathfrak{K}_2, \ldots.$$

Let r_n denote the degree of $\varphi_n(x)$. Then every element of $\mathfrak{K}_n$ can be represented in the form $g(j_n)$, where $g(x)$ is an integral function over $K(p)$ of degree less than r_n. Among the elements of $\mathfrak{K}_n$, r_{n-1} are roots of $\varphi_{n-1}(x)$. If we want every field of the sequence (4) to be part of the following one, so that, as in the case treated in § 16, these fields together constitute the desired algebraically closed field, we must still make a stipulation by which it is decided which of the roots of $\varphi_{n-1}(x)$ contained in $\mathfrak{K}_n$ is to be equal to j_{n-1}. Now it has indeed not yet been possible

[1] Theorie der algebraischen Zahlen I (Leipzig and Berlin 1908).
[2] Vol. 136 (1909), p. 183.

to represent $K(p)$, or any system of the power of the continuum, as a well-ordered system; but $K(p)$ can very easily be represented as an ordered system.[3] On the basis of such an order of $K(p)$, one obtains an order of the integral functions over $K(p)$ by stipulating that the function $g(x) = a_0 + a_1x + \cdots$ shall precede the function $h(x) = b_0 + b_1x + \cdots$ when, in the case $a_k = b_k$ $(k < n)$, $a_n \neq b_n$, the element a_n is earlier than b_n. One can then require that, if $g(x)$ is the first function over $K(p)$ which, for $x = j_n$, goes over into a root of $\varphi_{n-1}(x)$, then $g(j_n) = j_{n-1}$. This solves the problem.

As a second case consider the field $\mathfrak{K} = \mathbb{C}(z)$, obtained from the field $\mathbb{C}$ of complex numbers by adjoining a transcendental element z (the field of rational functions in one variable). Here the system of all integral functions of one indeterminate x over $\mathfrak{K}$ is no longer equivalent to a countable system; yet the construction of an algebraically closed extension of $\mathfrak{K}$ (without the principle of choice) is easily achieved using the means of analysis. Namely, let $\mathfrak{A}$ be the system of all series, convergent for sufficiently small values of r, of the form

$$\sum_{k=q}^{+\infty} a_k r^{k/n} \left(\cos\frac{k}{n}\varphi + i\sin\frac{k}{n}\varphi\right),$$

where r is a positive variable, φ a real variable, n an arbitrary natural number, q an arbitrary integer, $r^{k/n}$ is always to be taken positive, and the coefficients a_k, whose index k runs through all integers from q upward, are arbitrary complex numbers. Each such series represents, in its region of convergence, a single-valued function of r and φ. Two series are regarded as the same element of $\mathfrak{A}$ if, for all pairs of values r, φ in their common region of convergence, they converge to the same value. To two elements α, β of $\mathfrak{A}$ there belong, as is shown in the usual way, two further elements γ, δ, such that within the common convergence region the relations between $\alpha, \beta, \gamma, \delta$, regarded as functions of r and φ, are

$$(5) \qquad \alpha + \beta = \gamma, \qquad \alpha\beta = \delta.$$

If we let equations (5) serve as composition equations for $\mathfrak{A}$, then $\mathfrak{A}$ becomes a system with double composition, which is easily seen to be an algebraically closed field. To each element $R(z)$ of $\mathfrak{K} = \mathbb{C}(z)$ there corresponds in $\mathfrak{A}$ an element α such that, for sufficiently small values of r, $R(r(\cos\varphi + i\sin\varphi))$ becomes equal to α. The field $\mathfrak{K}$ is thus mapped isomorphically onto a subfield $\mathfrak{K}'$ of $\mathfrak{A}$; and since $\mathfrak{A}$ is algebraically closed, the elements of $\mathfrak{A}$ algebraic with respect to $\mathfrak{K}'$ form a field $\mathfrak{A}'$, which is algebraically closed and algebraic with respect to $\mathfrak{K}'$.

Isomorphic Mapping of a Field onto a Subfield. – Let Ω be an algebraic extension of the field $\mathfrak{K}$, let Ω' be a field between $\mathfrak{K}$ and Ω and different from Ω, let α be an element of Ω not contained in Ω', and let $\varphi(x)$ be the irreducible function over $\mathfrak{K}$ with root α. Then the number of irreducible factors of $\varphi(x)$ in the field Ω is greater than in the field Ω'; hence Ω and Ω' are not equivalent extensions. If $\mathfrak{K}$ is a prime field, then Ω and Ω' cannot be isomorphic, because in an isomorphic relation between Ω and Ω' every element of the prime field would remain unchanged, and Ω and Ω' would therefore be equivalent extensions of $\mathfrak{K}$. Thus:

1. *An absolutely algebraic field is not isomorphic to any proper part of itself.*

Since every subfield of an absolutely algebraic field is again absolutely algebraic, we have:

2. *Two subfields of an absolutely algebraic field, one of which is a proper part of the other, cannot be isomorphic.*

The property described in 2 is characteristic of absolutely algebraic fields. For if the field $\mathfrak{K}$ is not absolutely algebraic and x is transcendental with respect to its prime field $\mathfrak{P}$, then $\mathfrak{P}(x^2)$ is a proper part of $\mathfrak{P}(x)$, but the two fields are isomorphic. In contrast, 1 expresses no characteristic property of absolutely algebraic fields; for one sees very easily, for example, that a field of type $\mathfrak{A}_{c,t}$, where t is finite, cannot be isomorphic to any proper part of itself.

[3] Hensel, Neue Grundlagen der Arithmetik. This journal, Vol. 127, pp. 52, 53.

Among the fields that are isomorphic to a proper part of themselves, we have to distinguish two kinds. A representative of the first kind is the field $\mathfrak{R}_{c,1}$. It possesses infinitely many subfields, but, except for the prime field, all are of type $\mathfrak{R}_{c,1}$ by Lüroth's theorem. A field lying between a field $\mathfrak{R}_{c,1}$ and an isomorphic subfield is therefore always again isomorphic to $\mathfrak{R}_{c,1}$. Of another kind is the field $\mathfrak{A}_{c,t}$ for infinite t. By adjoining one transcendental element x, from $\mathfrak{A}_{c,t}$ one obtains a field $\mathfrak{A}_{c,t}(x)$ which again has transcendence degree t, but is plainly not algebraically closed, and so is not of type $\mathfrak{A}_{c,t}$. If, however, we algebraically extend $\mathfrak{A}_{c,t}(x)$ to an algebraically closed field, this field is again of type $\mathfrak{A}_{c,t}$. Here, therefore, between two isomorphic fields lies a non-isomorphic one. Whenever such a case occurs, we have two fields Ω_1, Ω_2 of different types, each of which is isomorphic to a part of the other. The subfields of Ω_1 therefore represent the same types as the subfields of Ω_2.

Contents.